



THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MECHANICAL ENGINEERING

MECH4429 Integrated Capstone Experience
Final Report
2025-2026

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Date:	1 st May 2026	

Abstract

The "Design, Build, and Fly" (DBF) project undertaken by the University of Hong Kong (HKU) Mechanical Engineering students is an aircraft design project which aims to conduct an iterative engineering approach to perform academic research, design, and model-manufacturing, developing a high-performing radio-controlled aircraft. This project presents the team's comprehensive design process in fulfilling AIAA DBF competition requirements, including payload handling, banner deployment and banner release, as well as flight endurance. The aircraft experimented innovative design choices such as a blended airfoil configuration of Clark-Y airfoil and Wortmann FX airfoil, 3D-printed empennage, and a curved-truss fuselage structure for optimal aerodynamic performance and structural integrity. Both advanced simulations and detailed prototyping guide component selection decisions such as the adoption of specific propulsion systems, avionics, and manufacturing processes. Ground and flight tests are scheduled to validate the aircraft's performance, with iterative improvements based on prototype assessments ensuring the design meets the mission and competition objectives. The final competition prototype adopted high-wing configuration, single Clark-Y as the airfoil, flat-plated empennage, tricycle configuration as the aircraft outline, and gripper mechanism as the banner deploy and release mechanism for the competition mission.

Acknowledgement

The team would like to express its gratitude to Dr. CK Chan and Dr. MWL Ko for their invaluable guidance, and Mr Zheng Qianli, the team's student advisor and pilot for his generous support. The team also acknowledges the support of the Department of Mechanical Engineering and the Innovation Wing at the University of Hong Kong, and companies from aviation industries (ANSYS, L'Voyage, and the Safety Collaborative), for providing essential resources, funding and facilities that enabled prototyping and testing. Lastly, the team appreciates the collaboration and efforts of all members, whose dedication has been crucial in the progress of this challenging and enriching endeavour.

Statement of Contributions

- The team designed, built, and flew 4 aircraft prototypes.
- The team achieved carrying 3 plastic ducks and 1 hockey puck as payload for mission 2, and stowed, deployed, towed and released a 4-meter-long banner for mission 3 during flight test.
- The team ranked 44th in proposal round, 28th in report round, and 32nd at the fly-off and achieved ground mission, mission 1 and mission 2. The team ranked 3rd in Asia, and 2nd in Hong Kong.

Individual Contributions of Each Member

Team Member	Work Distribution
Zhao Zehui	Team Leader, Chief Engineer, Aerodynamic Lead, Observer Crew
	Report: Section 1, 2.1, 2.2, 3.1, 3.2, 4.1 – 4.2, Flight Performance and Mission Specific Performance, Testing objectives & schedules, format checking, Complete aircraft performance, Competition Result and discussion, Conclusion
	Design: Main Wing trade studies and specification calculation Banner Mechanism Design, Manufacturing and Testing, Banner Mission Strategy, Banner Material Studies Aerodynamic Design Analysis (Airfoil, Stability, CFD), Multidisciplinary Optimization Framework for Performance Analysis
	Assistance: Landing gear assembly in prototype 1, assisted in fuselage prototype 1 manufacturing, empennage prototype 1 manufacturing, prototype 4 fuselage manufacturing
Fu Lok Hei	Avionics & Propulsion Lead, Administrative Officer
	M2 Sensitivity Analysis, Main Propulsion Configuration, Propulsion Optimization, Avionics and Wiring, Firewall and control systems in manufacturing, Flight Testing Schedule, Static Trust Test, Electronics Test, Flight Test, Pre-Flight Checklist, Propulsion System Performance, Complete Aircraft Performance
Gurung Srazeen	Project Manager, CAD Lead
	Report: Gantt Chart, Detail Design Introduction, Wing Design and FEA

	Design: Wing Model in Wing Helper, High Detailed CAD Model in SolidWorks. Wing Manufacture, Wing Attachment, Servo Placement in Wing.
Jun He Chen	Manufacturing Lead, Structural & Materials Lead, Safety Officer
	Report: Sensitivity Analysis Intro (Section 3.3), Mission 3 Sensitivity Analysis (3.3.2), Landing Gear Configuration (3.4.5), Landing Gear Design Analysis (4.8, 4.8.1-4.8.5), Landing Gear Structure (5.4, 5.4.1-5.4.3), Manufacturing Plan (6.1), Manufacturing Processes Investigated (6.1.1-6.1.4), Manufacturing Processes Selected (6.2), Manufacturing Milestones (6.3), Landing Gear Testing (7.2.3), Grammar Check and Formatting
	Design: Mission 3 Sensitivity Analysis, Landing Gear Trade Studies (Material, Configuration, Suspension, Wheels) and Specification Calculations, Main Landing Gear & Nose Landing Gear Design and Manufacture, Nose Gear Avionics, Landing Gear Testing, Manufacturing Plan
	Assistance: Fuselage Prototype 1, 2 & 3 Manufacturing, Empennage Prototype 2 CAD Design, Wing Prototype 1 Manufacture
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	Report: Fuselage Construction Configuration, System Design Requirements Table, Fuselage Structure Preliminary Design Stage, Fuselage Structure Final Detailed Design Stage, Fuselage FEA Analysis
	Design: Trade Studies (Fuselage Construction & Architectural Configurations, Payload Restraint System & Orientation), Fuselage Specification Calculations, Fuselage Design and Manufacture, Ground Mission & Mission 2 Strategic Design, Passenger and Cargo Compartment Restraint System Design and Manufacture, Overall Aircraft CAD Assembly
Leung Kwun Tung	Manufacturing Lead, Financial Officer
	Empennage Sizing Design, CAD Modelling, FEA Test and Manufacturing 3D Printing Material Studies & Technical Support

	Assistance in Empennage Control Surface, Cooperation with tail section of fuselage
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List of Variables

A	Ampere	L	Lift force
AIAA	American Institute of Aerospace and Aeronautics	LiPo	Lithium-Polymer
AC	Aerodynamic Centre	lb	Imperial Pound
AoA	Angle of Attack	lbf	Pound-force
AR	Aspect Ratio	m	meter
BEC	Battery Eliminated Circuit	M#	Mission Number
b	Wingspan	MAC	Average chord length
C_D	Coefficient of Drag	MTOW	Maximum Take-off Weight
$C_{D,0}$	Coefficient of zero-lift drag	mAh	Milliampere per hour
C_L	Coefficient of Lift	M1	Mission 1
CAD	Computer-Aided-Design	M2	Mission 2
CFD	Computational-Fluid-Dynamics	M3	Mission 3
cm	Centimetre	m	Meter
CFRP	Carbon Fibre Reinforced Polymer	mm	Millimetre
CG	Centre of Gravity	oz	Ounce
CLK-Y	Clark-Y	τ	Period
COTS	Commercial Off-the Shelf	P	Battery Power
CR	Cargo	PS	Passengers
ρ	Density of air	RAC	Rated Airplane Cost
DBF	Design Build and Fly	RPM	Revolution per Minute
D	Drag force	S_h	Horizontal Stabilizer Area
ζ	Damping Ratio	S_v	Vertical Stabilizer Area
E	Total Energy Consumption	S	Planform Area
η	Efficiency	STOL	Short Take-Off and Landing
ESC	Electronic Speed Controller	s	Second
ft	Imperial Foot	T/W	Thrust to weight ratio
FEA	Finite Element Analysis	t	Time required
f	Frequency	t_2	Time to half the Amplitude

F_1	Natural Frequency	V	Voltage
g	Gravitational constant	V_{cruise}	Cruise Velocity
GM	Ground Mission	V_h	Horizontal Stabilizer Volume Coefficient
GPS	Global Positioning System	V_v	Vertical Stabilizer Volume Coefficient
HKU	The University of Hong Kong	UBEC	Universal Battery Eliminated Circuit
in	inches	W	Watts
I	Current	WFX	Wortmann-FX 63-137
K	Oswald Efficiency	Wh	Watt-hour

1 Introduction

This design report details the aerodynamic and structural design, manufacturing, and testing of GRIFFIN, the bush model aircraft from the University of Hong Kong DBF team for the 2026 American Institute of Aeronautics and Astronautics (AIAA) Design, Build, Fly (DBF) competition. The competition aircraft consisted of a 1.25-meter wingspan that could carry a payload consisting of 3 passengers and 1 cargo, in a total of 0.23kg, and a 4-meter banner, in a total of 0.3kg to complete technical inspection, three flight missions and one ground mission. The MTOW of the aircraft is 2.42 kg.

The GRIFFIN is a single-motor tractor with a tricycle landing gear formation, a high rectangular wing, and a conventional thin-plate backward-tapered tail on a truss-structured fuselage. The single motor was selected to reduce the overall weight of the aircraft. The tricycle landing gear was chosen to provide a larger ground clearance for the banner towing mechanism needed for Mission 3 (M3), as well as for its inherent stability. The high wing was opted for to facilitate the design and integration of the payload and avionics compartments. The conventional tail was selected thanks to its structural simplicity and excellent stability characteristics.

In this competition, the team aimed to carry 3 passengers and 1 cargo in Mission 2 (M2) to maximize aircraft speed, and a 4-meter-long, 0.8-meter-wide banner in M3 to maximize banner-sizing score. The Griffin was constructed out of plywood, balsa, and CFRP strips and rods, while the banner mechanism was made of light-weight pongee fabric. The team designed and manufactured 4 prototypes. Prototype 1 carried 48 passengers and 16 cargoes alongside a 2-meter-long banner, whereas prototypes 2 to 4 carried 3 passengers and 1 cargo alongside a 4-meter-long banner.

The team ranked 44th in the proposal round, 28th in the report round, 32nd overall worldwide, 3rd in Asia and 2nd in Hong Kong. The team also completed the Ground Mission in 36 seconds, successfully attempted Mission 1, and completed 8 laps in Mission 2.

2 Management Summary

2.1 Team Organization Chart, Responsibility and Roles

The HKU DBF team is an entirely student-managed team from HKU consisting of 17 members, 6 of whom are seniors and the rest underclassmen. The team is financially supported by the HKU Innovation Academy, HKU Mechanical Engineering Department, Engineering Faculty, and by various grants and sponsorships. The team is also technically advised by alumni and academics in periodical design reviews.

The HKU DBF technical team is divided into 3 categories, where green indicates the academics and pilot; blue indicates the team leader, 6 technical divisions and 1 helper team, and yellow indicates the administrative division as seen in Figure 1 The 2025-2026 HKU DBF technical team division of duties. The team leader manages the team's societal functions like member intake, events, and direct communication with the Engineering Faculty and sponsorships. The leader also works on schedule planning with the project manager, technical administration with the administrative officer, design with the design officer, and finance with the financial officer.

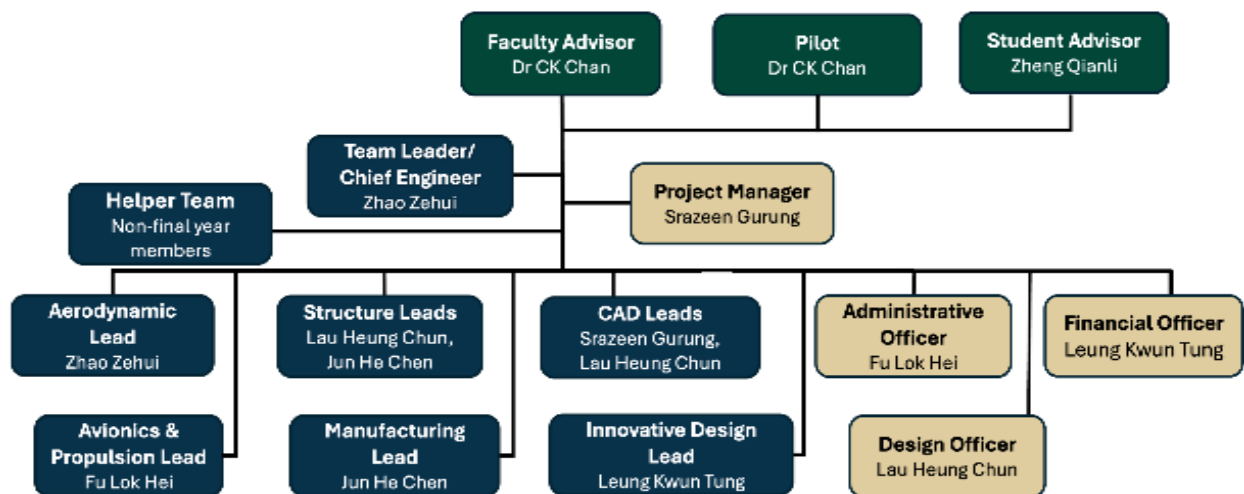


Figure 1 The 2025-2026 HKU DBF technical team division of duties

On the technical side, each sub-team has a team lead overseen by the chief engineer who manages the overall project. The chief engineer is responsible for maintaining and tracking the project timeline and communicates directly with the team leads setting up key milestones, meetings, and deliverables. All technical leads manage logistics and organize design reviews with the supervisor.

Project deliverables are set out by the chief engineer with all technical leads during regular design reviews and delegated amongst helpers in the team based on individual skills and interests. Members work closely with their team leads who report back to the chief engineer on deliverable updates. Any complications that arise or major design decisions that cause progress delays result in a full team meeting to analyze the advantages and drawbacks of the subsequent steps forward in the design. Table 1 shows the roles and responsibilities of each division.

Team	Tasks	Skills
Aerodynamics	Aerodynamic analysis for sizing and desired performance	Fundamental aerodynamics, parameter derivation, CFD simulation
Propulsion & Avionics	Motor and propeller selection, electronics selection for stability and control	Analysis of required and available power; electronics, stability and control theory
Structure	Design airframe and detailed structure, conduct testing to ensure structural stability	Structural analysis and FEA knowledge
CAD	Model design and modification	CAD and FEA knowledge
Manufacturing	Manufacturing the aircraft	Machine tools and material property analysis.
Innovative Development	Testing on various innovative materials	Knowledge on 3D printing techniques and 3D print materials

Table 1 Team Roles and Individual Skillsets

2.2 Background

This year's flight objective was to design, build, and test a banner towing bush plane, conduct charter flights to pay for the airplane in Mission 2, and start a banner towing business which was simulated in Mission 3. Mission 1 (M1) is a delivery flight that demonstrates the aircraft's flight capabilities without a payload, and the team has to fly 3 laps in a 5-minute time window. Mission 2 (M2) is a chartered flight that includes passengers and cargo, where the score will be added up and charged according to the number of passengers and cargoes carried. Mission 3 (M3) is a launch

flight that demonstrates the stowing, towing, deployment and release of the onboard banner. Finally, the Ground Mission (GM) is a timed demonstration of the aircraft assembly, payload installation, banner mechanism installation, and flight control checks. The competition was held at the Textron Aviation Employees' Flying Club in Wichita, Kansas, USA from the 16th to the 19th of April 2026.

The design of the aircraft was divided into three phases: conceptual, preliminary, and detailed design. During the conceptual design phase, the competition mission requirements were analyzed and mapped into design constraints to determine the overall aircraft configuration, together with geographical considerations which turns Griffin into a portable, detachable aircraft to be shipped to the US. The preliminary design phase included trade studies focused on aircraft sizing, performance, and stability. The detailed design included sub-system specification calculations, modelling and manufacturing. Flight tests and iterations were conducted to verify and improve the aircraft's performance.

2.3 Scheduling and Milestones

The Gantt Chart in Figure 2 was used to oversee the progression of the team and the timeline for objectives and milestones related to the development of the GRIFFIN and to provide a clear visualization of the past and the path forward for the team. The Gantt Chart structures the development around three primary prototypes. Prototype 1 (V1) was designed to validate the maximum passenger/cargo configuration (48:16 ratio). Following flight testing, insights from V1 informed the strategy for Prototype 2 (V2), which pivoted to a focus on the minimum viable configuration to maximize speed. Prototype 3 (V3) serves as a refinement of V2, with specific improvements to the cargo compartment and banner release mechanism and more importantly a semi-monocoque structure to improve the aerodynamics of the aircraft. Whilst the prototyping strategy has been sound, it should be noted that the manufacturing and testing phase of Prototype 1 took considerably longer than anticipated. This delay directly impacted the original schedule, leading to the cancellation of the planned Prototype 5 to ensure the team can focus resources on the remaining objectives.

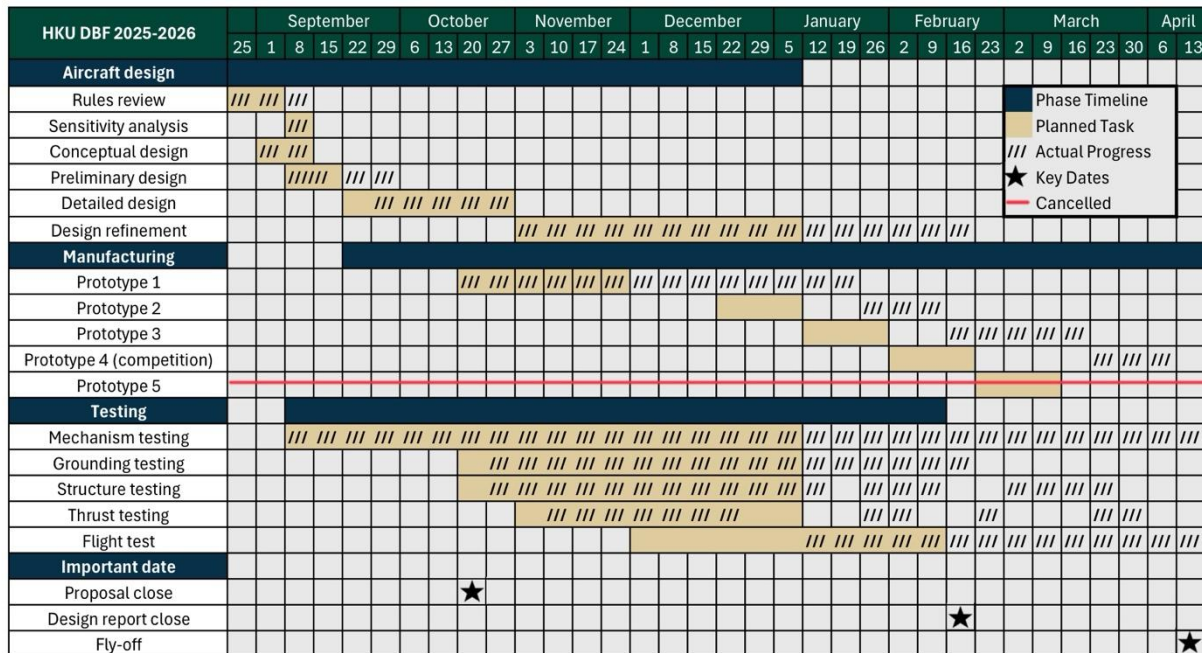


Figure 2 Gantt Chart with Planned Tasks and Actual Progress

3 Aircraft Design Methods

3.1 Conceptual Design

The team analyzed the competition rules to identify the key performance parameters that would maximize the overall flight score during the conceptual design phase. The design of the aircraft was divided into three phases: conceptual, preliminary, and detailed design. During the conceptual design phase, the competition mission scoring framework, design requirements in section 3.2, and sensitivity analysis in section 3.3 were analysed in depth, which then determined the design constraints for the overall aircraft configuration in section 3.4. These were then coupled with geographical considerations to help transform Griffin into a portable, detachable aircraft to be shipped to US.

3.2 System Design Requirements

As outlined in AIAA DBF Rules 2026 [1], the aircraft's general requirements along with M2 and M3 requirements, are outlined in Table 2 Subsystem Design Requirements, which guided the team

in the preliminary design analysis and final design of the GRIFFIN. Compulsory requirements and preferred desirable characteristics are labelled as “-C” and “-D” respectively.

Category	Label	Design Requirements	Req. No
Main Aircraft	Structures	The airplane maximum wingspan is 5 feet.	AC-1.1
		All structure, components, and payloads must be mechanically fastened.	AC-1.2
		Aircraft must be strong enough to handle payload weight.	AC-1.4
	Aerodynamics	Aircraft must achieve stable level flight in all flight missions.	AC-2.1
		Aircraft should maximise L/D to maximise efficiency.	AD-2.2
		Aircraft wingspan must be a maximum of 1.524 m.	AC-2.3
		Aircraft should be able to fly at an altitude of 50.3m AGL.	AC-2.4
	Propulsion	Total stored energy of the propulsion battery $\leq 100\text{Wh}$	AC-3.1
		The propulsion system must support the plane flying for 5 minutes	AC-3.2
		The maximum current rating of an arming fuse is 100 amps.	AC-3.3
	Performance	The aircraft must have sufficient T/W and W/S to take-off within the runway length of approximately 152.4 m (500 ft).	AC-4.1
General	Payload	The airplane must be capable of carrying a minimum number of three passengers and one cargo.	GC-1.1
		The passengers must be carried in one single passenger compartment on a single plane parallel to the horizontal plane while in flight.	GC-1.2
		There must be a separate compartment for passengers and cargo by a solid bulkhead and can only be accessed by one door for each.	GC-1.3

		Each door or hatch must remain rigidly attached to the airplane when opening and in the open configuration.	GC-1.4
		The passengers must be fully restrained in a vertical and upright orientation and prevent movement during all phases of flight.	GC-1.5
		The cargo must be carried in a separate cargo bay(s) that may be in front of, behind, or below the passengers.	GC-1.6
Banner	Structures	The banner mechanism must perform stow, tow, deploy and release event.	BC-1.1
	Performance	The aspect ratio of the banner must be within 5 and the minimum length of the banner must be 25.4 cm.	BC-2.1
		The banner must be remotely deployed in flight and remain in the vertical orientation position.	BC-3.1
		The banner should be deployed at the first downwind direction and released at the last lap.	BD-4.1

Table 2 Subsystem Design Requirements

3.3 Sensitivity Analysis

This sensitivity analysis set out to identify the parameters with the greatest impact on the aircraft's overall performance, helping guide design decisions and optimize aircraft capabilities across various mission scenarios. Sensitivity analyses were conducted on MATLAB [2] and Excel [3] by varying parameters such as the number of passengers, cargo number, and battery capacity for mission 2 (M2) as well as the lap number, banner length, and the rated airplane cost (RAC) for mission 3 (M3). The sensitivity of each parameter was then assessed using graphical representations of parameter variations, their corresponding impacts on performance metrics, and the steepness of the expected trends.

The ground mission (GM) score is determined by comparing the team's mission completion time against the fastest time achieved across all competing teams. To quantify this in a sensitivity analysis, the team established a baseline benchmark, making an educated assumption of the fastest

possible GM time, corresponding to the minimum payload configuration of 3 passengers and 1 cargo.

For mission 1 (M1), one point is to be awarded after a successful mission. For the purposes of this sensitivity analysis, Mission 1 was always assumed to be successfully completed.

3.3.1 Mission 2 Sensitivity Analysis

For M2 (see Figure 3), all parameters increase linearly with the Mission 2 score, except for battery capacity. The number of laps has the steepest gradient among all parameters, meaning this is the most sensitive factor for Mission 2's scoring. This is followed by the number of cargoes, number of passengers and battery capacity. Interestingly, the team needs to demonstrate the plane's capability to carry at least 3 passengers for each piece of cargo in the technical inspection.

To determine which parameter to prioritize, the team had to account for the effect of changing the number of passengers and cargoes simultaneously rather than changing the two variables individually. By combining the two graphs, an increase of 50% in the number of cargoes and passengers yields a 6.99% higher score compared to a 50% increase in lap count. Therefore, the team chose to maximize the number of payloads in prototype 1. In prototype 2 to 4, the team chose to maximize the number of laps by building a smaller aircraft to see its effect on the mission score

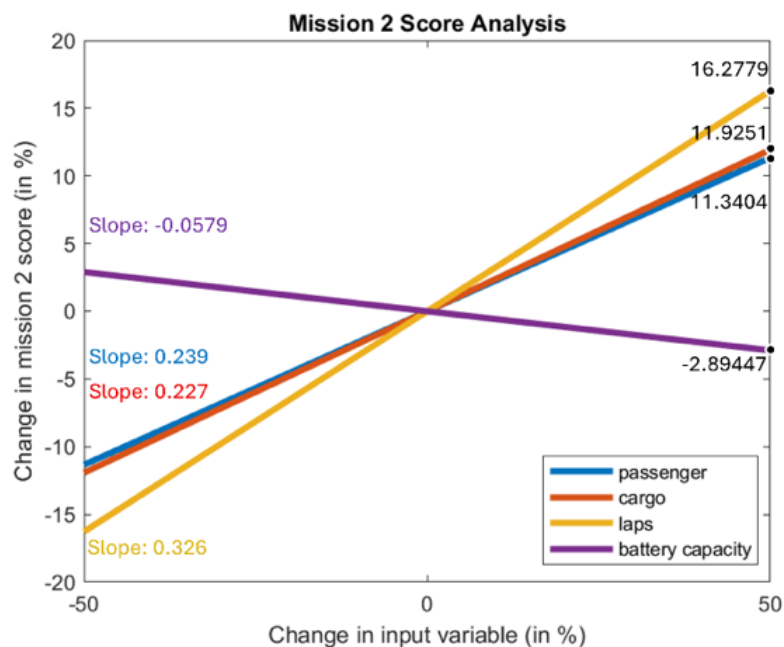


Figure 3 Mission 2 Score Analysis

3.3.2 Mission 3 Sensitivity Analysis

Figure 4 shows the sensitivity analysis for M3. Both the banner length (red) and number of laps (green) exhibit the same linear relationship with the M3 score when independently varied. This signifies that maximising either one yields a proportional benefit, although admittedly, increasing the banner length is more readily feasible if the team is willing to accept the simultaneous increase in drag. However, with the lightest payload configuration (1 cargo, 3 passengers) adopted in the final prototype, design emphasis ultimately shifted toward maximizing aircraft speed (and thus lap count) to benefit from the lighter airframe. With this change in the design approach, the banner length was finalized to a length that is not too long or too short in an attempt to balance the banner's M3 score contribution against drag penalties, as advised through aerodynamic analysis.

On the other hand, the RAC (orange), governed by the aircraft's wingspan, follows an inverse exponential trend where a lower wingspan yields a disproportionately higher score. Therefore, the team lowered the wingspan of the aircraft from 1.524 meters in the first prototype to 1.245 meters in the final design, which, in turn, lowered the RAC from 1 to 0.954. This reduction was accomplished while adhering to the competition requirements of achieving a minimum RAC of 0.9. With this RAC and wingspan, the team predicted that the final prototype would be capable of towing a 4-meter-long banner at a cruise speed of at least 17 m/s to complete around 7 laps in M3.

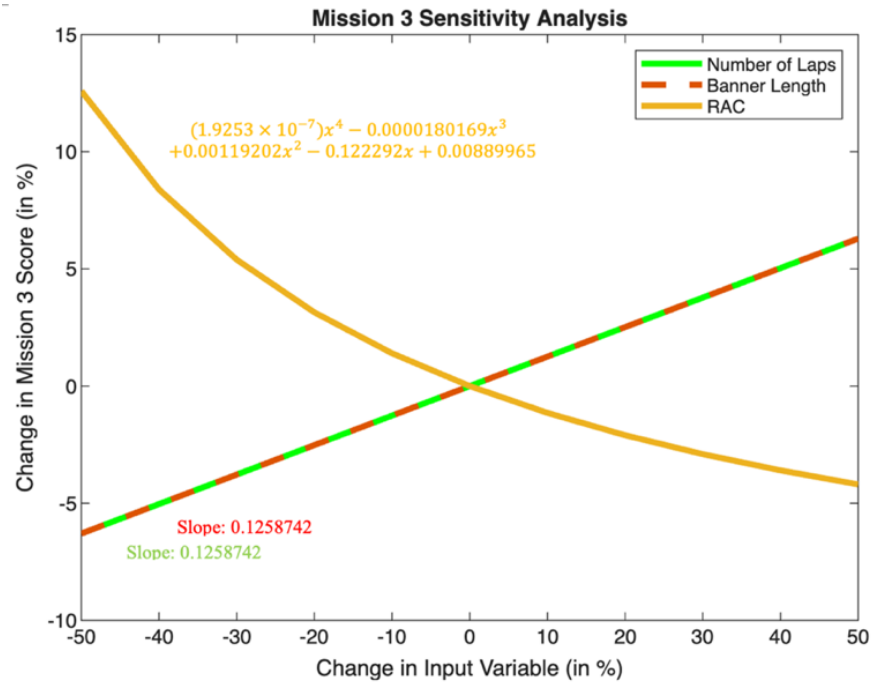


Figure 4 Sensitivity analysis illustrating a linear correlation between the M3 score and both lap count (green) and banner length (red), alongside an inverse exponential relationship with the RAC (orange)

3.4 Aircraft Configurations Decision Matrix

The characteristics of the aircraft configurations are identified and prioritized based on the 5 steps of a team-derived selection process as shown in Table 3.

No.	Step	Description
1	Identify Characteristics	The most desirable characteristics of the configuration choice are identified.
2	Weighting	The characteristics are given a weighting out of 100 (most important) based on their impact to mission performance.
3	Identify Options	Potential suitable configuration choices are examined further based on research and team experience.
4	Scoring	The potential choices are scored out of 1 (best performance) based on how well they perform in relation to each characteristic.
5	Total Score	The total score for each choice is calculated as the sum of the scores for the characteristics multiplied by their respective weights, with the

		choice of the highest total score being selected. The total score formular is shown below: Total Score = (Weight 1) x (Factor 1) + (Weight 2) x (Factor 2) + (Weight 3) x (Factor 3) ...
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Table 3 Configuration Selection Process

3.4.1 Main Aircraft Wing Configuration

The configurations considered for the wing planform were swept-back wing, rectangular wing and tapered wing as seen in Table 4 Wing Platform Configuration. The score given to each wing planform ranges from -1 to 1, where -1 is the least effective choice. While both the swept back and tapered wings provide better stability, the swept back wing reduced the most weight compared to rectangular and tapered wings, but was also the most time-costly to manufacture of all the wing planforms. With reference to Gudmundsson [4], a rectangular wing provides better aerodynamic efficiency than that of the other two planforms under low Re flight (less than 1×10^6). As Reynold's number is expressed as

$$Re = \frac{\rho V D}{\mu},$$

Where ρ is the air density, μ is the dynamic viscosity, V is the velocity of the air flow and D refers to the size of the object, a low Reynold's number indicates low speed (subsonic) flight, which is suitable for DBF aircraft design consideration.

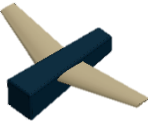
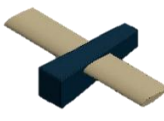
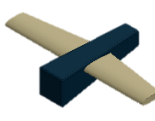
				
Wing Planform	Weighting	Swept Back	Rectangular	Tapered
Manufacturability	20	0	1	0
Aerodynamics	20	0	0.5	0
Flight stability	25	1	0.5	1
Lift	15	0.5	0.25	0.25
Drag	10	0.25	0.5	0.25
Complexity	10	0	1	0
Total Score	100	35	61.25	31.25

Table 4 Wing Platform Configuration

3.4.2 Tail Attachment Configuration

The tail attachment configuration was evaluated across three options as shown as Table 5: conventional, V-tail, and T-tail, using a weighted decision matrix with criteria of manufacturability (15%), weight (25%), structural rigidity (20%), drag (20%), and flight stability (20%). The conventional tail achieved the highest total score (+0.95 / +95%), excelling in weight, rigidity, drag, and stability as a result of simple attachment and decoupled control surfaces. The V-tail scored +0.59 (+58%), benefiting from a lower weight but penalized by complex fuselage integration and coupled pitch-yaw response. The T-tail ranked lowest (+0.455/ +45.5%), suffering from increased weight and structural stress at the horizontal-vertical junction, with minimal aerodynamic advantage at dedicated flying speed.

Based on the above scoring, the final selection will be conventional tail layout. This configuration provides the best overall balance of performance, manufacturability, and practicality for Mission 3 banner towing stability and repeated flight testing.



Empennage	Weighting	Conventional	V-tail	T-tail
Manufacturability	15	0.9	0.2	0.4
Weight	25	1.0	0.8	-0.3
Structural Rigidity	20	1.0	0.7	0.6
Drag	20	0.9	0.6	0.4
Flight Stability	20	0.9	0.5	0.6
Total Score	100	94.5	59.0	45.5

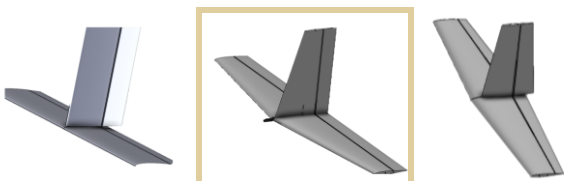
Table 5 Tail Attachment Configuration

3.4.3 Main Aircraft Tail Configuration

The empennage configuration was evaluated in Table 6 across three distinct planform shapes: rectangular, tapered, and trapezoidal. A decision matrix was constructed, taking into account the

following weighted criteria: manufacturability (12%), structural weight (24%), flight stability (36%), and drag characteristics (28%).

Rectangular scored the lowest (+30.8%) due to poor drag and stability despite excellent manufacturability and weight. Tapered achieved the highest score (+81.2%), offering the best balance of low induced drag, superior lift distribution, and good stability for low-speed banner towing. Trapezoidal type ranked at middle among them of 72.2%. As a result, the final selection of the main aircraft tail configuration is tapered planform for both the horizontal and vertical stabilizers, optimising the M3 performance while maintaining acceptable weight and manufacturability.



Empennage	Weighting	Rectangular	Tapered	Trapezoidal
Manufacturability	12	1.0	0.9	0.60
Weight	24	1	0.8	0.75
Flight Stability	36	0.4	1	0.8
Drag	28	-0.7	0.75	0.65
Total Score	100	30.8	81.2	72.2

Table 6 Main Aircraft Tail Configuration

3.4.4 Main Propulsion Configuration

For UAV aircrafts, there are 3 common configurations: tractor (motor at the front), pusher (motor at the back), and top-mounted (motor at the top). For dual motor aircrafts, the most common configuration is wing-mounted. In Table 7, these four configurations are compared according to the weighted criteria: manufacturability (10%), weight (15%: M2 related), stability (20%: M3 related), drag (20%: M3 related), ground clearance (10%), propeller efficiency (15%: M2 related), and noise and vibration (10%). A tractor design was selected for its ease of manufacturing, favourable balance between stability and aerodynamic drag, and sufficient ground clearance.

					
Motor	Weighting	Tractor	Pusher	Wing Mounted	Top Mounted
Manufacturability	10	0.7	0.2	0	0
Weight	15	0	0	-1	-0.2
Stability	20	0.6	-0.5	0.3	-0.5
Drag	20	0	1	0.5	1
Ground Clearance	10	0.8	-1	0	1
Propeller Efficiency	15	0.8	0.6	0.5	1
Noise and vibration	10	0.6	0.4	0.3	0.2
Total Score	100	45	15	11.5	34

Table 7 Main Propulsion Configuration (scale from -1 to 1, with 1 being highest)

3.4.5 Landing Gear Configuration

Table 8 shows a trade study conducted to select the optimal landing gear configuration for the AIAA competition. The tricycle landing gear achieved the highest score of 52.5 thanks to its superior performance across mission-specific critical criteria.

Although the taildragger configuration offers slight advantages in weight and cost, this configuration is prone to ground looping and exhibits dangerous directional instability during take-off; as the tail lifts, the centre of gravity shifts ahead of the main wheels, requiring constant rudder correction to counteract the swerving induced by propeller forces [14].

The tricycle gear was, thus, selected as the most balanced and reliable solution given the nature of the competition, providing exceptional stability during take-off and landing, intuitive and predictable ground steering essential for precise operations, and a structurally straightforward attachment point for the banner towing mechanism at the rear fuselage to minimize interference during the ground mission [16].

					
Landing Gear Type	Weighting	Taildragger	Tricycle	Bicycle	Quadricycle
Ground Handling/Steering	15	-0.25	0.75	-0.5	0.25
Landing Stability	15	-0.5	0.75	-0.25	0.5
Banner Towing Integration	15	-0.25	0.5	-0.5	1
Weight/Drag	20	0.5	0.25	0.75	-0.25
Cost	20	0.75	0.5	0.25	-0.5
Design/Manufacturing Simplicity	15	0.75	0.5	0.25	-0.25
Total Score	100	21.25	52.5	5	7.5

Table 8 Weighted performance comparison of four landing gear configurations across six criteria, with the tricycle configuration achieving the highest score of 52.5 (highlighted in yellow)

3.4.6 Fuselage Construction Configuration

In the conceptual design stage, the general fuselage construction configuration was determined. Through a trade study comparing the three primary construction methods, namely, the Truss Structure, Semi-Monocoque Structure, and Monocoque Structure, the Truss Structure was identified as the most optimal configuration for the competition requirements, as detailed in Table 9

The Truss Structure was selected based on its manufacturing simplicity, ease of repair, and superior weight efficiency. Given the project's constrained schedule and limited manpower, manufacturability and repairability were assigned the highest weightings in the decision matrix. In these categories, the Truss Structure configuration significantly outperformed the alternative methods. Weight efficiency and structural rigidity were prioritized as secondary factors. Compared to Semi-monocoque and Monocoque designs, the Truss Structure offers the highest weight

efficiency, as the alternatives require a greater number of structural members, such as formers and stringers, to maintain their shape.

While the Semi-monocoque configuration provides the greatest structural rigidity, Finite Element Analysis (FEA) validated that the rigidity of the Truss Structure is more than sufficient for the operational requirements of the aircraft. A detailed discussion of this analysis is provided in the Section 5. Consequently, the Truss Structure was finalized as the primary construction method for the fuselage.

Fuselage Construction Type	Weighting	Truss Structure	Semi-monocoque	Monocoque
Manufacturability	30	1	-0.5	-0.5
Repairability	30	1	-0.5	-1
Weight Efficiency	20	1	0	0.5
Structural Rigidity	20	0.5	1	0.5
Total Score	100	90	-10	-25

Table 9 A weighted decision matrix was utilized to evaluate three fuselage construction configurations against four primary criteria. Using a scale of -1 to 1, the Truss Structure achieved the highest score of 90 out of 100, establishing it as the most optimal construction method.

3.4.7 Banner Mechanism Position

Three positions of the banner mechanism were considered and compared in Table 10. The front position was the most desirable position as it is the closest to the CG of the aircraft, however, it was noted that this would affect the installation of the main landing gear. The back position would create a bigger moment arm for the aircraft which would negatively impact the overall. Thus, the middle position was chosen as the ideal compromise for the mechanism positioning.

				
Release	Weighting	Front	Middle	Back
Manufacturability	25	0.5	0.75	0.25
Complexity	20	0.25	0.75	0

stability	20	0.75	0.5	0
Release Success	35	0.75	0.75	0.75
Total Score	100	56.25	70	32.5

Table 10 Banner Mechanism Position

4 Preliminary Design

The preliminary design was an iterative approach used to optimize the engineering design for the main aircraft and the banner mechanism. A combination of simulations, sensitivity analyses, and trade studies in the conceptual design stage shaped the final preliminary design. This preliminary design was first workshopped using the iterative approach outlined in section 4.1. The design was then used to optimize the engineering design for the aircraft sizing as shown from sections 4.2 to 4.8, aerodynamic analysis in section 4.9, predictive performance mission model analysed in section 4.10, predictive performance in section 4.11, and uncertainties and assumptions of the model in section 4.12.

4.1 Design Methodology

The team's design approach combines methods from previous HKU DBF projects with new techniques learned in coursework and academic research. Using structured design and analysis, the team developed the final aircraft shown in Figure 5 Structured Design and Analysis Diagram. From the sensitivity analysis in section 3, it is determined that M2 varied the most when maximizing payload, and M3 varied the most when maximizing banner size and minimizing the RAC. As stated in AIAA DBF competition rules, RAC is expressed as

$$\text{RAC} = 0.05 * (\text{WS} / 12) + 0.75, \text{ where}$$

WS refers to the wingspan. Initial empty weight and design choices were based on similar aircraft used in past competitions. These choices shaped the first prototype, named GRIFFIN.

From there, the design focused on two key areas: power and wing loading. Aerodynamic analysis began with selecting the airfoil and defining the shape of the wing, tail, and fuselage. These were tested using XFLR5 software, and the resulting stability data were compared with previous aircraft.

The team then selected different motors and batteries, ran propulsion tests to collect real-world data, and used MATLAB to predict aircraft performance. Trade studies helped find the design that would score best under the given requirements and limits.

Once the preliminary design met all requirements from Section 3, the team created a SolidWorks model [6] to verify the mechanisms. These parts were planned to be made using laser-cutting and 3D printing. Based on lessons from the first prototype, the team intended to build and test five aircraft in total, trying different combinations and testing under various constraints.

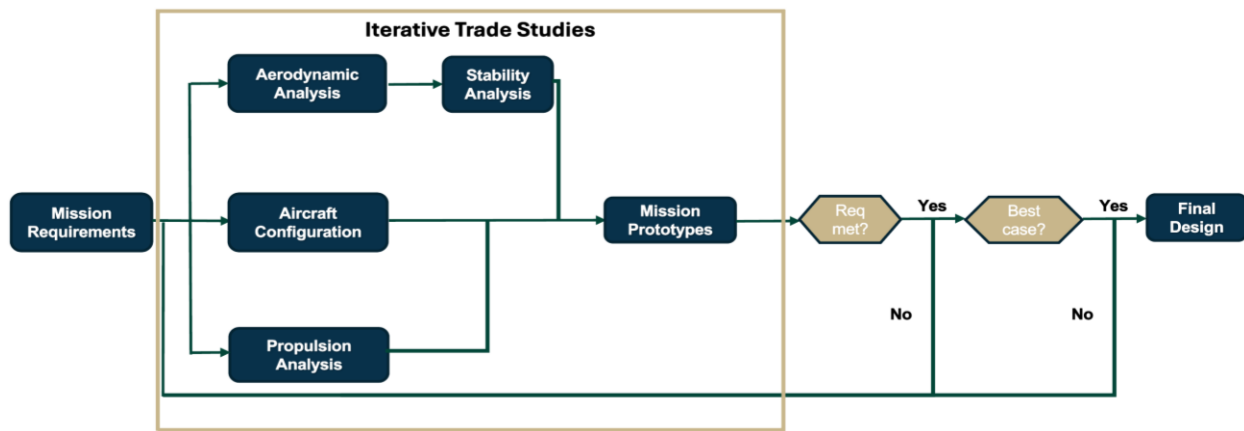


Figure 5 Structured Design and Analysis Diagram

4.1.1 Mission Model

The GRIFFIN's initial size was based on Durmuş [7], with the fuselage length set at 85% of the wingspan. This provided an early estimate of the aircraft's overall size and weight. The team also looked at past AIAA DBF aircraft designs as a weight reference, as shown in Table 21 Aerodynamic Design Comparison of previous DBF aircraft. Aircraft performance was estimated for each of the three flight missions using several key factors: empty weight, maximum takeoff weight (MTOW), wing area, wing aspect ratio (AR), and other variables related to lift and drag (discussed later in Section 4.9). The team set the limits and assumptions for these factors based on their own past aircraft and the official 2025–2026 competition rules.

Lift- and drag-related parameters came from historical data and XFLR5 simulations, which were then checked and updated through testing on a similar aircraft. The mission model (shown in Figure 6) was run in MATLAB using the equations from Sections 4.10 and 4.11. Finally, the team

determined the aircraft's size requirements by studying its aerodynamic performance during three key phases: takeoff and climb, cruise, and landing.

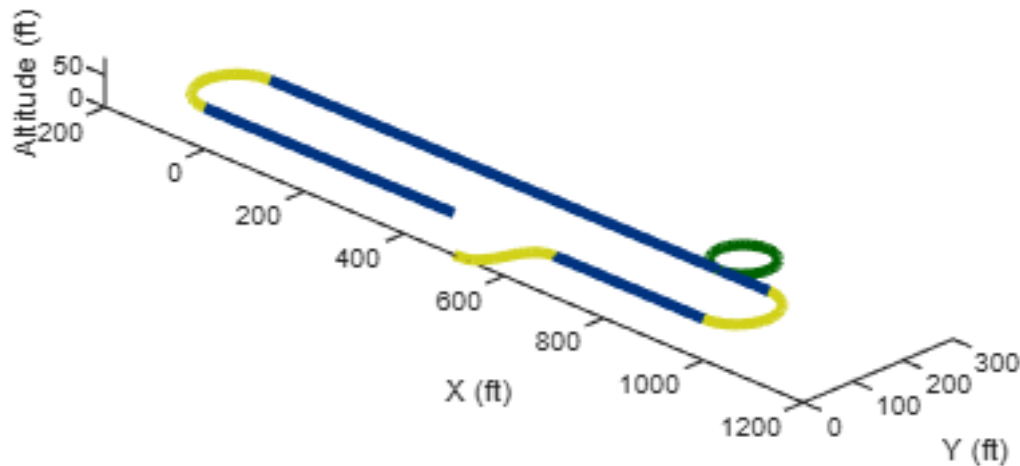


Figure 6 Mission Model

4.1.2 Take-off/Climb

At the take-off/climb stage, the throttle is set to 100%, with a ground roll friction coefficient of 0.6 between rubber and concrete (dry) material [8]. The aircraft accelerates to take-off velocity where it pitches up to reach the required α until sufficient lift is produced to accelerate into the air. The aircraft then maintains full throttle and the required α until it reaches the desirable altitude which ends the climb stage. For M1/2, the desirable altitude is 20 meters and for M3, it is 30.5 meters to give more reaction time to the pilot for any possible sudden weather changes during the flight. Although the take-off distance is not limited this year as mentioned in AIAA 2026 rules [1], the team considered STOL to reduce friction induced in the ground roll process and required T/W to achieve such performance.

4.1.3 Straight Flight

In straight flight, the throttle is adjusted dynamically to achieve the desirable airspeed by using 80% battery usage at maximum, while 20% is reserved for landing. α is adjusted to achieve level flight. If more than 80% of the battery is used, the configuration is invalidated.

4.1.4 Turns/Loop

In the turns/loop stage, the throttle is adjusted dynamically to achieve the desirable air speed for the turns. α will be adjusted accordingly, depending on the weather change and the target. It

increases until the maximum acceptable angle is achieved which includes sufficient room for the bank angle to produce upwards lift to balance the weight of the plane. This approach includes a simulation based on 2D simulation. However, this neglect impacts by complex situations, such as a loss of thrust with an inconsistent shift of α due to weather changes, an impact on the lift coefficient due to inconsistent velocity, and aircraft vibration which affects the aerodynamic performance. The team investigated a Multidisciplinary Design Optimization (MDO) framework, which includes safety factors and realistic considerations for estimating the aircraft performance, mentioned later in section 4.11.

4.2 Airfoil Selection

Reynolds numbers are a quantity that describes the state of the airflow the aircraft is experiencing under different velocities. For a subsonic, low-speed model aircraft, a Reynolds number ranging from 200,000 to 1 million were simulated for ten kinds of airfoil in XFLR5, with the results shown on the comparison graph on the software and exported into an excel form which was weighted based on the main requirements set out in the constraints diagram. This included lift coefficient which determines how much lift the aircraft can generate in flight; lift to drag in cruise, climb and turn and the pitching moment which shows how efficient the airfoil shape is in terms of maximizing the difference between the lift and drag force acted on the aircraft. With basic specifications of the main wing, fuselage and empennage specified, three airfoils were taken into 3D space, with data exported and simulated on MATLAB, which calculated the score of M2 and M3. The team investigated a hybrid of 65% CLARK-Y and 35% Wortman FX 63-137 (WFX), and the single use of CLARK-Y, NACA23012 and WFX. The hybrids of WFX and CLK-Y airfoils preserved the good lift creation for flaps and high sensitivity to stall motion in aileron. In prototype 1, the team adopted the hybrid airfoil configuration, as it is hoped that the hybrid airfoil configuration would provide higher speed and better sensitivity to stall motion which enhanced the stability of the aircraft compared to a single airfoil configuration. Though, the team has changed to adopting only the CLK-Y airfoil in prototype 2 to 4, as it eased the manufacturing process and reduced wing twist issue compared to that of NACA23012 and WFX airfoils.

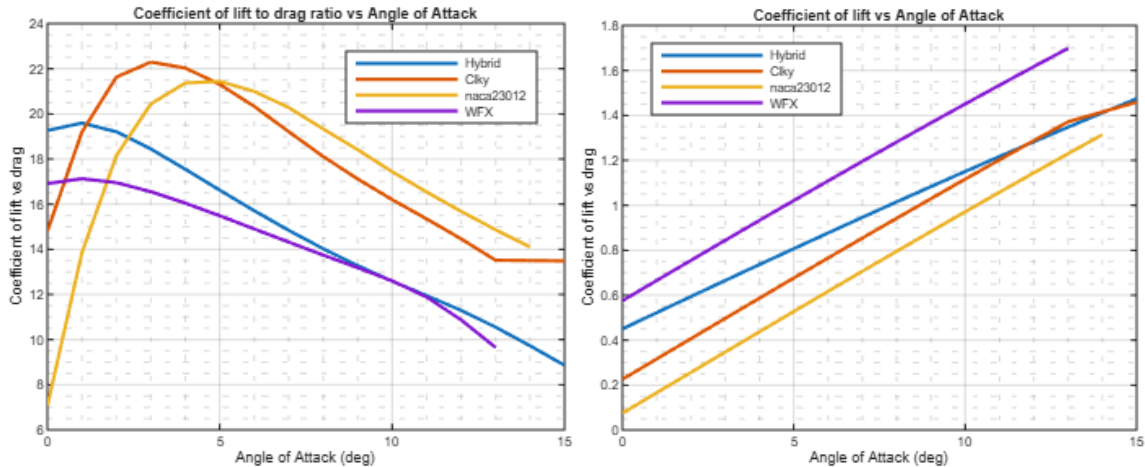


Figure 7 a) Coefficient of Lift-Drag Ratio vs Angle of Attack and b) Coefficient of Drag vs Angle of Attack

Characteristics	Weighting	Clark-Y	WFX-63 137	NACA23012	Hybrid
CL/CD for turn	30	1	0.5	0.75	0.25
CL/CD for cruise	25	0.5	0.75	0	1
Maximum Lift Coefficient	20	0.5	1	0.25	0.75
Manufacturability	25	1	0.25	0.25	0.5
	Total	67.5	55	33.75	60

Table 11 Comparison of Airfoil Performance

4.2.1 Empennage Sizing

The empennage plays a critical role in ensuring flight envelope integrity, maneuverability, and static/dynamic stability. Empennage sizing typically employs two primary approaches: rule-of-thumb methods and empirical formulae [9]. Lennon [9] recommends sizing the tail surfaces as a fixed percentage of the wing area for simplified RC aircraft design. For more precise calculations, the empirical method utilizes the horizontal tail volume coefficient (V_h) and vertical tail volume coefficient (V_v), defined as:

$$V_h = \frac{S_h l_h}{S_w MAC} \quad (1) \quad V_v = \frac{S_v l_v}{S_w b} \quad (2)$$

where S_h and S_v are the horizontal and vertical stabilizer areas, l_h and l_v are the respective moment arms, S_w is wing area, MAC is mean aerodynamic chord, and b is wingspan. Both coefficients are directly proportional to aircraft stability and control authority. With wing design

parameters (including MAC , l_h , l_v) already determined by the Wing Design Division, the stabilizer areas are solved as:

$$S_h = \frac{V_h S_w MAC}{l_h} \quad (3)$$

$$S_v = \frac{V_v S_w b}{l_v} \quad (4)$$

To ensure adequate stability margins, tail volume coefficients were set to $V_h = 0.9$ and $V_v = 0.07$ — values higher than typical ranges (0.6 and 0.04) — providing conservative stability for the banner-towing mission.

4.2.2 Control Surface Sizing

Control surfaces were sized using three stages, Lennon figure 2 [9] for preliminary design, XFLR5 for trim and turn deflection and then a pilot test. The team used historical values from HKUDBF2025 to guide initial sizing with the values shown in **Error! Reference source not found.** This was then tested in an actual flight test to verify reasonable deflection. The team also investigated the use of flaps in prototype 1 to increase the lift force, however removed them in the final design due to a large increase in drag.

These proportions (40% elevator, 50% rudder) are conservative and larger than typical single propeller aircraft (elevator 25–35%, rudder 30–45%) [9]. The oversized surfaces offer significant advantages in Mission 3: enhanced low-speed response, improved disturbance rejection from banner dynamics, and reduced pilot workload in gusty conditions. While larger control surfaces introduce minor increases in profile drag and hinge moment, the impact on endurance and lap count is limited in this low-speed regime. The stability and control benefits far outweigh the small aerodynamic penalty, making this sizing reasonable and effective for achieving stable banner towing performance.

	Chord %	Span %
Aileron	25	4.16
Elevator	40	100
Rudder	50	86

Table 12 Control Surface Sizing

4.3 Fuselage Structure: Introduction to the Preliminary Stage

In the preliminary design stage of the fuselage, there were two main iterations, Prototype 1 and Prototype 2. Prototype 1 was a double-deck fuselage with length at 1680 mm and width at 203 mm as shown in Figure 8 The side view of Prototype 1 CAD, illustrating the double-deck configuration with the upper deck designated for passengers and the lower deck for cargo. **Error! Reference source not found.** and Figure 9 The side view of Prototype 1, illustrating the double-deck configuration with the upper deck designated for passengers and the lower deck for cargo., whereas Prototype 2 was a single-deck fuselage with length at 1090 mm and width at 92.2 mm as shown in Figure 10 and Figure 11. The design progressed from a relatively larger, double-deck fuselage (manufactured in mid-January) aimed at meeting the highest possible payload target, to a relatively smaller, single-deck fuselage (manufactured in mid-February) configured for high speed, minimal payload, and optimized ground mission performance. This transformation formed the basis for Prototype 3 and Prototype 4, which all imitated Prototype 2, opting for a design that aimed at high-speed configuration.

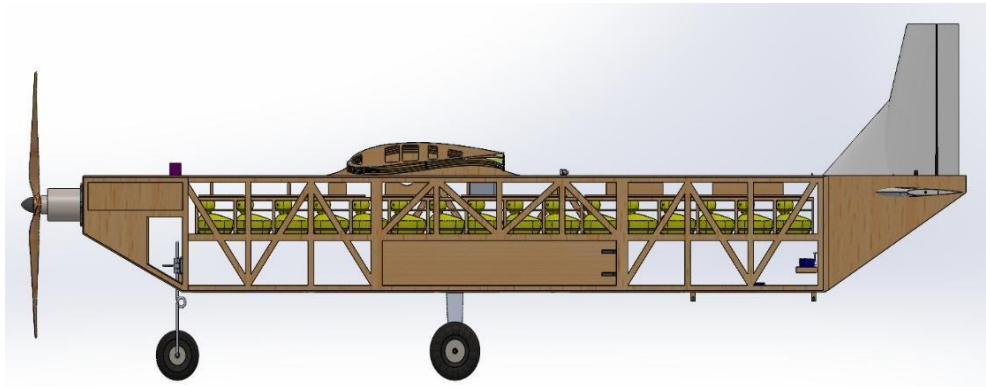


Figure 8 The side view of Prototype 1 CAD, illustrating the double-deck configuration with the upper deck designated for passengers and the lower deck for cargo.



Figure 9 The side view of Prototype 1, illustrating the double-deck configuration with the upper deck designated for passengers and the lower deck for cargo.

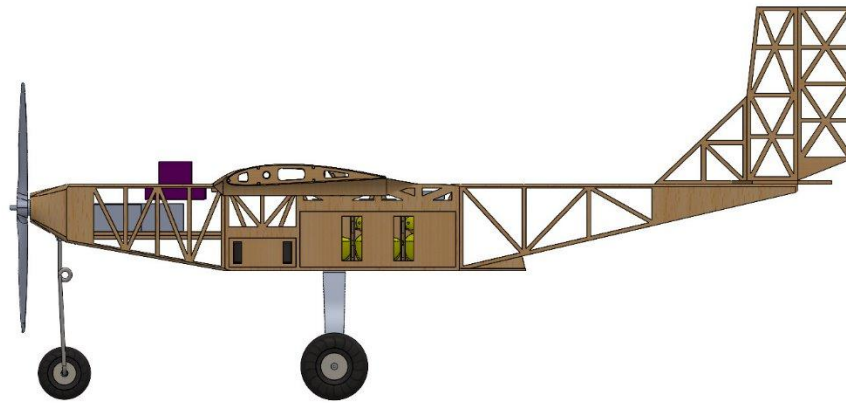


Figure 10 The side view of Prototype 2 CAD, illustrating the single-deck configuration with the passengers and cargo all on one single level.



Figure 11 The side view of Prototype 2, illustrating the single-deck configuration with the passengers and cargo all on one single level.

Initially, payload quantity was prioritized over the number of laps flew by the aircraft and ground mission efficiency (which requires rapid, timed loading and unloading). Consequently, a payload combination of 48 ducks and 16 hockey pucks was evaluated. However, upon the completion of Prototype 1, the manufacturability of the double-deck structure was discovered to be overestimated, while the complexity of the ground mission with this structure had been underestimated.

Following a new sensitivity analysis as covered in the previous chapter, a reduced payload combination of 3 ducks and 1 hockey puck was selected for Prototype 2, which was the minimum payload permitted by the competition regulations. Despite this significant reduction in scale, the

core design of the payload retention mechanism from the first prototype was retained and successfully integrated into the second iteration.

Detailed specifications of both prototypes are enclosed in the following sections.

4.3.1 Fuselage Structure: Prototype 1

In the first prototype, the retention system utilized two latches for the passenger compartment, located at the wing seat and the rear, and a side latch for cargo compartment. Within the upper deck (passenger compartment), the ducks were secured by a latch assembly equipped with internal pusher foam. As the latch closed, the foam compressed the ducks against the bottom panel, effectively restraining them with zero degrees of freedom, as illustrated in Figure 12 and Figure 13. For the lower deck (cargo compartment), the hockey pucks were inserted into the tight-fit slot constructed from structural plywood sticks and plywood panels. This interference fit similarly eliminated all degrees of freedom of the hockey pucks, as illustrated in Figure 14 and Figure 15. Consequently, the overall restraint system prevented payload translation and vibration in all flight conditions, satisfying the competition regulations.

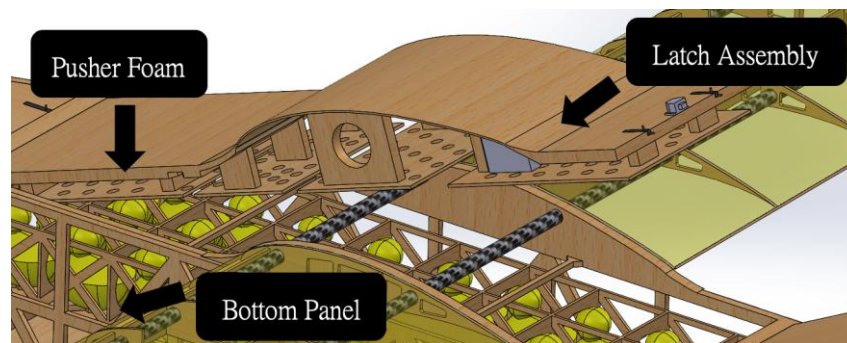


Figure 12 The CAD model of Prototype 1 of the passenger compartment, showing the labelled closing latch system with pusher foam attached beneath the latch assembly, the slots, the bottom panel of the passenger compartment, the wing seat and all 48 ducks.

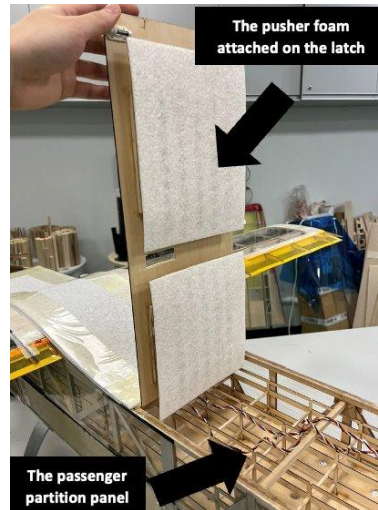


Figure 13 The passenger compartment of Prototype 1, showing the labelled closing latch system.

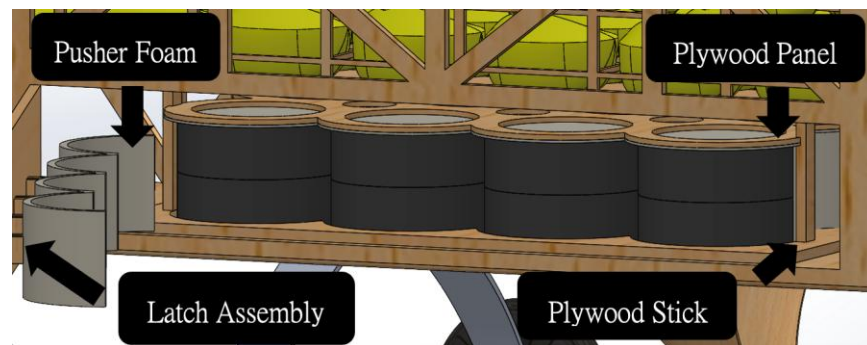


Figure 14 The CAD model of Prototype 1 of the cargo compartment, showing the labelled closing latch system with pusher foam attached beneath the latch assembly, the plywood panel and sticks which form the hockey pucks slot and all 16 hockey ducks.



Figure 15 The cargo compartment of Prototype 1, showing the slot for the 16 hockey pucks.

On the other hand, several areas for improvement were identified. First, actual loading practices revealed that the loading efficiency was significantly lower than anticipated. To illustrate, the

frequent adjustments required to position a large volume of ducks and pucks hindered ground mission performance. Second, dimensional variations among individual ducks necessitated additional foam padding, which increased both total weight and manufacturing complexity. Third, the double-deck configuration, intended to minimize fuselage length (the ratio of fuselage length to wingspan was greater than 1, deemed to be excessively long already), resulted in excessive unused internal volume on the lower deck, as shown in Figure 16, contributing to unnecessary structural weight. Finally, it was determined that a more integrated and overall interlocking system in the fuselage would be more preferable for optimizing load distribution and facilitating the manufacturing process, as shown in Figure 17.



Figure 16 An illustration of the excessive unused internal volume within the lower deck cargo compartment in Prototype 1.



Figure 17 An illustration of the inadequate empennage interlocking system in Prototype 1. The lack of integrated mechanical joints highlighted the need for a more robust interlocking mechanisms in subsequent designs to improve structural integrity.

Based on these observations, the team concluded that a reduced payload would provide a tactical advantage by increasing loading efficiency and thus the ground mission performance. Furthermore, the decrease in payload size could be compensated for by the more lap times achieved by the

smaller and compact aircraft. Following a new round of sensitivity analysis, Prototype 2 was developed. This iteration focused on a high-speed, nimble configuration designed to operate at the minimum required payload capacity.

4.3.2 Fuselage Structure: Prototype 2

Based on the aforementioned findings and further rigorous simulation, the team decided to move away from a high-payload strategy in favor of a simplified configuration consisting of only three ducks and one hockey puck. This significant reduction in payload quantity enabled a transition to a single-deck layout, where all components were situated on a single level, dramatically improving space efficiency and accelerating ground mission execution.

A side-access latch configuration, as shown in Figure 18, was adopted for the passenger compartment, providing superior accessibility while avoiding interference with the avionics bay on top of the passenger compartment, including the receiver, UBEC, and battery. This design choice was influenced by a post-Prototype 1 official Q&A release, which clarified that wiring is prohibited within the passenger compartment. Additionally, by relocating the latch from the top to the side, the need for a complex, curved latch that matched the wing seat contour was eliminated. This modification significantly facilitated and accelerated the manufacturing process. For the cargo compartment, the side-access latch design was retained from Prototype 1. This compartment was positioned forward of the passenger compartment (closer to the fuselage nose) as shown in Figure 19, to optimize the Center of Gravity (CG). Maintaining the CG forward of the Aerodynamic Center was a primary design priority to ensure longitudinal stability. Additionally, the construction material of the cargo compartment was transitioned from structural plywood sticks and panels to 5 mm balsa wood panels. These panels utilized an interlocking mechanism to form the compartment, reducing weight while maintaining structural integrity.

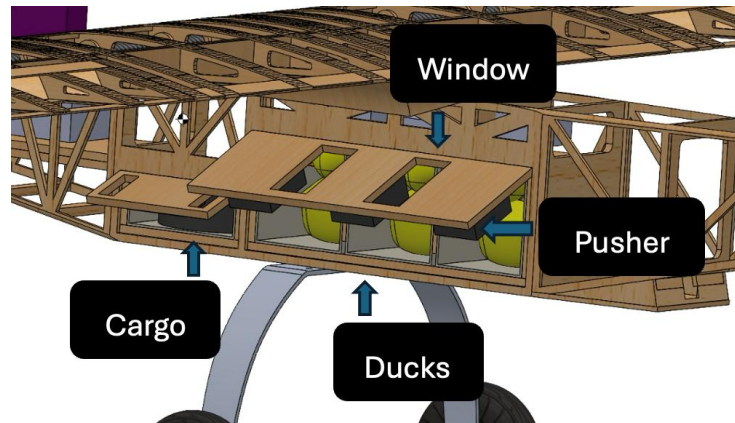


Figure 18 An isometric view of the prototype 2 CAD with the major components from both the passenger compartment and cargo compartment labelled and specified.

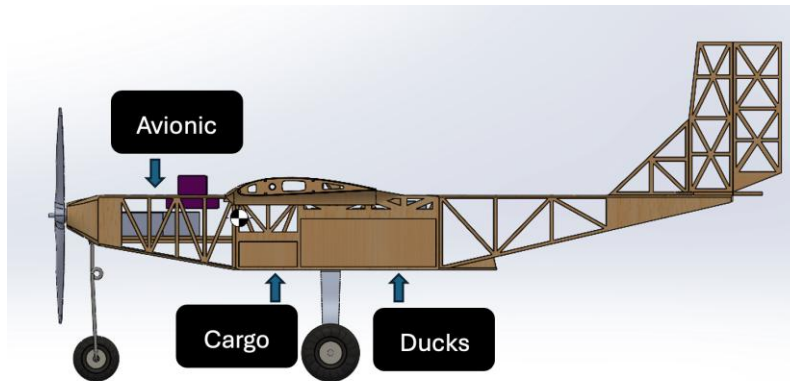


Figure 19 The side view of Prototype 2 CAD illustrating the primary internal compartments and their spatial orientation relative to the Center of Gravity (CG).

Flight testing of the initial prototype revealed significant aeroelastic fluttering, which was attributed to the wing being installed as two separate halves, as shown in Figure 20. Consequently, the Prototype 2 design required the wing halves to be joined into a single unit before being mounted onto a dedicated wing seat on top of the fuselage, as shown in Figure 21. To summarize, Prototype 2 featured side doors for both the cargo compartments, as shown in Figure 22, and the passenger compartment, as shown in Figure 23. This design retained the established latching mechanism from Prototype 1 but now pushes the ducks laterally to secure against the side wall panel, rather than vertically against the bottom panel as in the previous iteration.

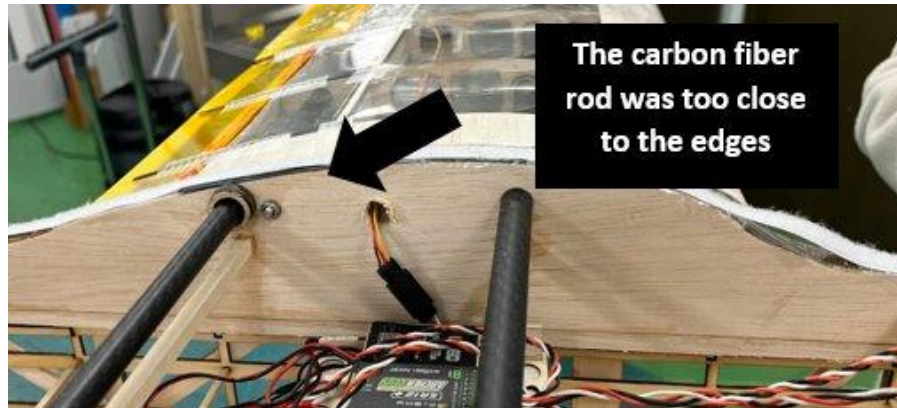


Figure 20 The installation of the wing as two separate halves in Prototype 1. It also shows the rear carbon fiber rod positioned in close proximity to the fuselage wing-seat edge. This configuration is deemed unsuitable for effective load transfer between the wing and the fuselage structure.



Figure 21 The wing halves are joined together into a single unit before being mounted on top of the fuselage in Prototype 2. It is different from joining laterally and separately on the side of the fuselage in Prototype 1.

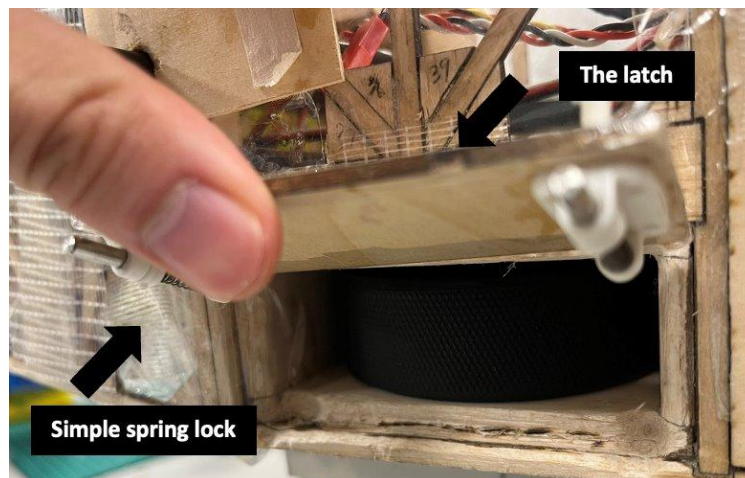


Figure 22 An illustration of the cargo compartment in Prototype 2 with the key components labelled.

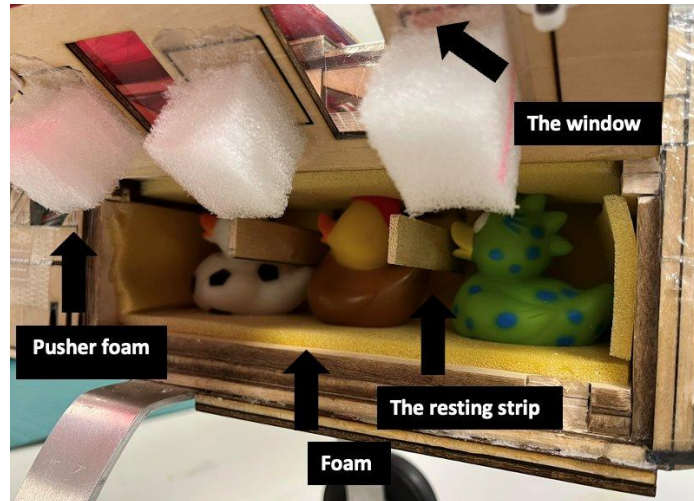


Figure 23 An illustration of the passenger compartment in Prototype 2 with the key components labelled.

4.4 Banner Sizing and Mechanism Design

Four design mechanisms were considered and tested for the banner stowing, towing and deployment stage. On the first prototype, a servo is placed on one side of the board, and two pillars are placed on the other side of the board as shown in Figure 24 Pillars and Board Mechanism. Two wires are tied from the servo arm to the hole on the two pillars. It is firstly manufactured using Plywood and Balsawood and fishing wire, and tested for stow stage, but it cannot restrain the axial movement of the banner, thus it will be fall on the ground when the aircraft flies in high speed. The second prototype is a 4-bar linkage bucket mechanism. The servo arm is connected to one of the linkages. When the servo arm moves from one side to the opposite side, it pulled the linkage bar which further pulls the other bars connected, thus extracted and retracted the bucket linkage. It provided strong rigidity to hold the banner; however, it occupied too much space underneath the fuselage and added more weight to the aircraft.

The cage mechanism is the third prototype which adopted balsa wood and aluminium strips as manufacturing materials as shown in Figure 25; however, the material for the cage design was too soft to hold the banner tight. The fourth mechanism is an aluminium rectangular tube where two holes are drilled on both sides of the tube, and a rubber band is tied on each side of the tube which wrapped around the banner as shown in Figure 26. Another side of the rubber band is tied with a hook that went through a metal pin that is connected to the servo arm and thus hold the

folded banner in place. Though, the rubber bands mechanism tended to entangle the banner wires during deployment. Lastly, the gripper mechanism is the fifth mechanism which was 3D-printed, has a sufficient rigidity to hold the weight of the banner, minimize the effect on the shift of CG, hold the banner firmly during the flight, restrain all movements of the banner, avoid wire entanglement during flight, and thus successfully deployed in the air. Table 13 Banner Deployment Mechanism indicates all considerations and the final selection of the deployment mechanism.

Moreover, two release mechanisms were considered. The first design is a spool mechanism which organized the banner wires but vibrated vigorously during flight. It increased the uncertainty of the success release rate of the banner, as the metal pin that went through the center hole of the spool which held the spool in place sometimes fell out of the hole during flight because of the vibration. The second mechanism considered is a badge reel design. It organized the wires inside the box and restricted the deployment speed to make the overall release process steady and reduce complexity of the process of making a towline for the banner. Table 14 Banner Release Mechanism show a comparison and selection of the release mechanisms.

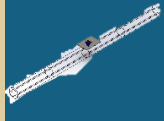
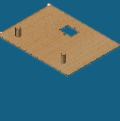
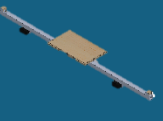
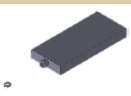
						
Deployment	Weighting	Gripper	Cage	Linkage Bucket	Pillars	Rubber bands
Manufacturability	20	0.75	0.25	0.25	0.5	0.5
Complexity	25	0.25	0.5	0	0.5	0.25
Occupation	20	1	1	0.25	0.75	1
Deploy Success	35	1	-1	0.75	-1	0
Total Score	100	76.25	2.5	36.25	-2.5	36.25

Table 13 Banner Deployment Mechanism

			
Release	Weighting	Badge Reel	Spool
Manufacturability	20	0.75	0.75

Complexity	25	0.75	0.5
Release Speed	20	1	0.5
Release Success	35	0.75	0.25
Total Score	100	80	46.25

Table 14 Banner Release Mechanism



Figure 24 Pillars and Board Mechanism



Figure 25 Metallic Cage Mechanism



Figure 26 Rubber Band Mechanism

4.5 Banner Material

The team investigated 5 types of banner materials in terms of their impact on drag force, rigidity, and density in Table 15. The density of each material is shown in Table 16 Banner Material Densities, where super-light-weight nylon is the lightest material, and pongee material had the best balance in density and rigidity. The team prioritized a balance in density and rigidity which maintains sufficient steadiness and reduces vortices along the edges. Mesh banner was the heaviest out of all the material which will create a huge drag force on the aircraft, thus greatly reduce the aircraft speed. The team selected pongee as the final material choice.

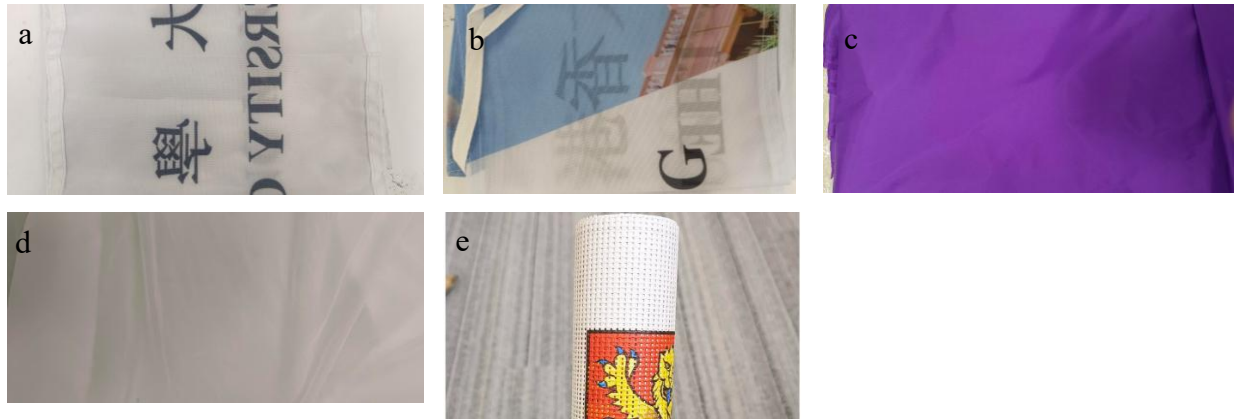


Table 15 a) Pongee b) Tricot c) Nylon d) Super light-weight Nylon e) Mesh fabric

Material	Pongee	Tricot	Nylon	light-weight Nylon	Mesh
Density (g/m^2)	78.11	92.76	87.88	24.41	385.7

Table 16 Banner Material Densities

4.6 Payload Container Design for Banner Weight Distribution Adjustment

To ensure the banner towed vertically during flight, 3 types of payloads were considered to increase the weight at the bottom edge of the banner rod as shown in Table 17 Trade Studies Regarding Payload Container Design for Banner Weight Distribution Adjustment. The first type of payload is a water-droplet-shaped container. It has a very smooth streamline as its outline; thus, it was the best shape of payload container to improve the aerodynamic system. The second type of payload container has a short cylinder outline. Both the first and the second type of the containers were estimated to be 3D-printed. The third type of the payload container was a fabric bag that includes payloads inside. The team prioritized space occupancy and easiness of adjusting the payload amount inside the container, as the size of the banner when it was stowed under the aircraft would affect the ground clearance of the aircraft, which is the distance between the banner to the ground. This would affect the take-off angle of the aircraft when it nosed up. The first and second containers were to be fixed on the rod using epoxy glue, thus, it was difficult to be replaced with new containers if the old one was damaged, or the quantity of the payload inside the container needs to be adjusted. Therefore, the fabric bag was chosen to be the final design of the payload container.

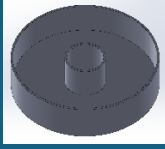
				
Deployment	Weighting (%)	Fabric Bag	Cylinder	Water Droplet
Space Occupancy	35	0.75	0.25	0.25
Aerodynamics	25	0.25	0	1
Easiness of Adjustment	40	1	0.25	0.25
Total Score	100	72.5	18.75	43.75

Table 17 Trade Studies Regarding Payload Container Design for Banner Weight Distribution Adjustment

4.7 Propulsion Optimization

The propulsion system was designed to achieve high cruising speed in M2 and high thrust in M3 to overcome the drag induced by the plane and the banner during the flight. First, a thrust-weight ratio of 1.1 was selected to achieve sport-flying performance, meaning the plane should produce at least 1.6kg of thrust. Table 18 listed the motor-propeller combinations which fulfilled the ratio requirements at a throttle lower than 50%:

Motor	Weight (g)	Max Lipo Cell	Power (W)	Max current(A)	Temperature at 100% throttle(°C/mins)	Remarks	Propeller	Max Thrust (gf)	RPM	Efficiency (g/W)
A-Scorpion 3017-600kv [20]	154 (include accessories)	6s	962 (Peak)	35A	N.A.	Sport Weight Range: 2.3-5.03kg	12x6E	N.A.	10900	N.A.
				43A	N.A.		13x6E	N.A.	10500	N.A.
T-MOTOR AT3530 Long Shaft KV580	298	4-6s	1263.33 (14*7)	65A	N.A.	N.A.	APC 13*8	4082	9678	3.48
					74°C/ 3mins		APC 14*7	4805	9496	3.80
T-MOTO	218	4-6s	966.63W (14*7)	50A	41°C/3 mins	N.A.	APC 13*6.5	3575	9252	4.66

SunnySky KV585	V3	159	6s	1625 (Max Continuous Power)	65A	59°C/3 mins	APC 14*7	4116	8762	4.26
						130°C/3mins	Weight for aerobatics: 1400g- 1800g	12x6.5	4320	11465

Table 18 Motor Comparison

Based on the weight, market availability, thrust requirement, result ground clearance and motor temperature, motor SunnySky X3120 V3 KV585 was chosen in prototype 2 and 3 while motor T-MOTOR AT3520 Long Shaft KV550 was chosen in prototype 4. 13x7 propeller was selected for prototype 2 and 3 based on its high thrust and relatively high efficiency, which is important when the plane needs to maximize speed and endurance. For prototype 4, selecting a 13-in propeller avoided extra ground clearance and additional weight from the landing gear, and it also maximised performance by increasing pitch compared with the recommended 13x6.5 propeller. For battery selection, the nominal voltage and the corresponding current was being substituted into Equation (1) and allowed a 20% safety margin for preserving battery health:

$$\text{Voltage}(V) \times \text{Battery Capacity (mAh)} = \text{Energy (Wh)} \quad (1)$$

Based on the calculation results, a 4500mAh 22.2V battery ($E=99.9 < 100\text{Wh}$) was chosen for Prototype 2 and 3, while 4000mAh 22.2V battery ($E=88.8\text{Wh}$) was chosen for prototype 4. Moreover, dividing the battery capacity by the battery density ($\sim 0.2\text{Wh/g}$) illustrated that the weight of the battery should be around 500g and 444g respectively. For ESC selection, a safety factor of 1.2 was applied to the maximum continuous current (65A for Sunnysky X3120 V3 KV585), which means that an ESC with a current rating larger than 78A could be considered. For prototype 4, since the maximum motor current was still below As a result, the 80A ESC could still be used. Moreover, an 80A fuse instead of a 100A blade fuse was chosen to protect the 80A ESC.

4.8 Landing Gear Design Analysis

Section 3.4.5 established the notion that the tricycle configuration is widely adopted for small-scale UAVs for its inherent stability and operational simplicity. However, realising and harnessing this configuration's full advantage over the taildragger requires careful attention to several specific design parameters and literature-derived constraints. This section serves to outline the parameters that governed the preliminary landing gear design across all prototypes.

4.8.1 Landing Gear Shape

Geometric discontinuities significantly influence stress concentration factors in landing gear struts. More specifically, sharp corners act as stress risers that may exacerbate local stresses beyond the material's fatigue limit, promoting crack propagation under the cyclic loading that landing gear components will inevitably experience. Therefore, it is recommended that a landing gear strut be curved with gentle fillets, which distribute the stress over a greater effective area. This reduces the strut's peak stress concentration factor and enhances its fatigue life. As a result, all landing gear struts across all prototypes were designed with generous curved fillet transitions to improve structural durability [24].

4.8.2 Nose Gear Positioning

In UAV model aircrafts, the nose gear should be positioned so as to maximise the airframe's wheelbase, that is, the horizontal distance between the main gear and the nose gear, for two reasons. Firstly, when the aircraft brakes, it tends to pitch forward about the main gear which acts as a pivot. Having a longer wheelbase serves to create a larger lever arm, and hence, a larger resisting moment acting against this forward pitching moment. If the nose gear is moved too far back along the fuselage closer to the CG, the aircraft might pivot over the main gear when braking, causing a nose strike [26].

Secondly, a longer wheelbase provides greater lateral stability against shimmy, crosswinds, and uneven terrain during taxiing. Once again, this is because a lengthier wheelbase results in a greater correcting moment against lateral disturbances. The key ratio for nose gear placements is called the static margin on ground, which is shown in Equation (1):

$$\left(\frac{d}{B}\right) \times 100\% \quad (1)$$

where:

- d is the horizontal distance from the main gear to the CG and,
- B is the wheelbase

It is generally advised to maintain a static margin on ground of around 15-20% for tricycle UAVs [9]. Hence, across all of the team's prototypes, the nose gear was placed as far forward as possible, while the main gear was positioned to achieve a static margin on ground of approximately 15% when fully loaded. This positioning placed 85% of the aircraft's weight on the main gear and 15% on the nose gear.

4.8.3 Main Gear Positioning

Following the design rationale introduced in Section 4.8.2, namely that the main gear should support 85% of the aircraft's MTOW, Equation (2) was used to calculate the wheelbase B and, consequently, the main gear position relative to the nose gear for each prototype:

$$\frac{F_m}{W} = \frac{B_n}{B}; \frac{F_n}{W} = \frac{B_m}{B} \quad (2)$$

where:

- $\frac{F_m}{W}$ is 0.85 (the ratio of total weight supported by main gear),
- $\frac{F_n}{W}$ is 0.15 (the static margin on ground/ratio of total weight supported by the nose gear)
- B_n is the distance from the nose gear to the CG and,
- B_m is the distance from the main gear to the CG
-

These parameters and ratios are further visualised in Figure 27. Since the aircraft does not carry fuel, only cg_{for} was used in these calculations as the aircraft's constant weight throughout the flight eliminated the need to account for a cg_{aft} [14].

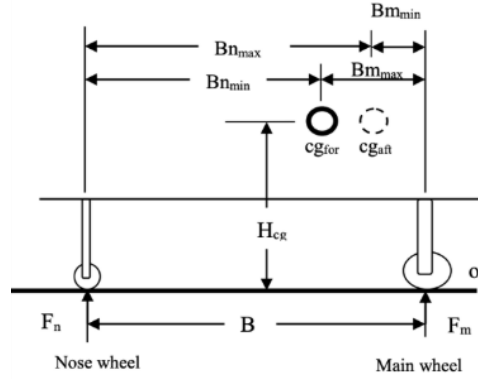


Figure 27 Visualization of design parameters involved in positioning the main gear relative to the nose gear

4.8.4 Track Width

The track width is calculated using a series of stability-based rule-of-thumb formulae seen in Equations (3), (4), and (5) with a landing gear stability factor of 6:

$$AR = \frac{b^2}{S} \quad (3)$$

where:

- AR is 6 regardless of what the actual aspect ratio is,
- b is the imaginary wingspan of the aircraft and,
- S is the wing area.

Reordering the above equation and substituting in 6 as the aspect ratio gives us:

$$b_{imaginary} = \sqrt{6 \times S} \quad (4)$$

Finally, the recommended track width is a quarter of the value $b_{imaginary}$ which scales the value down to UAV applications [14]:

$$Treadwidth = 0.25 \times b_{imaginary} \quad (5)$$

4.8.5 Ground Clearance and Angle of Attack

Based on the faculty advisor's recommendations, a propellor clearance of 2.54 cm was adopted. The corresponding fuselage clearance was then used along with the distance between the main landing gear and the tail sweep to determine the minimum take-off rotation ground clearance angle to prevent a tail strike, as shown in Figure 28. This clearance angle is computed by Equation (6):

$$\alpha_c = \tan^{-1}\left(\frac{H_f}{AB}\right) \quad (6)$$

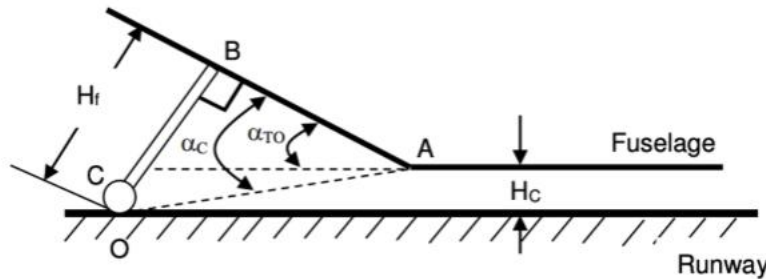


Figure 28 Visualization of geometric derivation used to calculate the minimum take-off angle of attack to prevent a tail-strike

where:

- H_f is the fuselage height and,
- AB is the distance between the main landing gear and the point where the fuselage sweeps up to the tail.

4.9 Aerodynamic Design Analysis

4.9.1 Wing Sizing

Wing configuration plays a crucial role in determining the lift and drag induced on the aircraft, stability of the aircraft, and how the aircraft manoeuvres in the air, such as roll. Different aircraft flight modes are analysed in later section to verify whether the aircraft flies stably. Under the

constraints implemented by the rules and the target of the aircraft performance the team wanted to achieve, the dimensional parameters of the wing can be calculated based on equations and literature references. The configuration of each part of the wing took references from the trainer plane configuration in the textbook by Andy Lennon [9], as shown in Figure 29. “C” refers to the total size of the chord, which is the cross-sectional distance of the wing. “A” refers to the total length of a half wingspan excluding the winglet. Two set of values of the dimensional parameters for main wing were calculated and examined in flight test, one for prototype 1 as shown in Table 19, and one for prototype 2 to 4 as shown in Table 20. The centre point is set at the half of the wing, which is also the centre point of the wing seat. The team took the maximum value for the wingspan allowed by the competition rules so that the wing would produce sufficient lift and rigidity to lift an 8-kg aircraft to the air and completed 6 to 8 laps in Mission 2. Later, the team focused on the aerobatic performance of the aircraft, including maximizing the speed and number of laps, so a much smaller wingspan and other dimensional parameters were taken. All dimensional relationships of the parameters kept unchanged for the main wing in all prototypes.

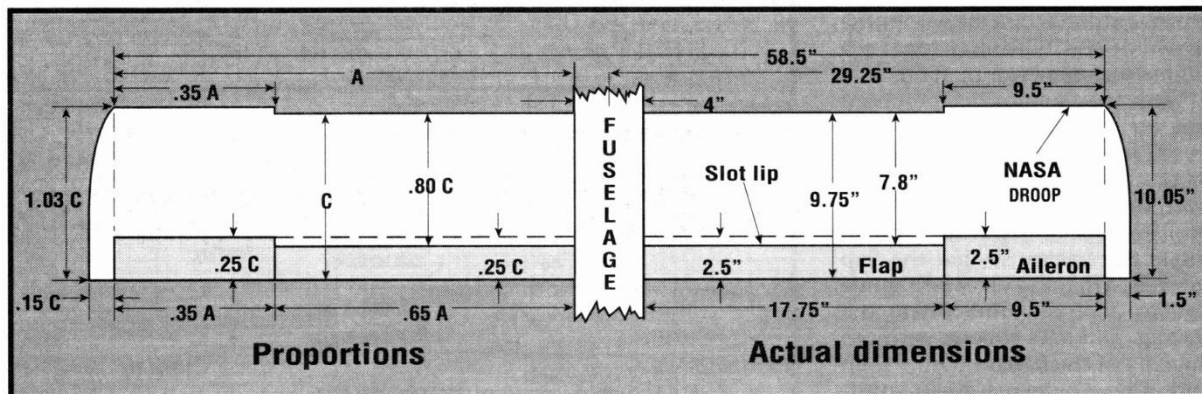


Figure 2.
Outline of the Swift's wing (576 square inches: aspect ratio of 5.94).

Figure 29 Trainer Plane Configuration in Andy L. The Basics of RC Model Aircraft Design

Dimensional Parameters of Main Wing of Prototype 1					
Parameters	Wingspan	Aileron	Flaps	Main Wing Chord length	Thickness
Size (m)	1.542	0.3	0.564	0.277	0.033
Distance from the center point (m)	0	0.198	0	N/A	N/A

Table 19 Dimensional Parameters of Main Wing of Prototype 1

Dimensional Parameters of Main Wing of Prototype 2 - 4					
Parameters	Wingspan	Aileron	Flaps	Main Wing Chord length	Thickness
Size (m)	1.250	0.313	N/A	0.208	0.025
Distance from the center point (m)	0	0.312	N/A	N/A	N/A

Table 20 Dimensional Parameters of Main Wing of Prototype 2-4

4.9.2 Lift and Drag Analysis

The lift and the drag coefficient extracted from XFLR5 were initially calculated and updated several times for several prototypes mainly focusing on prototype 1 (where the total weight was estimated to be 7.48 kg), and prototype 2 (where the total weight was 1.6 kg.), values which were input into the Multidisciplinary Design Optimization (MDO) framework. During initial sizing, the targeted C_L was calculated via (5) from the target mission speeds. The C_D was calculated with the parasitic drag coefficient, aspect ratio, lift coefficient, and Oswald's efficiency factor. The power of a propeller-driven aircraft in steady and level flights can be calculated using (8) and (9). The aerodynamic center of the aircraft in cruise was estimated as $\frac{1}{4}$ of the chord length with reference from Glenn Research Center [10].

$$C_L = \frac{2W}{\rho S W V^2} \quad (5) \quad C_D = C_{D,0} + K C_L^2 \quad (6) \quad K = \frac{1}{\pi AR \epsilon} \quad (7)$$

$$D = \left(\frac{1}{2} \rho V^2 S C_D\right) + \left(\frac{k W^2}{2 \rho S V^2}\right) \quad (8) \quad P = \frac{V D}{\eta} \quad (9) \quad T - D = \frac{M V^2}{R} \quad (10)$$

Referenced from the value of previous DBF aircraft in year 2024 and year 2020 [11][12], the initial aerodynamic design of the GRIFFIN was carried out and put into the MDO in 4.12 where the cruise speed and MTOW were particularly focused.

	UNSW2025	UOL2024	USC2020
Cruise Speed (m/s)	33.53	43.77	44.98
MTOW (kg)	9.98	2.88	5.23
W/S (kg/m^2)	25.05	10.4	28.9
Stall Speed (m/s)	N/A	11.82	17.2

Table 21 Aerodynamic Design Comparison of previous DBF aircraft

4.9.3 Stability Performance

After collecting the dimensional parameters of main wing and tail wing, with the control surfaces, a stability verification can be checked on the XFLR5 3D Plane simulation. Static stability and dynamic stability will be examined, and the data for 5 types of stability mode are collected, including Short period, which is a rapid oscillation in pitching movement; Phugoid, which is an oscillation with little damping along the pitching movement; roll, which is a movement controlled by the ailerons on the main wing and moved along the lateral direction; Dutch roll, which is a combination of roll, sideslip and yaw movement; and finally, spiral mode, which refers to changes to heading while turning, which increased the turning radius and behaves like a spiral[25]. This section investigated the value of 5 modes of the aircraft and verified whether the static and dynamic stability of the aircraft is ensured.

4.9.3.1 Static Stability

With the data of the empennage sizing as discussed in section 4, the main wing and the empennage of the aircraft was modelled on XFLR5. Then, the centre of gravity (CG) of the aircraft can be estimated using several parameters, including the weight of the individual parts of the aircraft, as mentioned later in section 5, weight of the payload, weight of the banner, weight of the avionics/electronic components and the initial estimation of the length of the fuselage and the wing. A static moment was estimated which confirmed that the preliminary CG was ahead of AC during cruise on XFLR5. Static Stability refers to the initial tendency of the body to return to its equilibrium state after being disturbed [25]. Static stability in aircraft corresponds to the longitudinal motion of the aircraft, which is also the pitching motion along the x-axis, as demonstrated in Figure 30. With a simulation performed on XFLR5, the performance of the aircraft in terms of static and dynamic stability can be known, and trimming can be performed to improve the performance of the aircraft. Then, the distance of the selective parameters which will affect the static stability of the aircraft were measured and recorded in Table 22. The data of the Short Period mode and the Phugoid mode is collected and shown in Table 23. The aircraft should be able to return to the equilibrium state of flight along the longitudinal direction without pilot's input, and such performance will be considered as good static stability.

Static Margin (%)	CG to LE	AC to LE	Main Landing gear to LE	Nose to LE	Tail to LE
15	-0.003m	-0.052m	-0.142m	+0.215m	-0.566m

Table 22 Distance Measurement for Static Stability

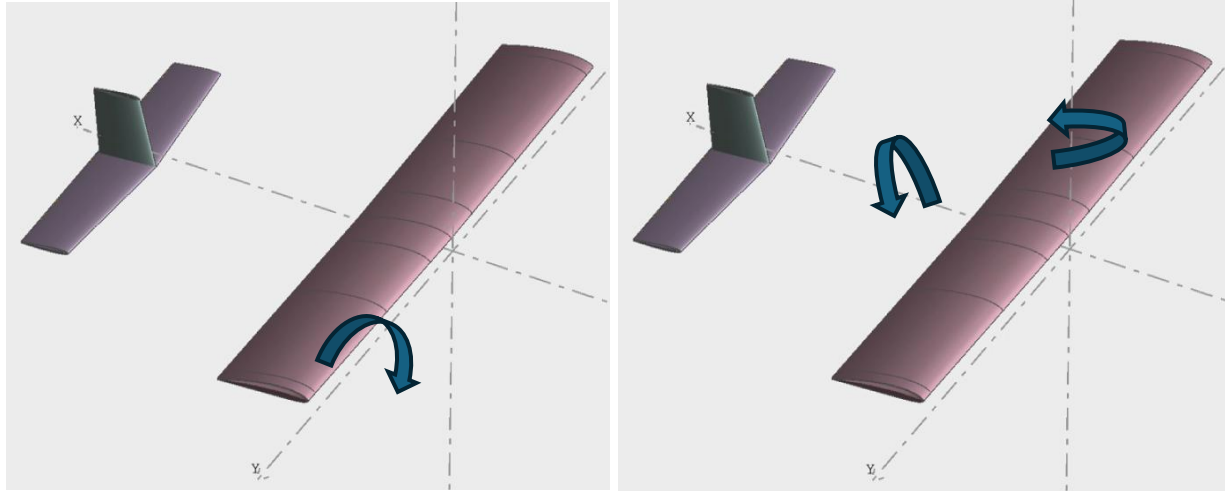


Figure 30 Aircraft Performance Simulation on a) Static Stability along the Y-axis and b) Dynamic Stability along the X and Z-axes

4.9.3.2 Dynamic Stability

Dynamic stability refers to how the aircraft responds to different movement with time, and how the response to basic pilot inputs may be determined [25]. With data obtained from XFLR5 [3], A root-locus plot for the complex roots of the modes is shown in Figure 31 Aircraft Dynamic Modes Root Locus. Each mode is plotted as a root-locus on the graph which shows its relationship between decay of rate of the motion, as indicated as σ in real axis, and the frequency of oscillation, as indicated as ω in the imaginary part. The aircraft mode solution is expressed in equation $x(t) = e^{\sigma t}$ (11), and the eigenvalue is expressed in equation $\lambda = \sigma + i\omega$ (12).

$$x(t) = e^{\sigma t} \quad (11)$$

$$\lambda = \sigma + i\omega \quad (12)$$

The real part of the eigenvalue is obtained by:

$$\omega_n = 2\pi F_1 \quad (13)$$

$$\sigma = \zeta \omega_n \quad (14)$$

$$\omega = \omega_n \sqrt{1 - \zeta^2} \quad (15)$$

An aircraft mode is stable if the real part of the root is on the negative side, and the frequency of the oscillation is positively proportional to the magnitude of the imaginary part. The position of each aircraft mode was calculated based on the data shown in Table 23 Dynamic Stability Data for Longitudinal and Lateral Motion where F_1 is the oscillation frequency determined by the real and imaginary number of the eigenvalue by equation $F_1 = \frac{\sqrt{\sigma^2 - \omega^2}}{2\pi}$ (16).

$$F_1 = \frac{\sqrt{\sigma^2 - \omega^2}}{2\pi} \quad (16)$$

f is the natural frequency, ζ is the damping ratio, t_2 is the time to half magnitude of the oscillation period, and τ refers to the rate of the disturbance decaying exponentially, which is calculated by:

$$\tau = -\frac{1}{\sigma} \quad (17)$$

Mode	ζ	f (Hz)	F_1 (Hz)	t_2 (s)	τ
Short Period	0.668	6.407	4.765	0.193	0.156
Phugoid	0.007	0.0712	0.072	1455	13.90
Dutch Roll	0.217	2.160	2.109	6.900	0.463
Roll	N/A	31.91	N/A	0.022	0.031
Spiral	N/A	N/A	N/A	30.46	N/A

Table 23 Dynamic Stability Data for Longitudinal and Lateral Motion obtained from XFLR5

With damping ratio in an acceptable range of short period and phugoid mode and all locus points lying on quadrant II of the coordinate plane as shown in Figure 31, the aircraft are said to be capable of maintaining good stability in both oscillating and non-oscillating flight conditions.

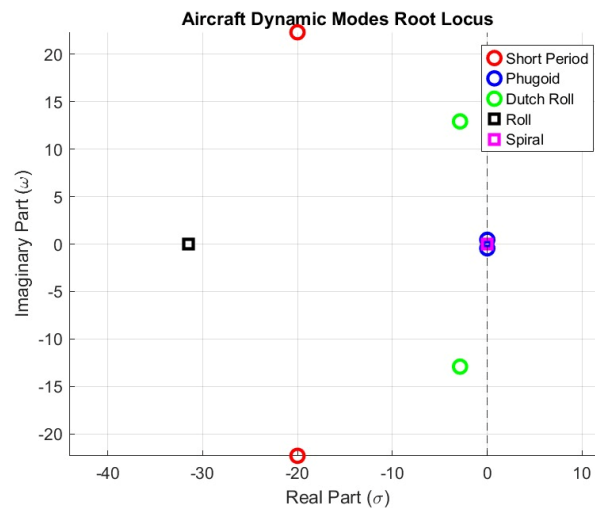


Figure 31 Aircraft Dynamic Modes Root Locus

4.9.4 CFD Analysis

A computational-fluid-dynamic analysis was conducted on ANSYS to estimate the aerodynamic performance of the aircraft during flight, assuming 32m/s in cruising. The simulation adopted Spalart Allmaras as the turbulence model, with an initial velocity of 32 m/s and an Angle of Attack of 2 degree upwards. The estimated Reynolds number in the model is around 4.68×10^5 . As prototype 2 and prototype 4 are the same, 3 simulations are conducted for 3 prototypes, and they compared in terms of static pressure distribution, dynamic pressure distribution, and lift and drag force. In static pressure distribution, the region where the airflow arrows intensify, or colour brightened means the airflow stagnates the most, and the region where the colour shifts to cold colour, such as blue and green, means the airflow moves the fastest. In dynamic pressure distribution, reg region refers to regions where the air flows the fastest. As shown in Figure 32. In prototype 1, a huge pressure distribution occurs at the leading edge of the vertical stabiliser, and the back of the banner gripper mechanism. These created a significant vortex due to a sudden change of wind direction and the pressure difference after the airflow flew from the inclined leading edge of the vertical stabiliser to the horizontal, flat-top surface of the vertical stabiliser, as well as the inclined surface of the crawling for the banner mechanism. Therefore, the air stagnates the most in these areas, as well as propeller, landing gear and some selective regions of the top surface of the wing.

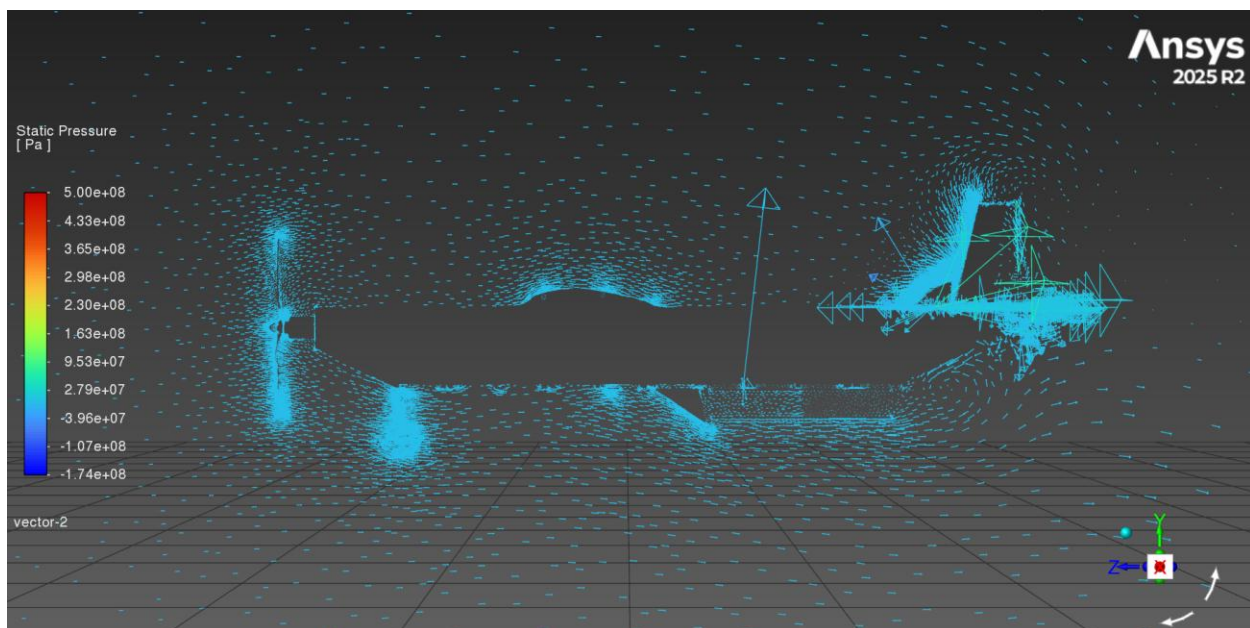


Figure 32 A CFD Analysis on ANSYS for Prototype 1, in terms of static pressure distribution

As the team later emphasizes strategy on maximizing the speed of the aircraft, later prototypes focus on lightweight design and minimizing the payload for mission 2 and maximizing the number of laps in mission 3. Another CFD simulation conducted on ANSYS was conducted for prototype 2 and 4. As shown in Figure 33, the static pressure distribution on prototype 2 is shown. The top surface of the nose of the fuselage, the leading edge and the trailing edge region of the fuselage experienced the most stagnated airflow, which means these areas has the biggest room of improvement for increasing the speed of the aircraft. Moreover, the dynamic pressure distribution of prototypes 2 and 3 are also compared. The result of the estimated lift force and drag force for prototype 2 and prototype 3 are collected, as shown in Table 24. In terms of dynamic pressure distribution, the air flow faster on the top surface of the wing, the edges of the nose of the fuselage, and the side panel of the fuselage for prototype 2 as shown in

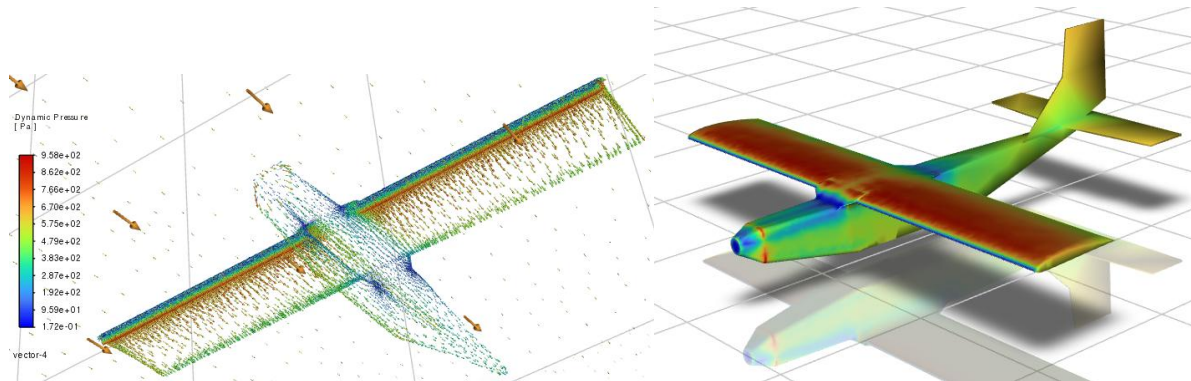


Figure 34. For prototype 3 as shown in Figure 35, the air flow faster at every turning edge of the curve of the fuselage. However, the airflow detached from the boundary layer of the surface into the air, which produced a wake [23]. This is due to the sudden change of the velocity at the turning edge of the side panel of the fuselage, when the air returned to freestream regions, the velocity decreased and thus created a differential pressure region between the two regions. This created significant wake that tabulates the air flow around the aircraft. Therefore, the drag force collected from the simulation for prototype 3 is larger than that for prototype 2.

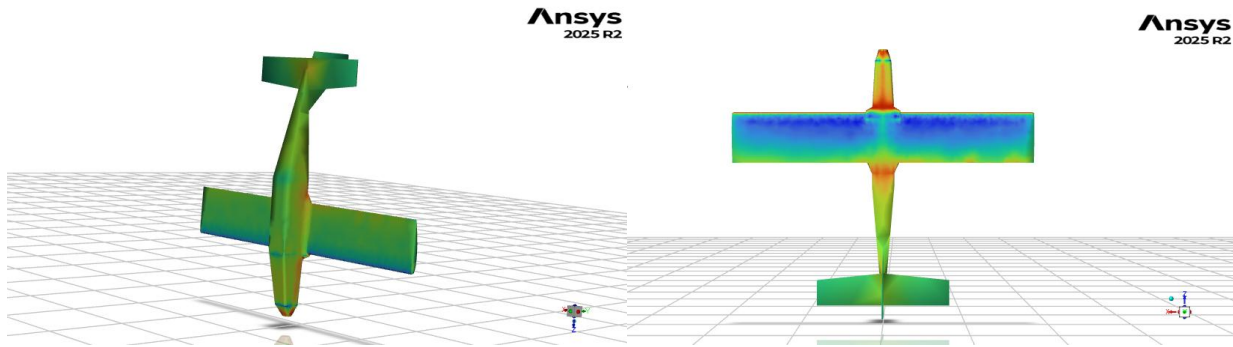


Figure 33 a) CFD Analysis ANSYS-Side View, b) CFD Analysis on ANSYS-Top View for Prototype 2 in terms of static pressure distribution

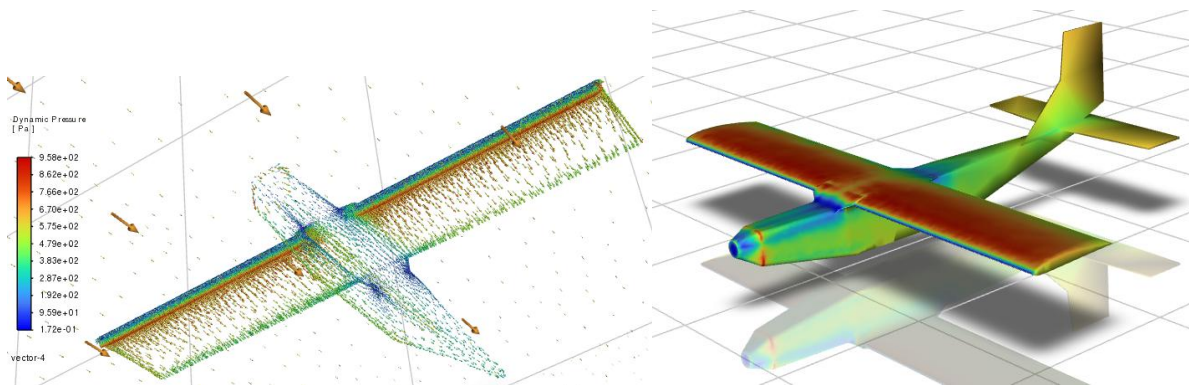


Figure 34 CFD analysis result of dynamic pressure distribution on Prototype 2 on ANSYS

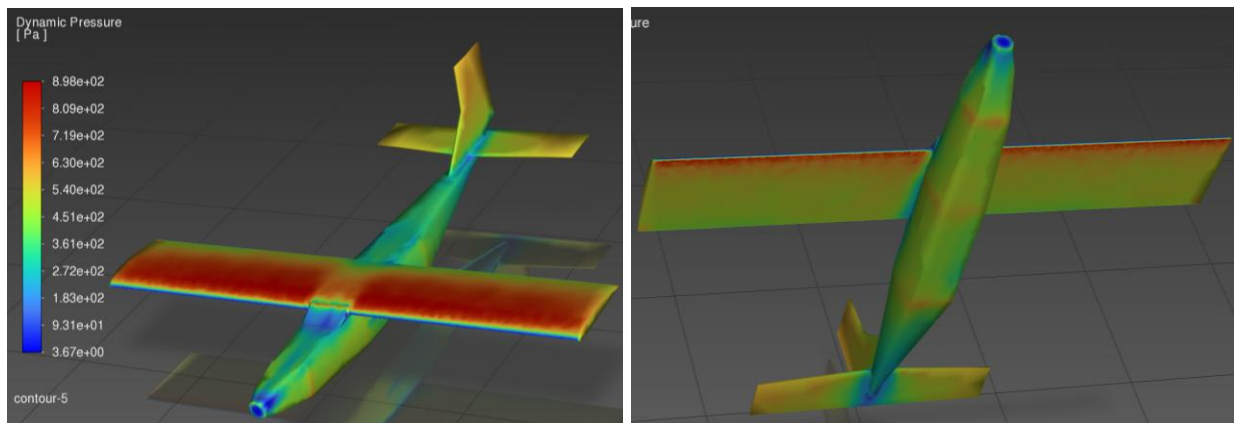


Figure 35 CFD analysis result of dynamic pressure distribution on Prototype 3 on ANSYS

Features	Prototype 2	Prototype 3
Side panel shape	Flat	Curved/round
Lift force (N)	41.5	34.72
Drag force (N)	6.5	7.15

Table 24 Lift and Drag data of prototype 2 and prototype 3 obtained from the analysis result of CFD simulation self-conducted on ANSYS

4.10 Mission Prediction Analysis

A parametric MDO framework was developed in MATLAB using a deterministic grid-search approach to explore the design space systematically to obtain the predicted performance of the preliminary design that maximizes mission score while satisfying aerodynamic and performance constraints. Several assumptions for the battery selection and empty weight of the aircraft under different conditions were made to provide a realistic scenario in the simulation, with an assumed velocity range based on previous aircraft performances. Then the flight path was estimated by examining the flight motion for three stages: take-off/climb, cruising, landing.

The highest possible score for each individual mission was estimated and was compared with the optimized score case where all missions' requirements were considered to determine the highest score case for the GRIFFIN. For Gound Mission (GM), the time for the fastest possible case was estimated as 25 seconds, based on experience from the team. In the iterative optimization simulation, the ground handling (GM) time can be expressed by formula $GM = 25 + \text{passenger} \times (n - 3) \times 8 + \text{cargo} \times (n - 1) \times 8$ (18):

$$GM = 25 + \text{passenger} \times (n - 3) \times 8 + \text{cargo} \times (n - 1) \times 8 \quad (18)$$

where:

- n represents the number of passengers or cargo units, depending on the term,
- 8-seconds is the estimated time required to load or unload one passenger or one unit of cargo,
- An additional buffer time is included to account for folding the banner and stowing it beneath the aircraft.

The fastest mission time is estimated as 25 seconds, which corresponds to the fastest possible scenario involving 3 passengers and 1 unit of cargo. For any additional passenger beyond the first three passengers, the term $(n-3)$ indicates the increment of the time needed, and for any additional cargo unit beyond the first one, the term $(n-1)$ captures the increment of the time needed. Each such additional unit adds 8 seconds to the total GM time, reflecting the extra loading and unloading effort required.

For Mission 2 (M2), the mass of the aircraft, the flight time and the number of laps, the flight velocity and the number of payloads, and the battery required were considered to estimate the performance of the aircraft. The velocity of the aircraft was estimated to be in a range of 26.79 m/s to 47.62 m/s, according to previous aircraft performances. Using an iterative approach by keep inputting different combinations of the number of passengers and number of cargoes into the mission score formula, the highest M2 marks were calculated. For M3, the same parameters were estimated based on the lift coefficient and the drag coefficient simulated in XFLR5 and wind testing. A value within the range of aspect ratio of the banner from 1 to 5, and a range of banner length from 0.254 meters to 4 meters was selected in each iteration. The drag coefficient of the banner and a data reference of velocity and number of laps was taken from a flight test using a commercial plane of 2 kg with a 3300 mAh battery and a 3 kg-f motor as shown in Figure 36. Using a similar iterative approach, the highest mark for M3 was calculated. Then, an optimization loop was simulated, by iterating the score calculation and putting them into the score formula provided by AIAA 2025 rules. The simulation went through 100 iterations and thus estimated the top 100 score cases for the team. A trade study was then conducted based on the team's constraints and requirements to determine the final prototype design.



Figure 36 Commercial Plane Flight Test

4.11 Predicted Mission Performance

From the predicted mission performance simulated on MATLAB, the top 100 score cases are as indicated in Figure 37. The “Bulk Group” in green colour corresponded to score cases where number of passengers more than 33 and number of cargoes more than 11 were chosen to be the amount of payload carried in Mission 2. The “Light Group” as orange triangles corresponded to score cases which 3 passengers and 1 cargo was chosen to be carried on the aircraft in Mission 2.

The rest of the combinations of payloads carried in Mission 2 were indicated as “Other Top Cases”. The comparison was taken as the ratio of the total score and the aircraft mass, since the size of the aircraft determines the time taken for manufacturing, and the battery and motor selection, and banner mechanism design. Since bulk groups were the highest score cases, the first prototype adopted 48 Passengers and 16 cargoes. Through examination after manufacturing and testing process, the light group was selected as the final aircraft design as it took the least time to manufacture and flew faster than other design cases, while the score deviation is mild compared to the bulk group, with a difference of 0.08. The predicted aircraft performance was shown in Table 25.

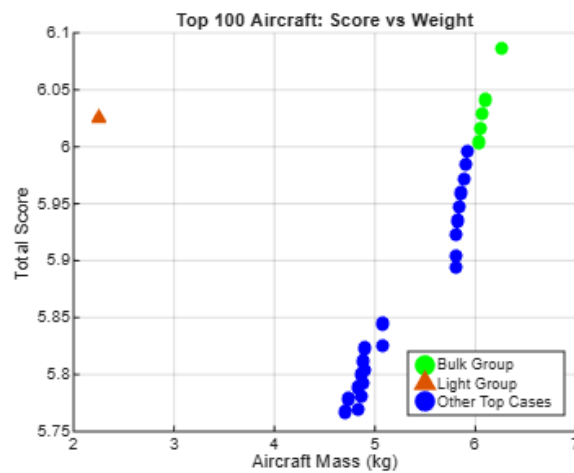


Figure 37 Top 100 Predicted Score Case

Predicted Aircraft Performance	
Total Score (out of 7)	6.026
Wingspan (m)	1.25
Cargo	1
Passengers	3
M2 V_{cruise} (m/s)	32
M2 Laps	9
M3 V_{cruise} (m/s)	18
M3 Laps	5
Battery (Wh)	75.50
Banner Length (m)	4

GM Time (sec)	30
M2 Mass (kg)	2.083
M3 Mass (kg)	2.679
Banner Aspect Ratio	5.0

Table 25 Predicted Aircraft Performance

4.12 Uncertainties and Assumptions

This score simulation was based on assumptions referenced from previous aircraft designs, and mass properties on CAD assembly while all the data were extracted from previous aircraft experiences and simulations on XFLR5. The battery selection and the weight assumption were based on rough estimation. These may not reflect the actual situation during flight thoroughly due to complex geographical factors in the airfield environment. The highest possible cases for GM, M2 and M3 were estimated based on the team's experience and a range of velocities based on previous aircraft performances which may vary in the future.

5 Detailed Design

Building upon the parametric optimization and configuration trade studies established in the preliminary design phase (Section 4), the detailed design phase translated the selected conceptual configurations into fully defined, manufacturable subsystems. While Section 4 focused on iterative

aerodynamic analysis, mission modelling, and performance prediction. Section 5 documents the final geometric dimensions, structural architectures, material selections, and subsystem integrations that characterize the competition-ready aircraft.

The detailed design phase addressed two critical objectives. First, it refined the structural design of each major airframe component: wing, fuselage, empennage, and landing gear. Ensuring they had to withstand the load cases identified in Section 4 then it validated structural integrity through Finite Element Analysis.

The outcomes of this section directly inform the manufacturing plan (Section 6), which translates these digital models into physical prototypes.

5.1 Wing Structure

The wing design followed a rigorous iterative trajectory, transitioning from initial conceptualization in WingHelper to high-fidelity modelling in SolidWorks and validation through Finite Element Analysis (FEA). Over 12 distinct versions and numerous sub-iterations were developed, incorporating feedback from faculty advisors and findings from preliminary tests. These iterations focused on refining the aerodynamic profile, specifically through airfoil selection and wingtip twist optimization, while optimizing the internal structure for strength and manufacturability. The design evolution shifted from the payload-maximizing philosophy of the first prototype to a weight-optimized, "light and fast" approach in subsequent versions, successfully balancing high-lift coefficients with structural efficiency and torsional rigidity.

5.1.1 Prototype 1 (High-Payload Configuration)

Prototype 1 was engineered with the primary objective of maximizing payload capacity, necessitating a robust structural framework capable of supporting significant passenger and cargo weight. From Table 19, the wing featured a 1.524 m span with a constant 0.277 m chord and a rectangular planform.

For the aerodynamic profile and control, the ribs and ailerons were constructed from 3 mm balsa wood, chosen for its superior strength-to-weight ratio (density approximately 125 kg/m³). To manage the high take-off weight and resultant g-loading, the primary load-bearing members consisted of two 12 mm diameter carbon fibre rods with 2 mm wall thicknesses. The front rod, placed at 25% of the chord, extended the full wingspan. The rear rod was limited to the inboard

section only, positioned to create at least 12 cm of separation from the front rod. This arrangement allowed for top-loading cargo without structural interference. Secondary spars, constructed from 3 mm balsa wood, provided additional support. Plywood webbing was inserted between the secondary spars to increase both vertical load capacity and torsional stiffness. All of this was designed on WingHelper as shown in Figure 38.

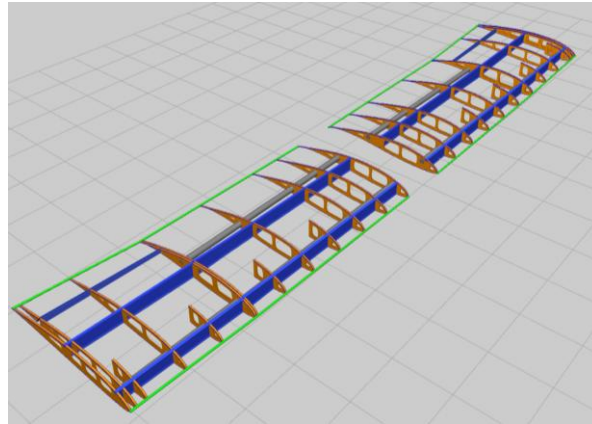


Figure 38 Model of wing for Prototype 1 modelled in WingHelper

Although FEA was bypassed for this initial build due to schedule constraints, the inclusion of professor-recommended sandwich ribs and reinforcement rings proved successful, as the prototype maintained its structural integrity throughout maiden flight tests.

5.1.2 Prototype 2 (Lightweight, High-Speed Configuration)

Following lessons learned from V1, Prototype 2 underwent a complete structural redesign. The shift to a minimal payload configuration (three passenger ducks, one cargo puck) eliminated the need for the heavy dual-spar system. The structural design was simplified to minimize manufacturing errors and reduce weight.

First, the basic wing properties were defined in Table 20. A detailed model was then developed in WingHelper as shown in Figure 39 and Figure 40.

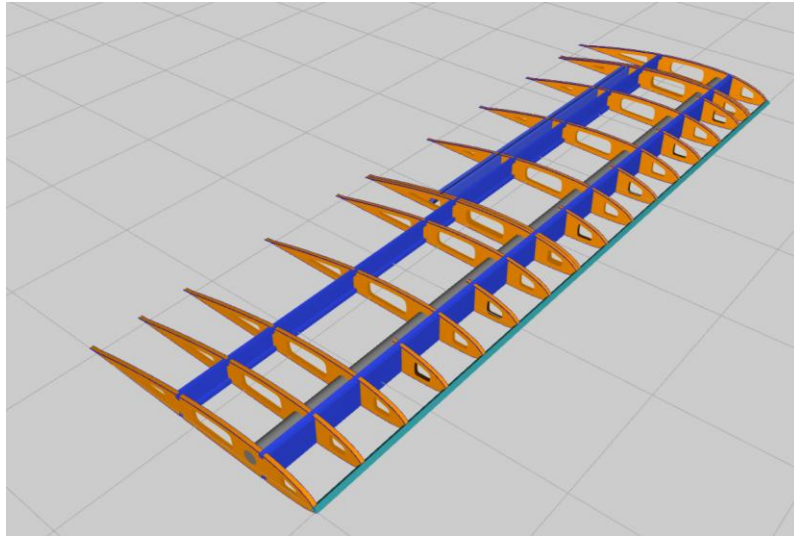


Figure 39 Half wing skeleton model of prototype 2 in WingHelper. Orange represents the wing, grey represents the carbon fibre rod, blue represents the secondary spar and aileron hinge, and teal represents the leading edge.

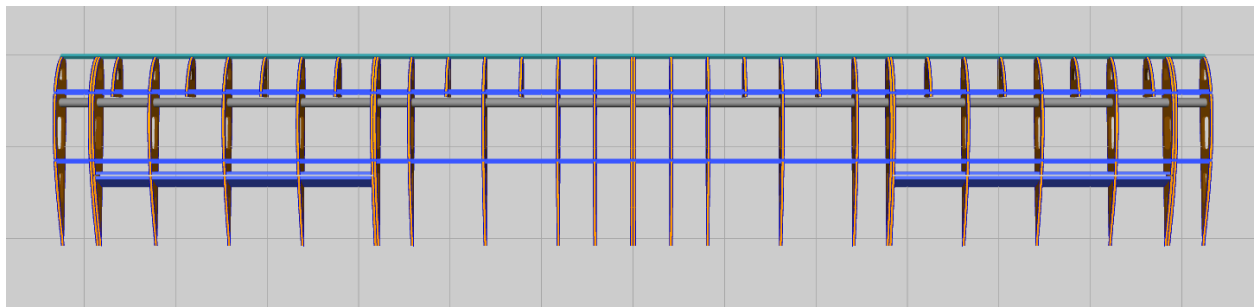


Figure 40 Top view of the wing model of prototype 2 in WingHelper

Subsequently, a high-resolution model was created in SolidWorks as shown in Figure 41 for FEA validation and to assemble a full digital model of the aircraft.

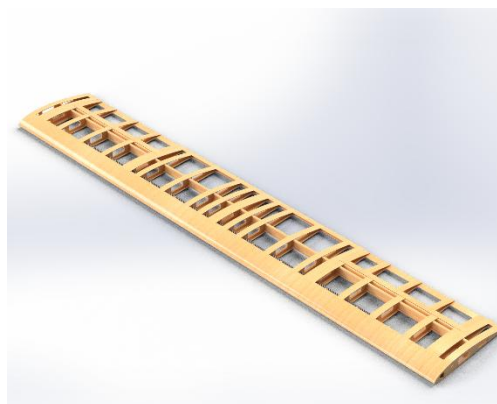


Figure 41 High resolution picture of the wing designed in SolidWorks.

Primary Structure:

The ribs and ailerons were still constructed from 3 mm balsa wood as it was still regarded for its strength-to-weight ratio. A single 1 mm thick, 880 mm long carbon fibre main spar served as the primary load-bearing member. The spar was located at 25% of the chord, aligned with the aerodynamic centre. A rear spar, positioned at 62.5% of the chord, supported the aileron hinges.

Rib Spacing and Reinforcement:

Ribs were spaced at 40 mm intervals, reduced from the 60 mm spacing used in V1. This tighter spacing improved the accuracy of the aerodynamic profile and reduced skin buckling between ribs. To prevent local crushing around the carbon fibre spar, 3 mm thick laser-cut plywood reinforcement rings were bonded to each rib at the spar hole interface as shown in Figure 42. Finite Element Analysis confirmed that these rings reduced peak stress concentration at the hole by 81.9%, validated in FEA section in section 5.1.6.

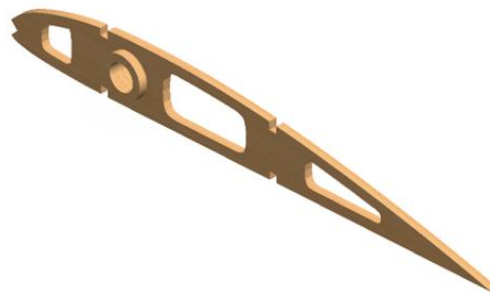


Figure 42 Rib with reinforcement rings

Secondary Spars and Webbing:

As shown in Figure 43, two additional secondary spars, constructed from 3 mm balsa wood, served a dual purpose. First, they provided structural webbing for the main spar, distributing vertical shear loads. Second, they acted as alignment guides during assembly, ensuring ribs remained perfectly positioned. The first secondary spar was located at 20% of the chord to support the D-box leading edge sheeting. The second secondary spar was placed at 55% of the chord for localized support. Unlike V1, where webbing was inserted between spars, V2 attached webbing either in front of or behind the secondary spars. This proved easier and more consistent to assemble, resulting in better vertical load distribution with fewer manufacturing errors. Moreover, the aileron hinge also acted as a spar to prevent further twisting.

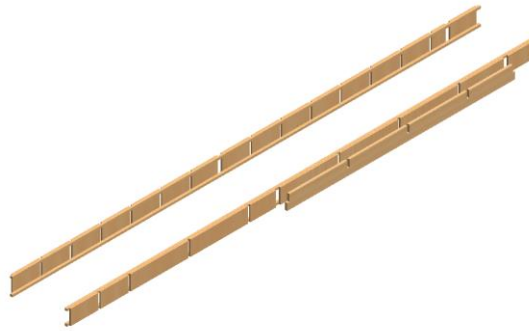


Figure 43 CAD model of secondary spars and webbing

Torsional Stiffening: D-Box and I-Beam

To maximize torsional stiffness, critical for maintaining control authority during banner towing in Mission 3, a classical D-box was formed by covering the leading edge with 1 mm balsa wood, extending from the leading edge back to the first secondary spar at 20% chord. This closed-section structure significantly resisted wing twisting under aerodynamic and banner-induced loads.

Additionally, a chordwise I-beam was created by attaching a vertical 1 mm balsa strip along each rib, extending from the rear of the D-box to the trailing edge. This reinforcement transformed the open structure into a more rigid assembly. The combination of the D-box and I-beam shown in Figure 44, improved torsional stiffness by 39% compared to an unsheathed frame, as validated by in FEA section in section 5.1.6.

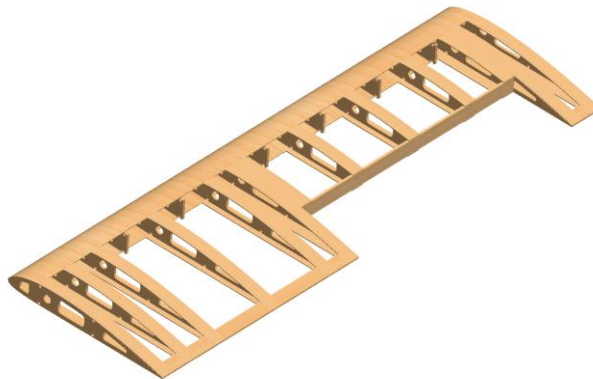


Figure 44 Sheeting of the wing skeleton

Skin and Finish:

The wing was covered with Monokote heat-shrink film, which provided aerodynamic smoothness, contributed additional torsional stiffness, and served as the hinge material for the ailerons. A lightweight ply-balsa sandwich was used for the inboard ribs to accommodate last-minute modifications to the wing-fuselage attachment interface.

5.1.3 Prototype 3 (Refinement)

Prototype 3 further refined the V2 design with two key modifications. First, a 1 mm offset was introduced in the rib dimensions to account for the thickness of the balsa sheeting as shown in Figure 45, ensuring that the final external geometry after skin application perfectly matched the intended aerodynamic profile. This eliminated the slight profile distortion observed in V2.



Figure 45 Comparison of Airfoil Size. Prototype 3 is shown above and Prototype 2 below, illustrating the smaller chord airfoil of Prototype 3.

Second, the vertical I-beam strips were extended to fully cover the first two and last two ribs of each wing half as shown in Figure 46. This provided enhanced structural rigidity at the critical wing-root junction (where bending moments are highest) and the wing-tip region (where aileron control surfaces are located). Furthermore, a 2mm plywood was glued with the first rib to form a partial sandwich structure to increase the strength of the connection



Figure 46 Half-Wing of prototype 3 with the extended vertical strip circled in red

5.1.4 Competition

No new wing was designed for the competition prototype, as the Prototype 3 wing remained serviceable. The wing from Prototype 3 was adopted as the final design, as it had proven its performance and structural integrity through multiple flight tests. Schedule constraints precluded the manufacture of a dedicated V4 wing.

5.1.5 Spare

A minor but significant upgrade was made to the wing in WingHelper as shown in Figure 47, this time using the rib combs as the spar and webbing to accelerate the manufacturing process if needed.

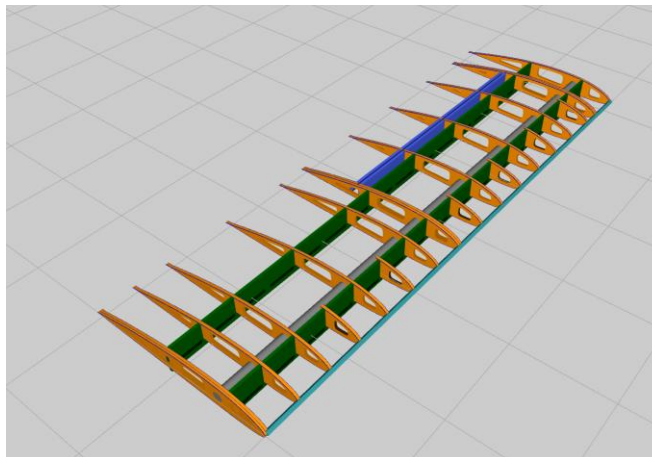


Figure 47 Half-wing model of spare in WingHelper, with rib comb (green) replacing the secondary spars and webbing

5.1.6 Structural Validation through Finite Element Analysis

The structural integrity of the optimized V2/V3 wing was validated through Finite Element Analysis conducted in SolidWorks Simulation from the detailed cad model. The analysis focused on stress distribution, deflection under extreme flight conditions, and torsional stiffness. Both models shown in Figure 48 were used for FEA.

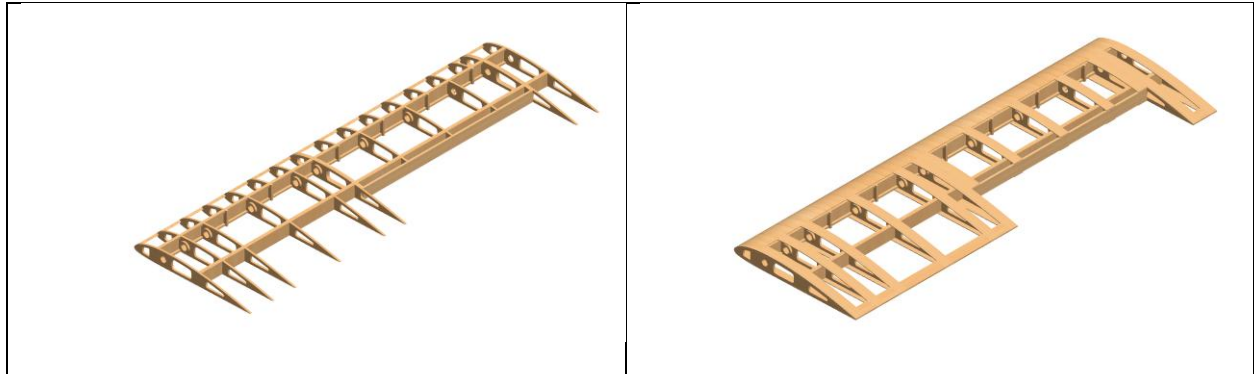


Figure 48 CAD model of half-wing in SolidWorks, with the left showing the wing skeleton and the right showing the wing with sheeting, both without carbon fibre rod

5.1.6.1 Load Cases and Boundary Conditions

A limit load factor of 6g was selected based on past team flight test data and conservative industry practice for remotely operated aircraft. The maximum take-off weight (MTOW) for Mission 3 was 2.70 kg, resulting in an ultimate load of 16.2 kg applied as a distributed pressure over the wing planform. The wing root was constrained at the fuselage interface.

For torsional stiffness evaluation, a 10 Nm torque was applied to the semi-span model, simulating aerodynamic twisting moments during high-speed turns and banner-induced yaw.

5.1.6.2 Stress Analysis and Factor of Safety

The analysis as shown in Figure 49 revealed a maximum von Mises stress of 174.2 MPa at the wing root, located near the carbon fibre spar penetration through the inboard rib. Compared against the 600 MPa ultimate tensile strength of the carbon fibre main spar, this yielded a robust Factor of Safety (FoS) of 3.44.

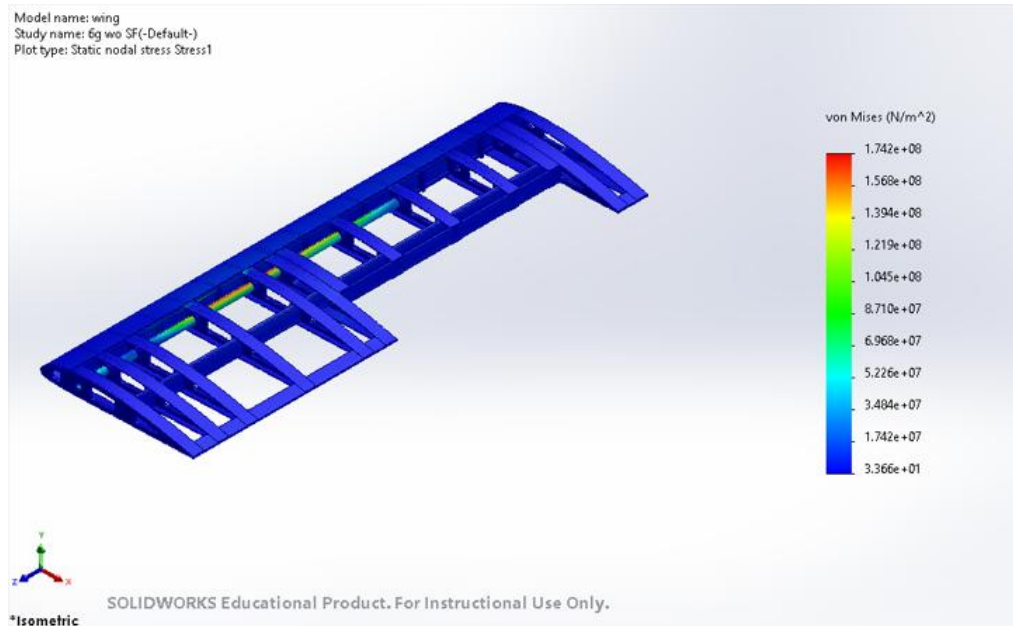


Figure 49 FEA stress test of half-wing under 6g MTOW load

The peak tip deflection under 6g loading was recorded at 42.6 mm over a semi-span of 625 mm as shown in Figure 50, representing 6.8% deflection. This was well within the margin required for maintaining control authority and aerodynamic stability, as deflections below 10% of semi-span generally produce negligible effects on lift distribution and aileron effectiveness.

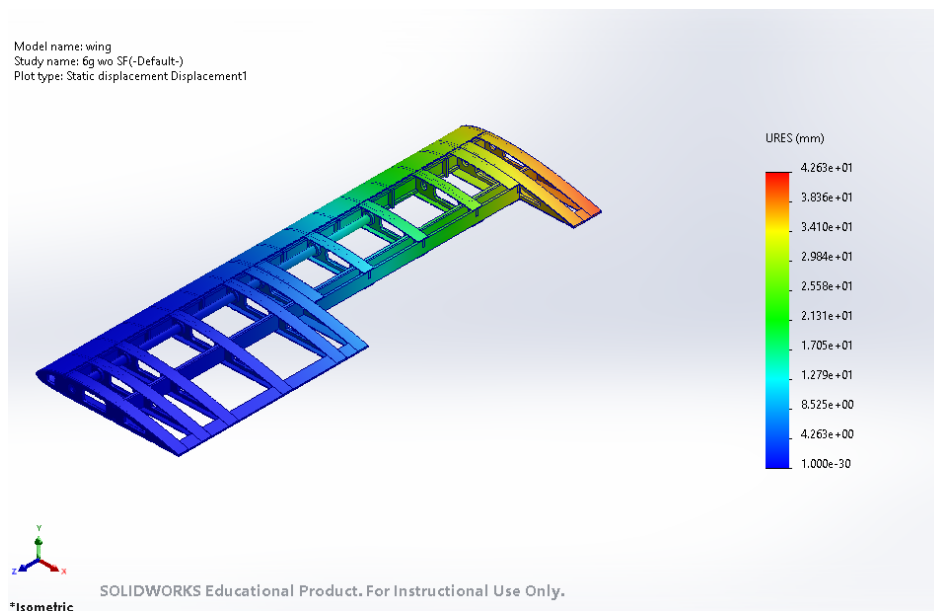


Figure 50 FEA displacement test of half-wing under 6g MTOW load

5.1.6.3 Reinforcement Ring Effectiveness

A sub-model analysis compared rib stress with and without the 3 mm plywood reinforcement rings around the spar hole as shown in Figure 51. The unreinforced rib exhibited a peak stress of 1.6×10^5 Pa at the hole interface, indicating a significant stress concentration. With the reinforcement ring bonded in place, the peak stress dropped to 2.9×10^4 Pa, representing an 81.9% reduction. This confirmed that the rings effectively distribute the load from the main spar into the rib structure, eliminating the risk of crushing or crack propagation.

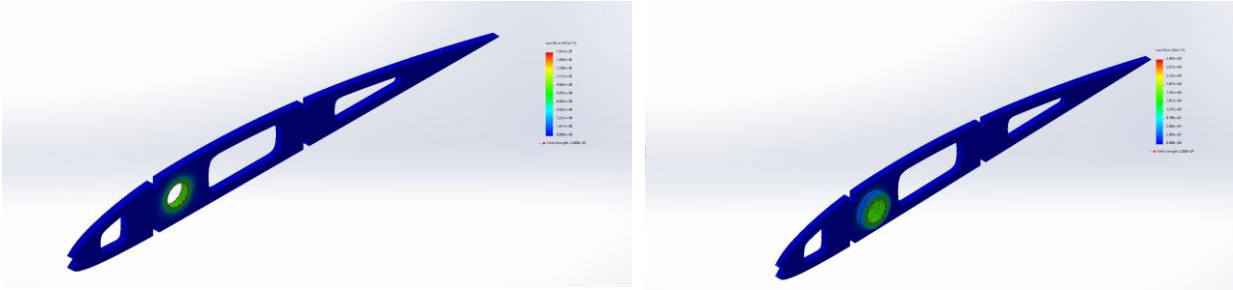


Figure 51 (Left) FEA analysis of ribs without rings, showing a peak stress of $1.6e5$ Pa. (Right) FEA analysis of ribs with rings showing a peak stress of $2.9e4$ Pa

5.1.6.4 Torsional Stiffness Validation

Torsional stiffness was evaluated by comparing tip deflection under 10 Nm torque for two configurations: an unsheathed frame (ribs and spars only) and the complete structure (D-box + I-beam) as shown in Figure 52. The unsheathed frame exhibited a tip deflection of 88.6 mm. The complete structure reduced tip deflection to 53.8 mm, a 39% improvement in torsional stiffness. This validated the effectiveness of the closed D-box and chordwise I-beam configuration.

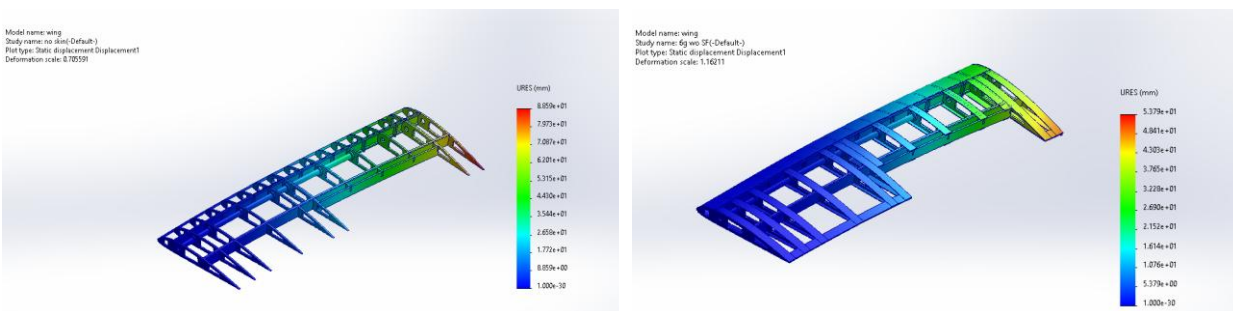


Figure 52 FEA of Wing under 10 Nm torque. (Left) without sheeting the skin deflection is 88.6 mm. (Right) with sheeting it shows a deflection of 53.8 mm

5.2 Fuselage Structure

5.2.1 Fuselage Structure: Introduction to the Final Detailed Design Stage

In the detailed design stage, Prototypes 3 and 4 will be introduced, with Prototype 3 serving as the backup aircraft and Prototype 4 designated as the final competition aircraft. Both iterations were developed using Prototype 2 as a baseline, maintaining a primary focus on high-speed configuration performance.

Prototype 3 utilized an elliptical, rounded-edged cross-section, different from the rectangular cross-section of earlier iterations, as shown in Figure 53 and Figure 54. This design aimed to increase airspeed based on the theoretical better aerodynamic efficiency of elliptical bodies. However, flight testing and airspeed sensor data indicated negligible performance gains compared to Prototype 2. Furthermore, pilot feedback suggested degraded handling characteristics, particularly during Mission 3 when the banner was deployed. Consequently, the rectangular cross-sectional shape was deemed superior for stability. Prototype 4 was subsequently developed based on the geometry of Prototype 2 rather than Prototype 3, as shown in Figure 55 and Figure 62. Detailed Computational Fluid Dynamics (CFD) comparisons are provided in the CFD Analysis chapter, whereas the current section focuses on structural design.

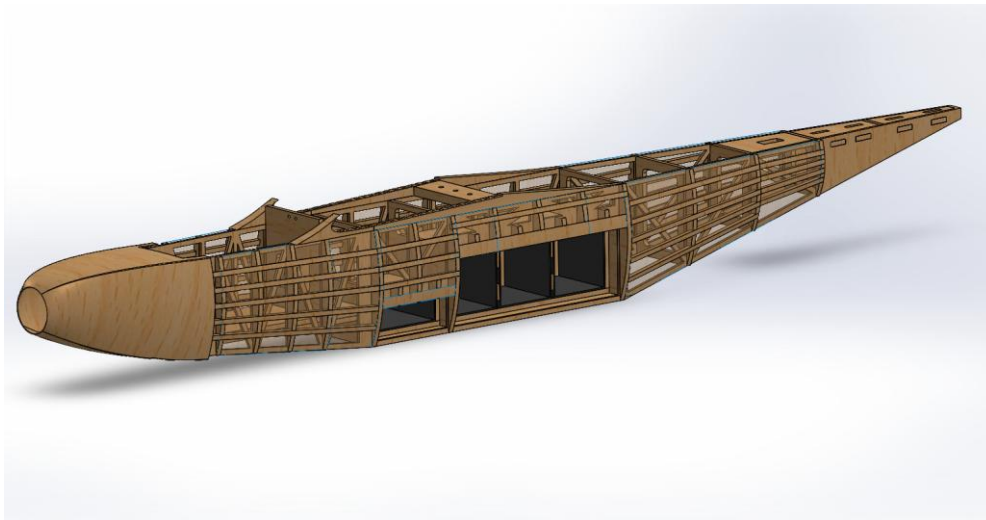


Figure 53 The isometric view of Prototype 3 fuselage

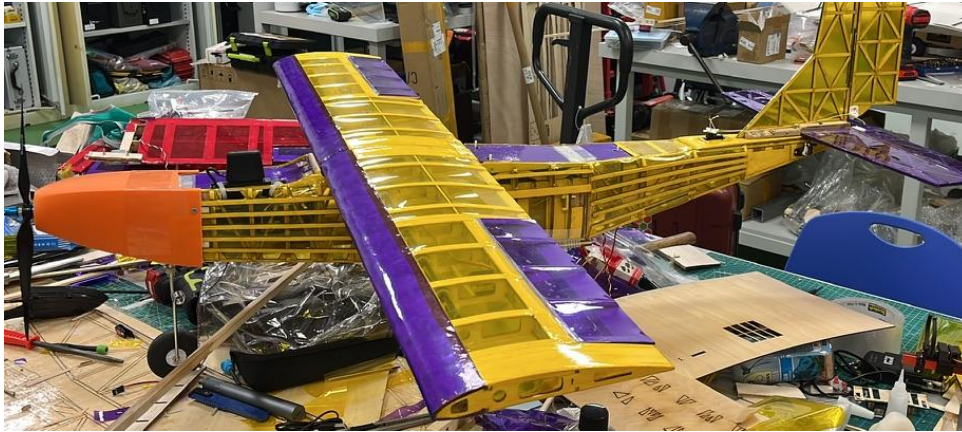


Figure 54 The Prototype 3

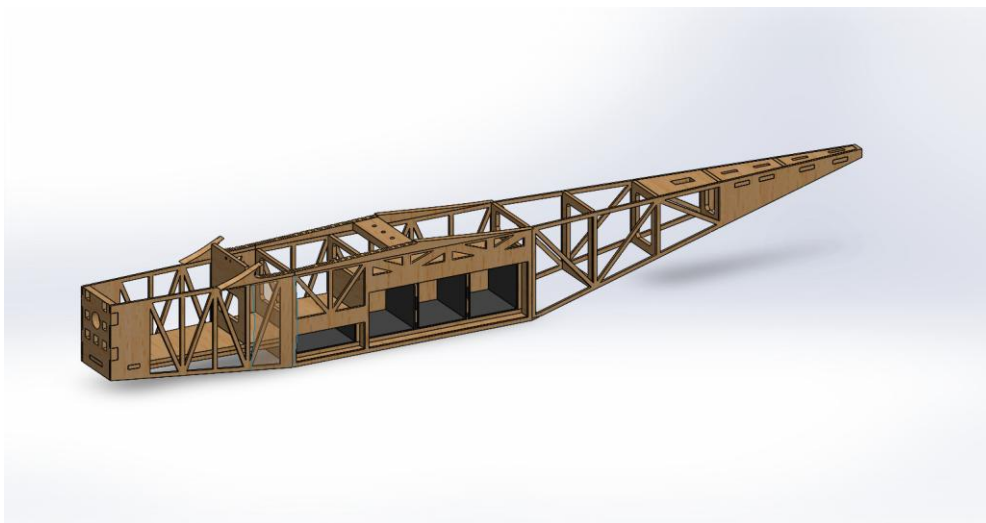


Figure 55 The isometric view of Prototype 4 fuselage

5.2.2 General Structure Commonalities: Prototype 3 and Prototype 4

Both prototypes were designed to accommodate three passenger units, one cargo unit, and all necessary avionics components. The internal layout was dictated by two primary constraints: wiring was prohibited from passing through the passenger compartment as required in the competition regulations, and the center of gravity was required to remain forward of the aerodynamic center for longitudinal stability. To satisfy these requirements, the heavier cargo compartment was positioned forward of the passenger compartment, with the avionics bay located at the furthest forward position.

Both fuselages extensively utilized interlocking tenon-and-mortise joints to enhance structural compactness, load distribution, and manufacturing efficiency. In Prototype 3 and 4, front bulkhead (Prototype 3) and the firewall (Prototype 4) were designed to interlock with the side truss panels and the avionics panel using the tenon-and-mortise joints, as indicated in Figure 56 and Figure 57. This represented a significant advancement over Prototype 2's manufacturing plan for the connection between the side truss panels and firewall, which relied solely on CA glue, epoxy resin, and triangular bracing blocks. This newly employed interlocking approach created a more robust connection for the motor mount and increased the overall torsional rigidity of the fuselage. The tail tapering section of both Prototype 3 and Prototype 4 also retained the interlocking middle panel design established in Prototype 2, as indicated in Figure 58 and Figure 59.

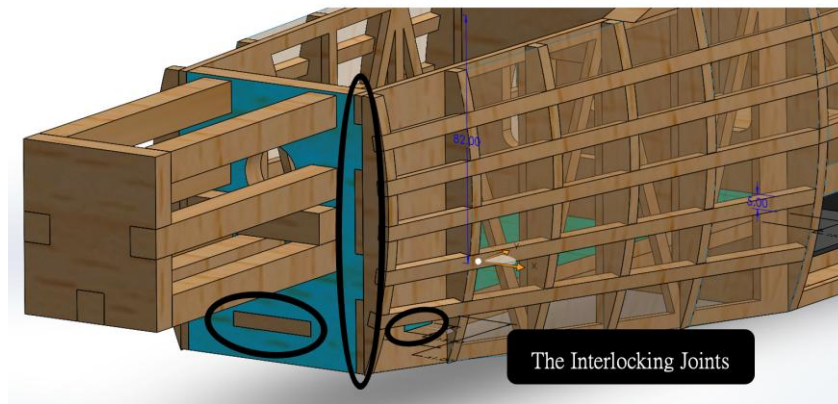


Figure 56 The demonstration of Prototype 3 interlocking joints (circled), with system panels highlighted in blue. This illustrates the one of integrated mechanical joints designs used throughout the assembly.

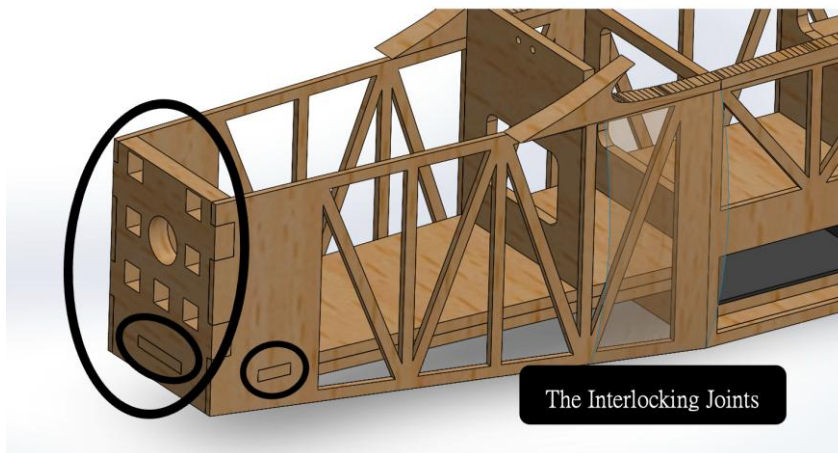


Figure 57 The demonstration of Prototype 4 interlocking joints (circled), with system panels not highlighted in blue but at the same location as Prototype 3. This illustrates the one of integrated mechanical joints designs used throughout the assembly.

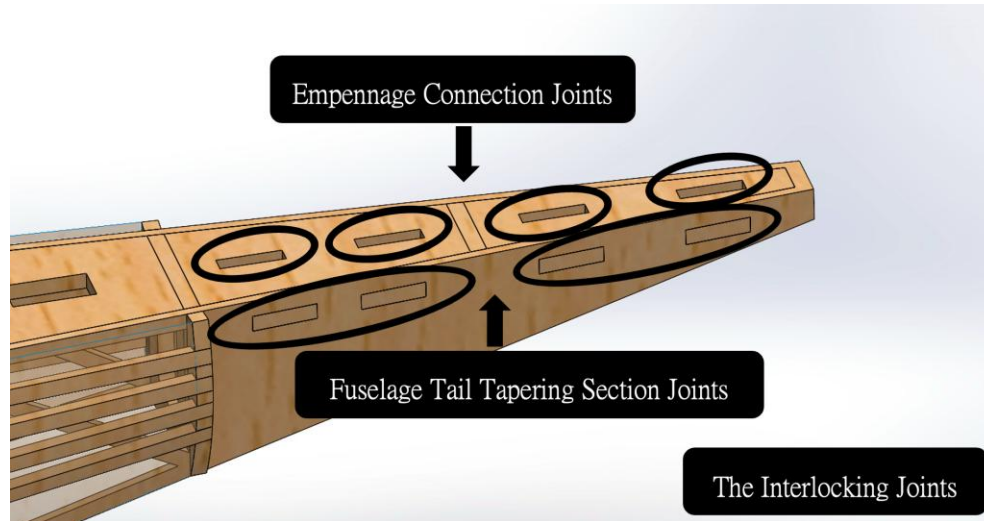


Figure 58 The demonstration of Prototype 3 interlocking joints (circled). This illustrates the one of integrated mechanical joint design used in the tail section.

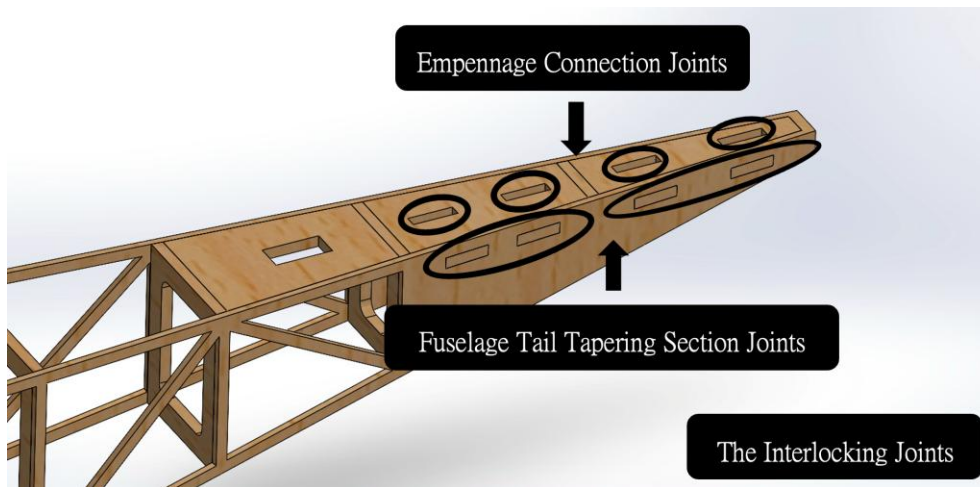


Figure 59 The demonstration of Prototype 4 interlocking joints (circled). This illustrates the one of integrated mechanical joint design used in the tail section.

5.2.3 Internal Structural Layout Commonalities: Prototype 3 and Prototype 4

The internal layout was governed by a decision matrix evaluating passenger and cargo placing orientations.

Regarding the passenger interior configuration, it was selected based on three criteria, fuselage cross-sectional area, fuselage length, and loading efficiency, with the fuselage cross-sectional area being the most critical due to its direct influence on parasite drag. Among the three configurations

considered, Three-Abreast, One-Abreast, and Two&One-Abreast, the Three-Abreast arrangement was heavily penalized as it would require approximately 178 mm of width including the retention mechanism, which was inefficient for a low-payload, high-speed design. While the One-Abreast configuration required the longest longitudinal length to accommodate three ducks in series, this increase remained within acceptable limits and even provided beneficial fuselage length for empennage effectiveness and stability. Regarding loading efficiency, the One-Abreast configuration excelled with the side-latch access design, as each duck could be loaded individually against the compartment wall without requiring alignment between multiple ducks per row. Furthermore, the One-Abreast arrangement integrated seamlessly with the pushing latch restraint mechanism by presenting a simple, uniform interface that eliminated sloping surface complications. Based on these advantages, the One-Abreast configuration was selected. In the table below, the three configurations were illustrated at the top view with the aircraft nose facing left.

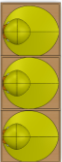
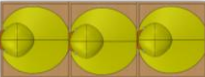
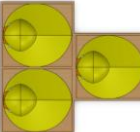
				
Passenger Interior Configuration	Weighting	Three-Abreast	One-Abreast	Two&One-Abreast
Fuselage Cross-sectional Area	50	-1	1	0
Fuselage Length	10	1	0	0.5
Loading Efficiency	40	0.5	1	0
Total Score	100	-20	90	5

Table 26 A weighted decision matrix was utilized to evaluate three passenger interior configurations against three primary criteria. Using a scale of -1 to 1, the One-Abreast configuration achieved the highest score of 90 out of 100, establishing it as the most optimal passenger interior configuration.

In terms of the cargo interior configuration, three orientations were considered, Configuration A with the rounded edge facing upward parallel to the longitudinal axis, Configuration B with the hockey puck flat surface facing upward and Configuration C with the rounded edge facing upward perpendicular to the longitudinal axis.

These configurations were evaluated against three criteria using a decision matrix in the table below, with all cargo accessed via side doors as previously established. Although Configuration B required the greatest fuselage width, it offered superior structural rigidity by itself due to its large flat contact surface generating a portion of friction for part of its retention. Configuration C demanded the same fuselage width but required more complex structural provisions due to the reduced contact area and corresponding friction. Furthermore, Configuration B also excelled in loading efficiency, which was of the greatest importance. In addition, practical loading operations proved more difficult for Configuration A compared to Configuration B and C. Given that only one hockey puck would be carried and the avionics compartment shared a similar width requirement, the width penalty of Configuration B did not significantly impact the overall design a lot. Based on these advantages, Configuration B with the flat surface facing upward was selected as the most optimal cargo interior configuration. In the table below, the three configurations were illustrated at the side view with the aircraft nose facing left.

				
Cargo Interior Configuration	Weighting	A	B	C
Fuselage Cross-sectional Area	40	1	0	0
Ease of Structural Requirement	10	0.5	1	0
Loading Efficiency (Through A Side Latch)	50	0	1	1
Total Score	100	45	60	50

Table 27 A weighted decision matrix was utilized to evaluate three cargo interior configurations against three primary criteria. Using a scale of -1 to 1, the Configuration B achieved the highest score of 60 out of 100, establishing it as the most optimal cargo interior configuration.

5.2.4 Fuselage Structure: Prototype 3

Due to time constraints in the design phase, a full semi-monocoque redesign was not feasible for Prototype 3. Instead, an elliptical profile was achieved by attaching semi-circular laser-cut side ribs perpendicular to the existing truss panel, as shown in Figure 60. Balsa longeron strips were then inset into the pre-cut slots on these ribs before applying the plastic film as shown in Figure

60. This accelerated the entire design and manufacture schedule of the fuselage by without having the need of a total structural redesign on SolidWorks and AutoCAD because a truss structure and a full semi-monocoque structure are pretty different. However, this aerodynamic approach predictably increased weight and unpredictably introduced uncertainty to flight, eventually leading to the selection of the validated rectangular profile (from Prototype 2) for Prototype 4.



Figure 60 The ribs and the balsa longerons strips that form the elliptical profile.

Additionally, Prototype 3 featured an extended motor box to accommodate a streamlined cowling and to simplify motor shaft fitment, as shown in Figure 61. In Prototype 4, this extension and the rounded ribs were removed to reduce weight and manufacturing complexity because the flight data did not justify the additional mass and additional manufacturing process.

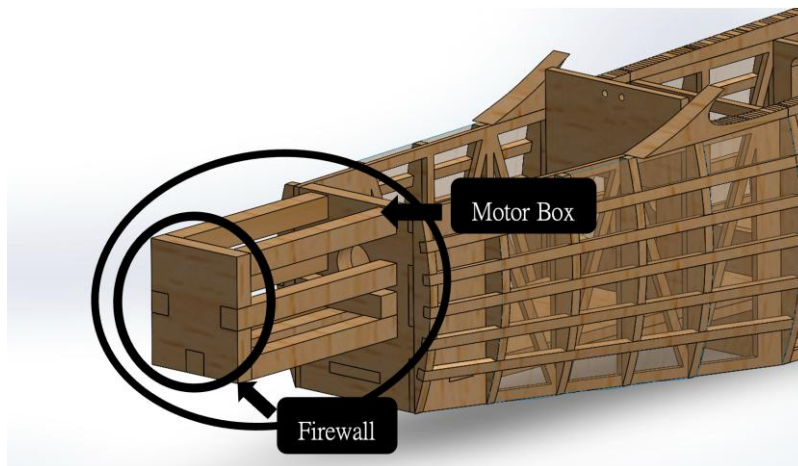


Figure 61 The motor extension box in Prototype 3, which is used to accommodate a more streamlined cowling profile and avoid the tedious work of cutting an overlong motor shaft.

5.2.5 Fuselage Structure: Prototype 4

Prototype 4, as shown in Figure 62, maintained the structural logic of Prototype 2 but featured an enlarged nose compartment to accommodate the higher-capacity propulsion battery selected for competition. This higher-capacity battery was initially used during the design of Prototype 3, consequently, both Prototype 3 and Prototype 4 utilized the same expanded nose compartment geometry, with both sizes greater than that of Prototype 2. In addition, following faculty advisor recommendations for high-speed manoeuvres, the triangular bracing blocks reinforced on the truss side panels were increased in size, compared to Prototype 2 (the baseline) as shown in Figure 63, to provide a higher safety margin for structural load handling, as shown in Figure 64.



Figure 62 The side view of Prototype 4. This final competition version utilizes a rectangular cross-section for the fuselage.

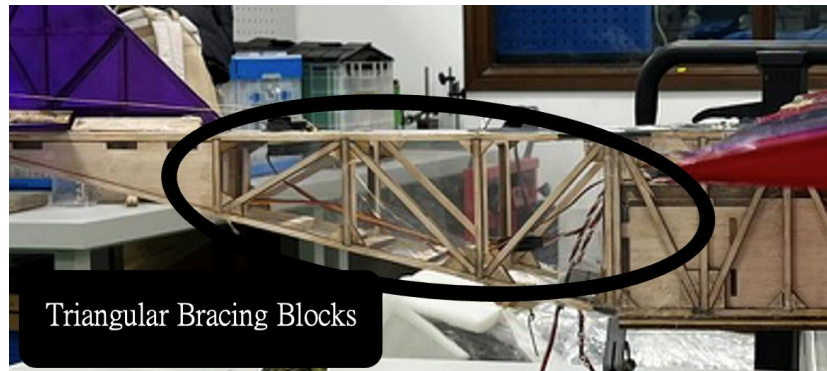


Figure 63 The triangular bracing blocks in Prototype 2. Compared to this earlier design, the blocks in Prototype 4 were increased in size to provide a greater safety margin for the competition.

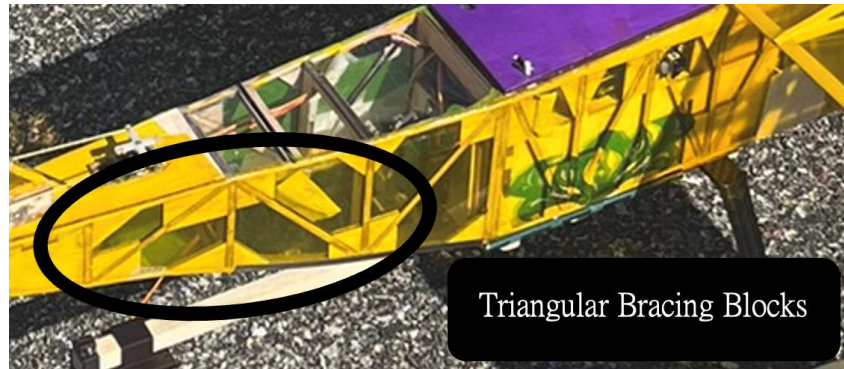


Figure 64 The greater size of triangular bracing blocks reinforced on the truss side panels to provide a higher safety margin for structural load handling under a high speed configuration.

5.2.6 FEA Analysis: Prototype 4

To assess the fuselage's capability to efficiently transfer loads from the wing, tail, avionics, and payload, a finite element analysis was performed to evaluate structural integrity under critical loading conditions. The model was subjected to a 5g loading condition at a required high-speed configuration with a 1.75 safety factor at maximum payload. Results indicated a peak stress of 4350 kPa in the longerons just forward of the wing mount, as shown in Figure 65, with a maximum deformation of 0.661 mm at the aft end of the fuselage, as shown in Figure 66. Considering Balsa's ultimate strength of approximately 20000 kPa, the design provided adequate strength for the 5g load case with the applied safety factor.

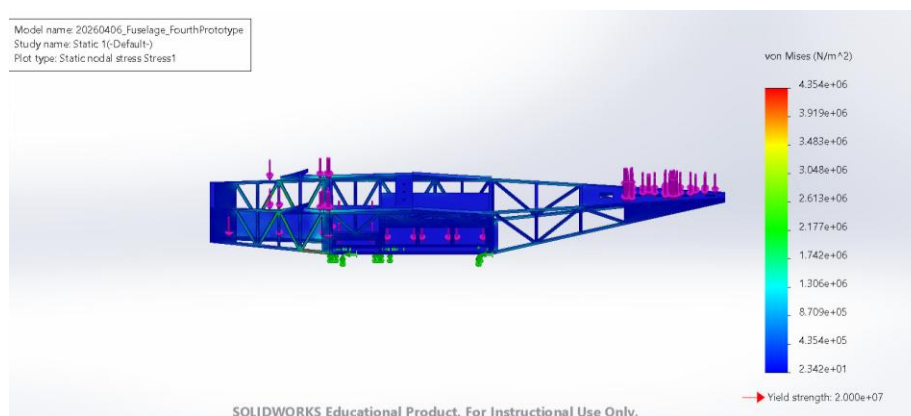


Figure 65 Finite element analysis (FEA) of the Prototype 4 fuselage under a 5g load factor and a safety factor of 1.75 for the high-speed configuration model. The Von Mises stress distribution is shown, ranging from a minimum of $2.34 \times 10^{-1} \text{ N/m}^2$ (blue) to a maximum of $4.35 \times 10^6 \text{ N/m}^2$ (red). The material yield strength is indicated at $2 \times 10^7 \text{ N/m}^2$.

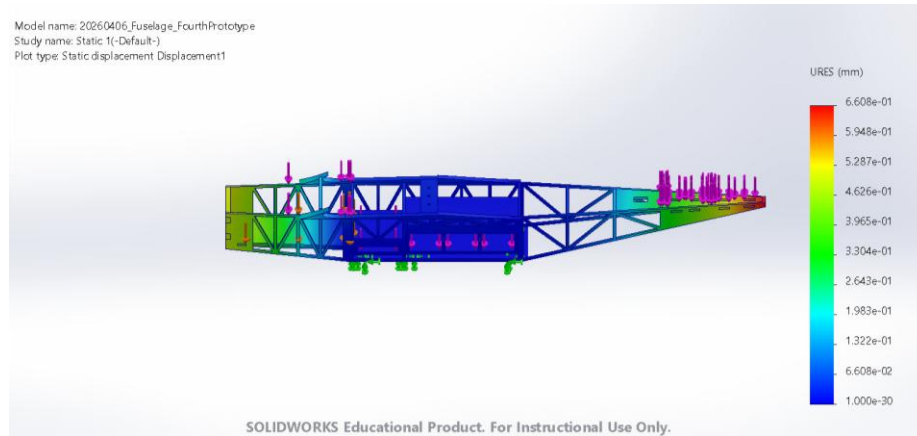


Figure 66 Finite element analysis (FEA) of the Prototype 4 fuselage under a 5g load factor and a safety factor of 1.75 for the high-speed configuration model. The maximum displacement is shown at 0.661 mm (red).

5.2.7 Payload Restraint System

Both prototypes utilized a side-access pushing latch mechanism, which were the same as prototype 2. For the passenger compartment, the ducks were secured in a foam-padded box where a side latch with pusher foam forced the ducks against the opposite wall, eliminating all degrees of freedom. Wooden strip separators were added between the ducks to prevent longitudinal shifting during high-acceleration flight. For the cargo compartment, the hockey puck was housed in an intentionally oversized box to facilitate rapid loading during ground missions. Upon closing the side door, a pusher foam block converted the initial clearance fit into a secure interference fit for flight. Inspection windows allowed for visual confirmation of retention without compromising the mechanical integrity of the latch.

5.3 Tail Structure

The empennage plays a critical role in maintaining the aircraft's flight envelope, manoeuvrability, and stability. In order to reduce the difficulty and time in manufacturing, the initial development of the empennage utilised a pre-prototype approach centred on advanced 3D printing techniques.

5.3.1 Pre-prototype

Bambu ASA Aero is preferred, a foaming 3D printing material designed to reduce weight while maintaining structural volume. The empennage featured grid-structure ribs (Figure 67) and reinforced with CFRP tube (Figure 68) to handle an aerodynamic load from manoeuvring.

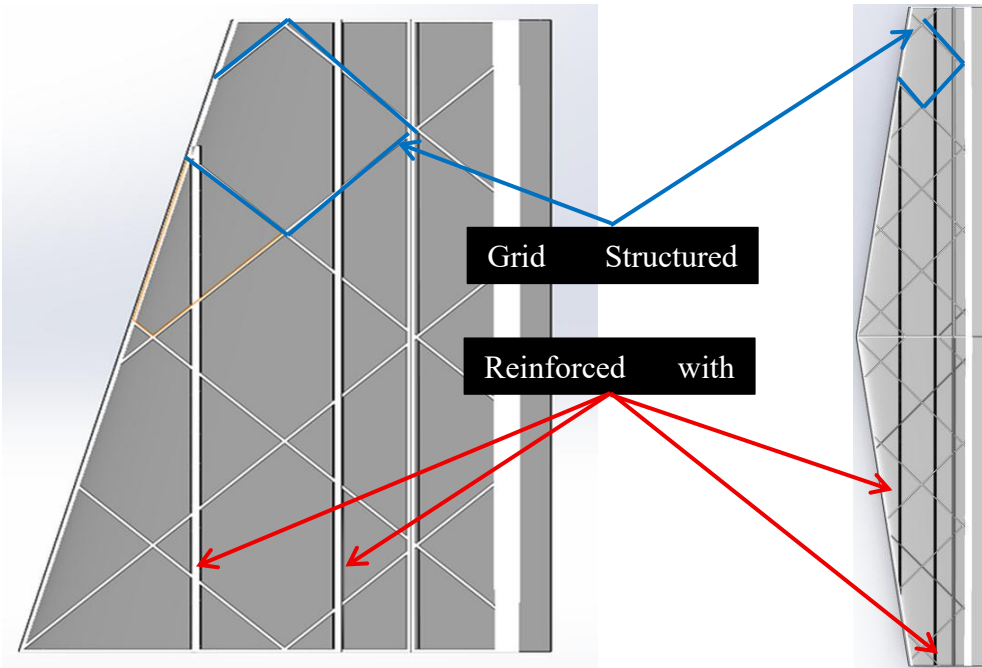


Figure 67 Pre-prototype Vertical Stabiliser

Figure 68 Pre-prototype Horizontal Stabiliser

The 3D printing phase of the empennage was marked by several critical technical failures that directly impacted both the timeline and the structural viability of the initial design. One of the most persistent issues is seams failure, where the bond between printed layers proved too fragile, leading to components that could not withstand any load. This is also compounded by thermal contraction; as the ASA Aero material cooled, it would warp or deform, causing the nozzle cannot extruded at the right place.

Mechanical errors further disrupted production including nozzle knocking. During the printing process, the printer nozzle would physically collide with the partially completed part, leading to immediate layering failure and forcing the team to restart prints that had already been running for hours. Extrusion and adhesion issue cause inconsistent material flow required frequent printing parameter testing and printer maintenance, while poor bed adhesion caused the foundational bottom layers to peel away from the print surface, rendering the entire build unstable.

These cumulative failures explain the reason to pivot to a contingency plan using more traditional materials like Balsa and Plywood for later prototypes. By switching to a hybrid of 3D printing and manual CFRP reinforcement, the team was able to bypass the fragility of pure 3D-printed seams while still benefiting from the complex internal geometries they had originally designed.

5.3.2 Prototype 1

Due to the manufacturing delays, the team pivoted to as contingency plan for prototype 1, which involved adapting previous 2025 AIAA DBF design. As the prototype 1's design strategy is to utilise the headroom of fuselage length and wingspan. The control surface has to be extending from the original design which as shown as Figure 69



Figure 69 Prototype 1 Empennage cum 2025 AIAA Prototype airborne

5.3.3 Prototype 2 to 4

The internal architecture of Prototype 2 to 4 moved toward a robust truss-style framework, which provided a high strength-to-weight ratio compared to the solid-fill methods that failed in earlier 3D printing attempts.

5.3.3.1 Extended Dorsal Fin

The dorsal fin is a critical component of the empennage's silhouette, acting as a forward extension of the vertical stabilizer that blends into the fuselage. In Prototype 2 to 4, this fin was extended to improve yaw stability and manage airflow transitions.

5.3.3.2 *Stabiliser Profile*

The Horizontal Stabiliser presents a broad, stable profile with a span of 549 mm and an aspect ratio of 4.0. Its geometry is characterized by a taper ratio of 0.8, narrowing from a root chord length of 137 mm to a tip chord length of 110 mm. With a volume coefficient of 0.9 and a wing area ratio of 0.289, this component provides the significant pitch authority required for the mission, supported by a substantial control surface ratio of 0.4.

The Vertical Stabiliser utilises a taller, more compact truss framework to manage yaw stability. It features a span of 230 mm and a lower aspect ratio of 1.5, which is better suited for directional control. Similar to the horizontal component, it maintains a taper ratio of 0.8, though it starts with a slightly larger root chord of 153 mm that tapers down to 123 mm at the tip. This vertical assembly is notable for its extended dorsal fin, which creates a long, graceful transition into the fuselage, and a high control surface ratio of 0.5, ensuring highly responsive rudder authority.

5.3.3.3 *FEA Analysis*

Although the tail assembly is not the primary structural component of the aircraft, FEA (Figure 70) is desired to evaluate potential enhancements and identify reinforcement strategies for the current design.

The static analysis applies a uniform distributed load on the horizontal stabilizer to represent a 2.5g tail lift condition, which typically accounts for approximately 10–20% of the total aircraft lift. For the $Sh = 0.075 \text{ m}^2$, assuming the tail contributes 10–15% of total lift, a conservative total load of 10 N is applied (distributed evenly across the surface, directed upward or downward depending on the positive or negative g case) to simulate the aerodynamic force experienced in flight. For the vertical stabilizer, a side load of 5 N is applied to replicate yaw torque induced by banner towing or crosswinds.

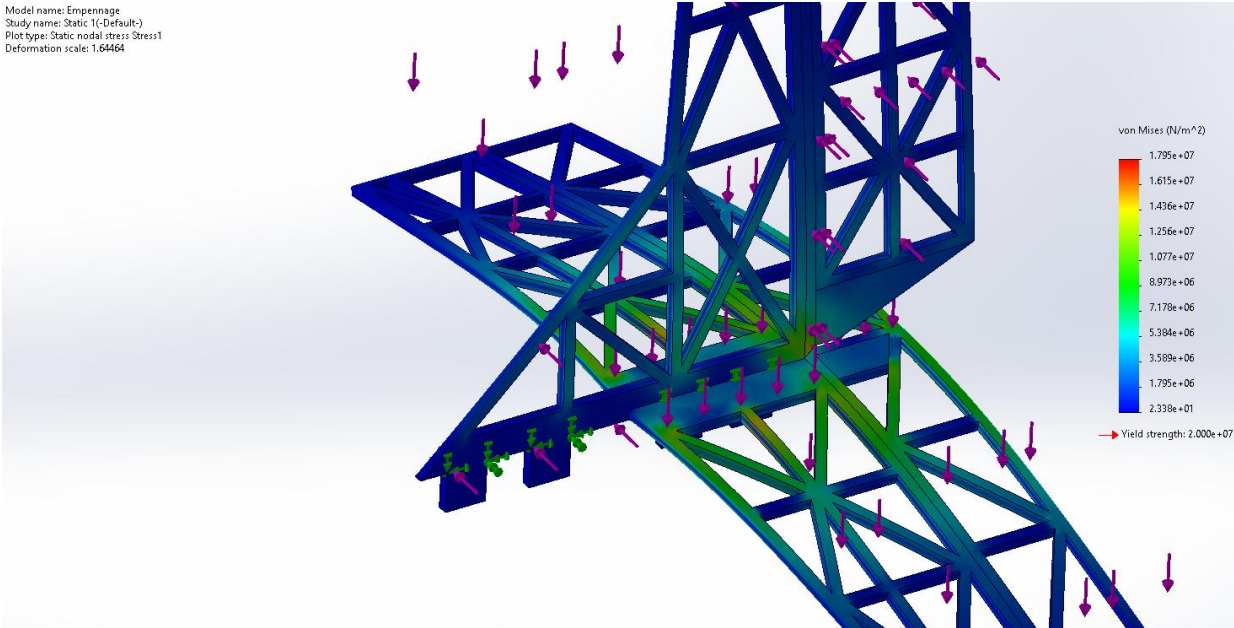


Figure 70 Von Mises Stress at Maximum UDL

For the Horizontal Stabiliser, the FEA results (Figure 71 & Figure 72) showed a maximum deflection of 3.343 mm. The peak stress recorded was 16.15 MPa, which resulted in a calculated safety factor of 1.24. This confirms that the internal truss and balsa skin remain within their elastic limits, as the stress is below the 20 MPa yield strength benchmark.

The Vertical Stabilizer was subjected to the same loading conditions, revealing a maximum deflection of 2.340 mm. It experienced a higher maximum stress of 17.95 MPa, leading to a much tighter safety factor of 1.14 (Figure 71 & Figure 72). While both structures technically passed the structural test.

These load cases enable detailed examination of stress distribution and deformation in the truss structure, mortise-and-tenon joints, and epoxy-bonded interfaces, facilitating identification of optimal reinforcement methods—such as additional carbon fibre doublers and adjustments—to improve durability and stability during Mission 3 operations.

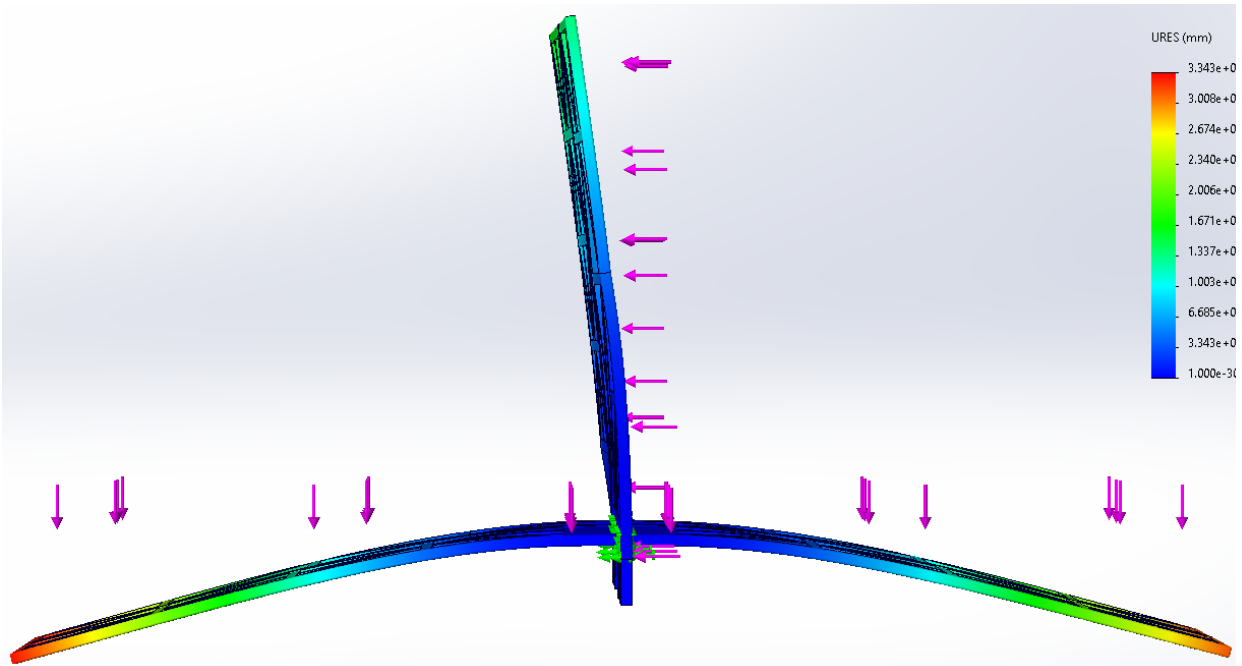


Figure 71 Prototype 2 FEA Test with MAX Displacement at 3.343mm & 2.340mm for Horizontal Stabiliser & Vertical Stabiliser respectively

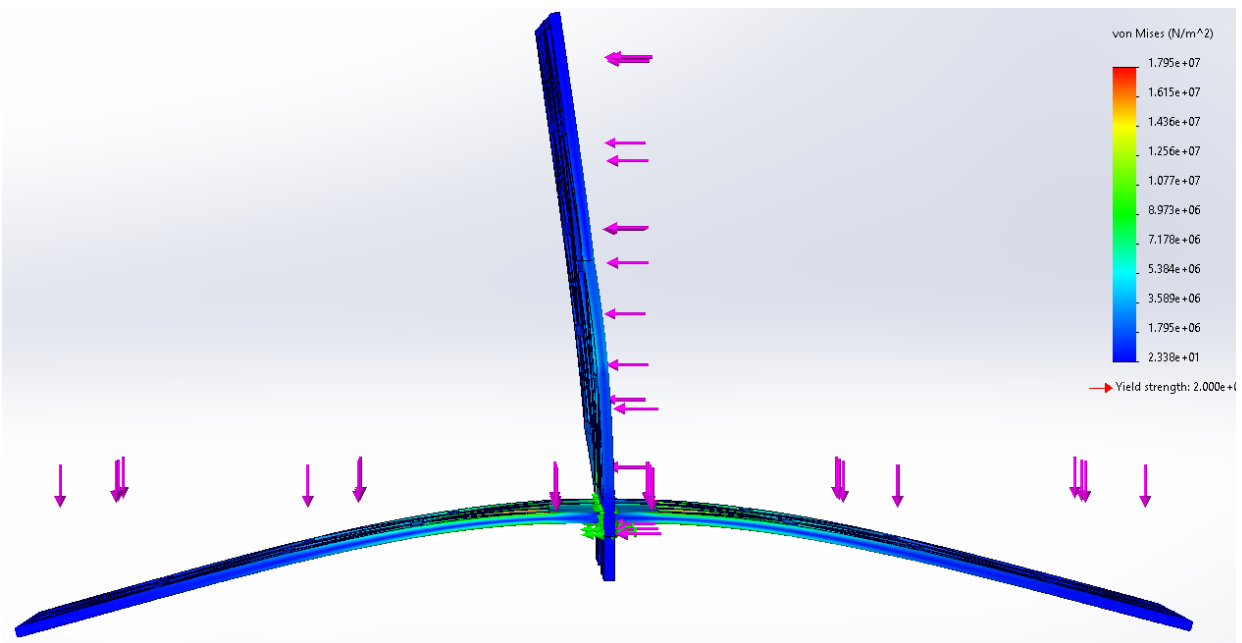


Figure 72 Prototype 2 FEA Test with MAX Stress 16.15MPa & 17.95MPa for Horizontal Stabiliser & Vertical Stabiliser respectively

5.4 Landing Gear Structure

The landing gear went through four distinct prototypes following shifts in the team's preferences, structural constraints, and material considerations. For prototypes 1 and 2, the team placed significant emphasis on producing rapid iterations and low-cost builds with the adoption of 6061-T6 aluminium struts. For prototypes 3 and 4, the focus shifted toward achieving the sufficient aerodynamic efficiency, damping, and strength needed for a competition-level performance, resulting in the use of carbon fiber struts and reinforced fairings. As a result, the design, manufacturing, and validation of the main and nose landing gear assemblies across all prototypes, including static margin calculations, FEA results, and material trade-off rationales, were all heavily documented to help guide the team's progress.

5.4.1 Main Landing Gear: Prototype 1 & 2

5.4.1.1 *Overall Design*

The overall CAD design of the main landing gear for prototypes 1 and 2 is shown in Figure 73. Although landing gear prototypes 1 and 2 featured different dimensions tailored to their respective fuselage geometries, the overarching design approach and manufacturing process remained identical. Both prototypes utilized a commercially sourced 6061-T6 aluminium alloy strut featuring gently curved fillets to reduce stress concentrations, incorporated 6 mm diameter axle holes, and were bandsaw-cut and manually bent to achieve their respective propeller clearances (71 mm for prototype 1 and 50.8 mm for prototype 2) and fuselage clearances (192 mm for prototype 1 and 159 mm for prototype 2).



Figure 73 Isometric view of the main landing gear assembly for prototypes 1 and 2, illustrating 6061-T6 aluminium strut integration with polyurethane foam wheels

As illustrated in Figure 74 and 75, dimensional differences between the two prototypes eventually arose from divergent structural requirements. Prototype 1 had an MTOW more than three times that of prototype 2, so consequently, its landing gear featured a thicker strut (4 mm vs. 3 mm), a higher mass (213 g vs. 172.37 g), and a wider mounting face taper (40 mm vs. 30 mm). The track width was determined to be approximately 321 mm for prototype 1 and 312.4 mm for prototype 2 based on the stability-based empirical calculation in Section 4.8.4 [14]. The greater ground clearance for the prototype 2 landing gear was necessitated by two factors: (1) accommodating the banner mechanism prototype beneath the fuselage, which, at that time, was 9 cm tall; and (2) preventing tail strikes during take-off at angles of attack up to 13 degrees, given the reduced fuselage clearance imposed by the banner mechanism's integration.



Figure 74 Dimensional overview of the prototype 1 main landing gear assembly with major parameter annotations
(unit: mm)

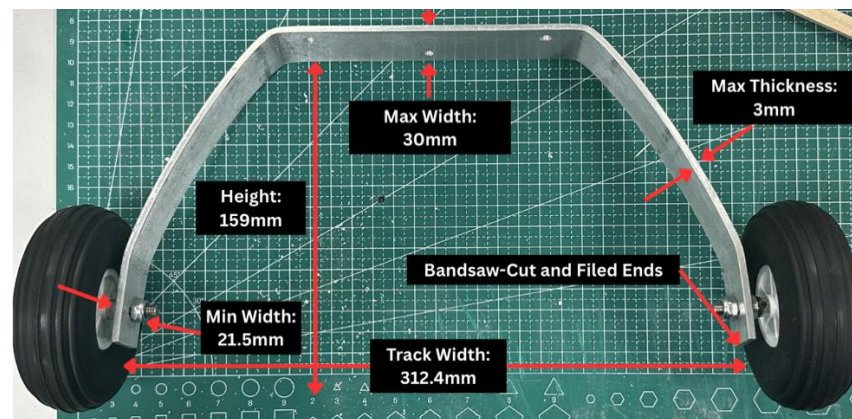


Figure 75 Dimensional overview of the prototype 2 main landing gear assembly with major parameter annotations
(unit: mm)

5.4.1.2 Material Choice

6061-T6 aluminium was initially selected for the first two prototypes because in the early stages of design and manufacture, it appeared to be the best overall compromise compared to the other options considered. It offered a strong strength-to-weight ratio, excellent ease of fabrication, the best cost and availability, and respectable stiffness. Although it had neutral damping and slightly negative fatigue resistance compared to carbon fiber composite and 4130 steel struts, its balanced performance across both structural and practical criteria made it a logical, low-risk choice for early-stage prototyping where iteration speed and budget were critical.

5.4.1.3 Major Design Choices Made

In Section 4.8.3, theory dictates that the main landing gear should be positioned to achieve a static margin on ground of 15%. However, during the development of prototypes 1 and 2, the integration of the banner mechanism spanning nearly the entire aircraft length was found to induce a significant CG shift. To compensate for this added weight and associated pitching moment, the main landing gear was repositioned to achieve a static margin on ground of 36% in its unloaded state. This adjustment followed calculations showing that, as the banner mechanism was integrated, the resulting CG shift would bring the static margin on ground to the target 15% in the plane's loaded state.

5.4.1.4 Design Validation

In Figure 76, one can see that this configuration was validated with FEA analyses (5g load factor and 1.5 safety factor), which confirmed the landing gear's ability to withstand 64% of the aircraft's weight in the unloaded state (36% static margin on ground) and 85% of the aircraft's weight in the fully loaded state (15% static margin on ground). Once validated, this design choice ensured that the aircraft stands firmly in both payload-loaded and payload-unloaded states, while still directing the majority of static and landing impact loads through the main landing gear upon touchdown.

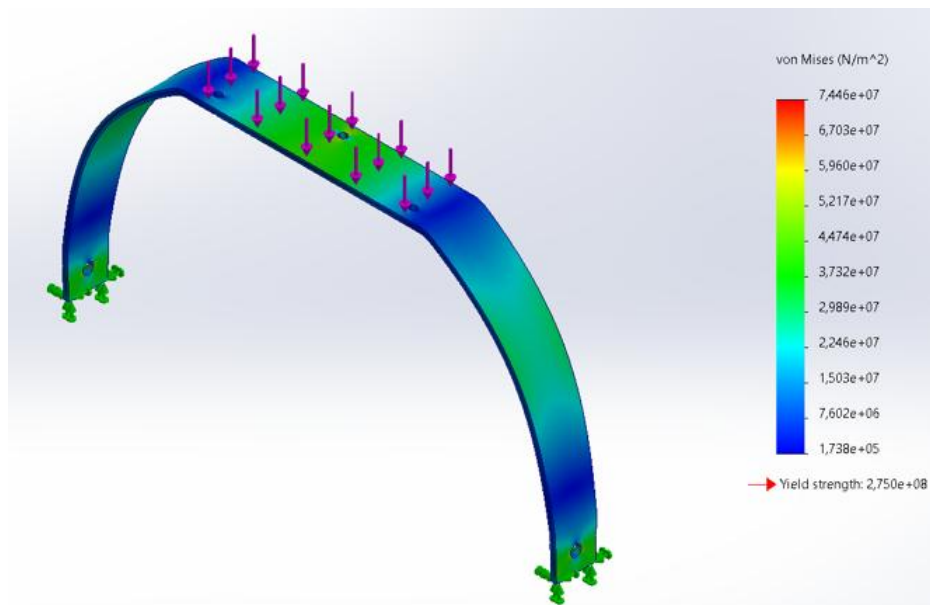


Figure 76 FEA of the prototype 1 and 2 main strut CAD under end constraints and distributed loading, showing von Mises stress distribution from $1.738 \times 10^5 \text{ N/m}^2$ (low, blue) to $7.446 \times 10^7 \text{ N/m}^2$ (high, red), with yield strength marked at $2.75 \times 10^8 \text{ N/m}^2$.

5.4.2 Main Landing Gear: Prototype 3 & 4

5.4.2.1 Overall Design

The main landing gear CAD design and final manufactured assembly for prototypes 3 and 4 are shown in Figure 77 (a) and (b), respectively, and remained dimensionally identical across both prototypes, as the fuselage design was virtually unchanged. The assembly has a mass of 55 g and utilizes a commercially sourced carbon fiber strut featuring curved fillets to reduce stress concentrations, as was discussed in Section 5.4.1.1. The 4 mm thick strut with 4 mm axle holes was drilled under running water and manually bent to achieve a propeller clearance of 90 mm and a fuselage clearance of 160 mm, as illustrated in Figure 78. The ground clearance was minimized to save weight while still accommodating the updated, smaller banner mechanism beneath the fuselage. The strut width tapers from a 20 mm mounting face at the fuselage interface to 15 mm at the wheel axle, and the strut features a track width of approximately 295 mm.

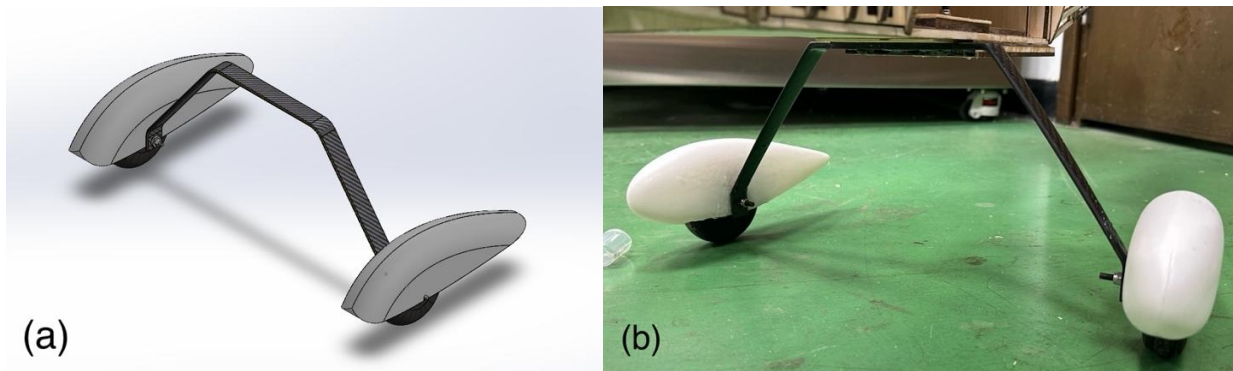


Figure 77 Transition in the preliminary design process from (a) the CAD concept to (b) the physical prototype of the UAV landing gear assembly, illustrating carbon-fibre strut integration with aerodynamic fairings.

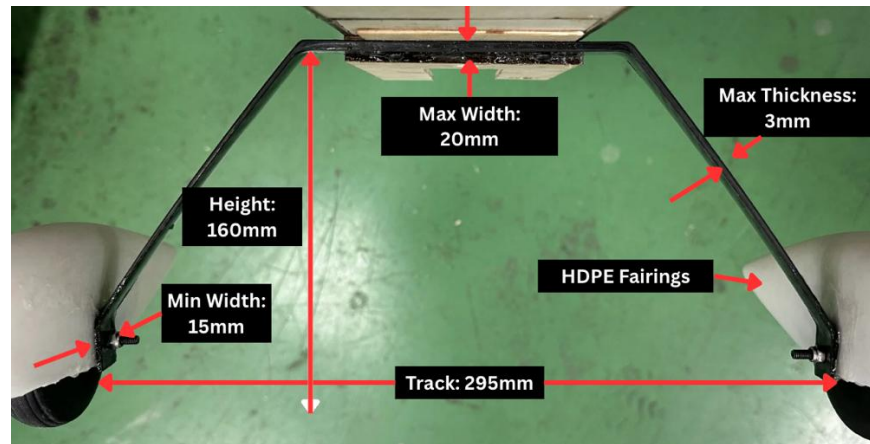


Figure 78 Dimensional overview of the prototype 3 and 4 main landing gear assembly with major parameter annotations (unit: mm)

5.4.2.2 Material Choice

While prototypes 1 and 2 used aluminium for its balanced performance during early developmental stages, a change in material was made for prototypes 3 and 4. As the testing phase concluded and the design process focused more on finalising the official competition prototype, the team's priorities shifted toward equipping the aircraft with a landing gear that exhibits improved damping, lower aerodynamic drag, and higher strength. A subsequent trade study identified carbon fiber as the more satisfactory choice, as commercially available struts are not only lighter than aluminium but also offer a higher strength-to-weight ratio for lighter airplanes.

5.4.2.3 Major Design Choices Made

The team also opted for carbon fiber because the specific strut purchased was marketed to feature a filleted leading edge, as shown in Figure 79, which reduces parasitic drag. Moreover, the carbon fiber strut was purchasable at custom dimensions that satisfied the parameters required by the team's calculations in Section 4.8, enabling rapid prototyping in the weeks leading up to the competition. In contrast, commercially available aluminium struts were consistently larger than required, necessitating subtractive manufacturing to balance weight savings against retained strength. While carbon fiber introduced increased manufacturing complexity, requiring most drilling to be done under water, its overall benefits aligned well with the performance-driven goals of the final prototypes, helping to justify the increased cost and complexity of the assembly.

Between prototypes 1 and 2, the design focus switched from carrying more payload in M2 to completing more laps in M2 and M3. As a result, an additional step was taken to incorporate high-density polyethylene (HDPE) fairings over the main wheels of the 3rd and 4th landing gear prototypes, as observed in Figure 79, to reduce aerodynamic drag as much as possible and achieve a higher average aircraft cruise speed. This yielded noticeable positive results in subsequent CFD and sensitivity analyses, which greatly improved the team's predicted performance.

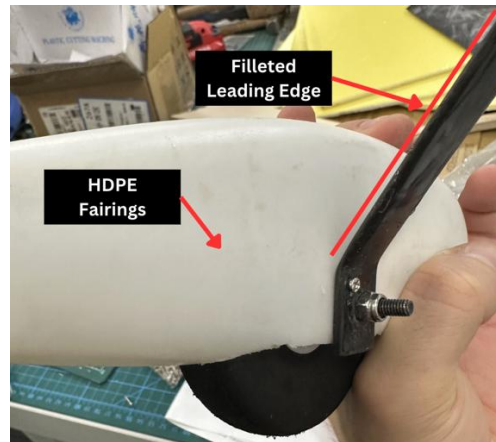


Figure 79 Close-up of final landing gear assembly, showcasing the filleted leading edge of the strut and the incorporation of HDPE fairings

5.4.2.4 Design Validation

Ultimately, an FEA analysis was conducted on the above design under maximum expected dynamic loading. A 5g landing factor was applied to the static load carried by the main gear (approximately 70% of the aircraft's 2.6 kg mass, excluding landing gear weight), corresponding to the highest anticipated M2 loads. With an additional 1.5 safety factor, the resulting ultimate test load was 133.9 N. In Figure 80, the von Mises stress distribution remains below the yield strength of the supplier's carbon fiber strut ($3.5 \times 10^9 \text{ N/m}^2$), confirming adequate strength margins.

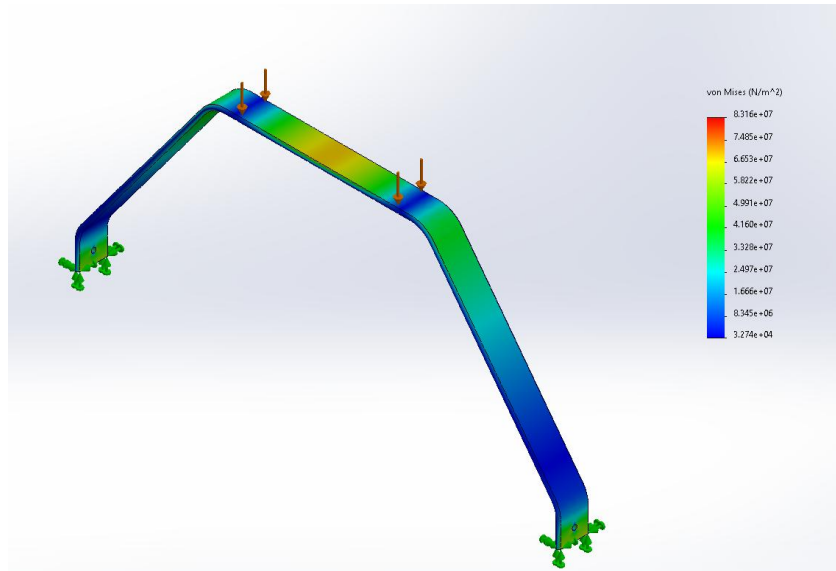


Figure 80 FEA of the prototype 3 and final prototype carbon fiber strut CAD under end constraints and distributed loading, showing von Mises Stress distribution from $3.274 \times 10^4 \text{ N/m}^2$ (low, blue) to $8.316 \times 10^7 \text{ N/m}^2$ (high, red), with yield strength set as $3.5 \times 10^9 \text{ N/m}^2$.

5.4.3 Nose Landing Gear

Figure 81 shows the nose landing gear design for all prototypes. This design comprises a lightweight 4 mm diameter stainless steel rod connected to a servo linkage via a metal pushrod. For a small UAV, the simplicity, reliability, and lower cost of an integrated coil spring nose gear outweigh the performance advantages of a more complex oleo strut, which is better suited to heavier airplanes where its higher efficiency and damping-to-weight ratio become critical [17]. The nose wheels are polyurethane molded foam: lightweight, narrow to reduce drag, and chosen for adequate energy absorption. The overall nose gear height of 210 mm in the final prototype matches the fuselage and propeller clearances established by the main gear, ensuring consistent ground contact at all three wheels of the plane.



Figure 81 CAD model of the nose gear assembly, illustrating wheel-rod-servo attachment integration

5.5 Banner Mechanism Structure

To maintain rigidity and ensure the successful deployment and release of the banner, a newly designed mechanism as shown in Figure 82 consists of two grippers for holding the banner during stow and tow stages which were controlled by servos for banner deployment. A badge reel was stored behind the passenger compartment for banner release as shown in Figure 85. This prevented soft wires from being stuck on the hooks on the aluminium tube, and ensured the banner was securely fastened.

The 3D printed gear-driven gripper mechanism as shown in Figure 83, and the servo, fixed by a 3D-printed adjustable fixture were attached to the two ends of a rectangular aluminium tube. Each gripper mechanism consists of five parts, a pair of grippers, a servo holder, a mounting device that combines the servo, servo holder and grippers together, and a small arm that fix the position of the mounting device onto the aluminium tube. The tube went through three wooden U-shape grooves which connected a plywood board. Two wooden slots fixed underneath the fuselage allowed the board to go inside, locked by a snap-mechanism made of wood and carbon fibre strip. The left edge of the banner is fixed by a 1-meter-long carbon fibre rod with a payload hanging at the bottom of the rod. To avoid the towline wire being cut by the sharp edge of the carbon fibre rod, small

strips of glass fibre tape were repeatedly stuck onto the edges. The team folded the banner repeatedly in half as a folding method for the ground mission. Three shapes of payloads were considered to be attached to the bottom edge of the banner rod in order to keep the banner vertically towed during the flight.

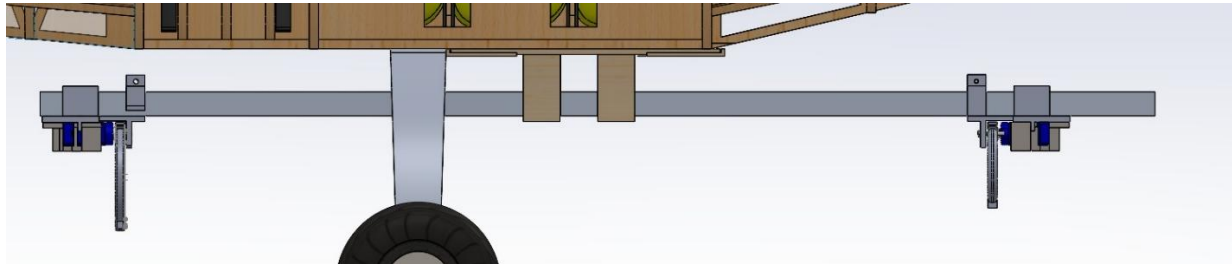


Figure 82 Banner stow-and-deployment mounting device

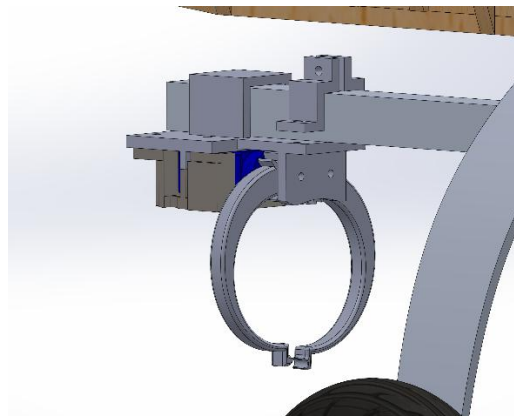


Figure 83 Gripper Mechanism

Moreover, for the release mechanism, badge reel with a length of 3m was connected to the towline of the banner. The end point of the metal wire from the badge reel was tied into a ring knot, which circled around the pin of the servo release mechanism. When the servo arm pulled the pin upwards, the ring released and thus released the towed banner. The badge reel connection with the towline line of the banner is shown in Figure 84.

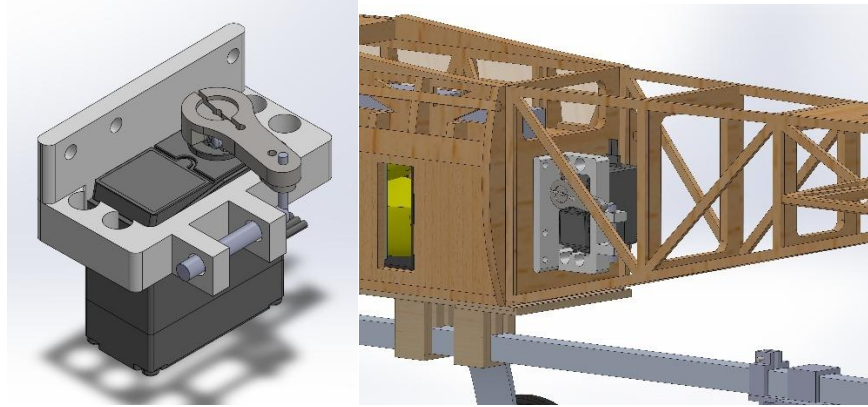


Figure 84 Release mechanism a) overview and b) position on the aircraft

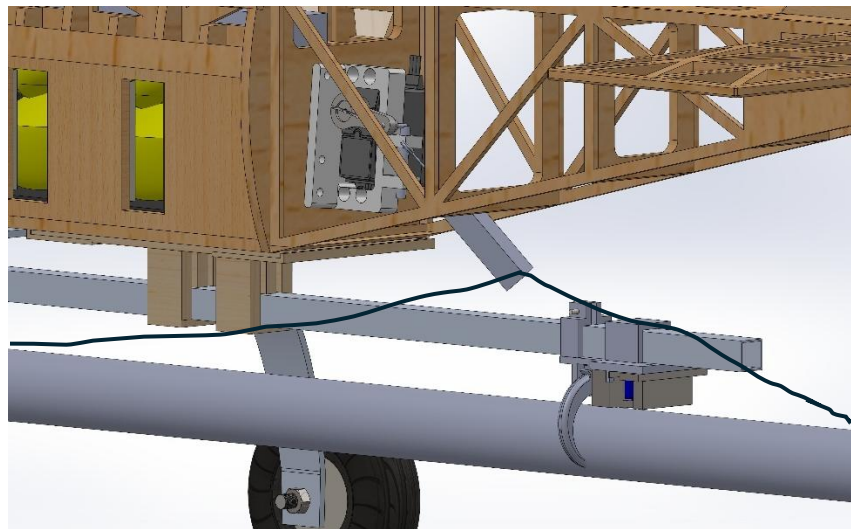


Figure 85 Badge Reel Connection with the Towline of the Banner

5.6 Avionics and Wiring

The masses of GRIFFIN's avionics components are summarized in Table 28, while their installation locations of prototypes 2 and 3 are illustrated in Figure 86 and Figure 87. The prototype 4 wiring diagram was not illustrated as the construction of Prototype 4 followed Prototype 2, and Prototypes 2 to 4 share the same overall avionics wiring architecture; however, several components were replaced, and their positions rearranged to improve performance, accessibility, and compliance with competition rules.

To minimize aerodynamic drag, the front compartment was designed to be as compact as possible. The motor was mounted on the firewall inside the nose cover, with the battery and ESC secured immediately behind it. An 80 A fuse holder was fixed to the side wall of the compartment. In Prototypes 2 and 3, a GAONENG 4500 mAh 6S 70C Li-Po battery was selected based on its low

internal resistance, high C-rating, and low weight. However, this battery is a lithium high-voltage type ($6 \times 3.8 \text{ V} = 22.8 \text{ V}$) rather than a standard 6S lithium battery, resulting in a total energy of 102.6 Wh, which exceeds the competition energy limit of 100 Wh. Consequently, a Spektrum 4000 mAh 6S 50C propulsion battery was selected for Prototype 4 to meet competition requirements.

Based on preliminary design calculations, a compact 80 A Hobbywing Skywalker ESC was selected. The UBEC with switch integrated, receiver battery, and receiver were installed under the wing in Prototype 2 and relocated to behind the wing in Prototypes 3 and 4. An external switch of the control system from the UBEC was mounted on the fuselage side wall to allow safe disconnection of power.

The FrSky Archer Plus SR12+ receiver was replaced by the Jeti Duplex REX 7 receiver in the final two prototypes to improve signal reliability, as signal loss was observed during testing with the FrSky system. Moreover, servo Y -cables were used in prototype 3 and 4 for data collection and also because of the reduction of receiver channels. For the banner release mechanism, a Dsservo digital 35 kg servo was selected because of its high torque capability.

Prototype 2 Avionics components	g	Prototype 3 Avionics components	g	Prototype 4 Avionics components	g
EOLO 13x7 Propeller	26	EOLO 12x6.5 Propeller	23	EOLO 13x7 Propeller	26
SunnySky X3120 V3 KV 585 motor	184.4	SunnySky X3120 V3 KV 585 motor	184.4	T-MOTOR AT3520 Long Shaft KV550	237.8
80A Hobbywing Skywalker UBEC (ESC)	86.5	80A Hobbywing Skywalker UBEC (ESC)	86.5	80A Hobbywing Skywalker UBEC (ESC)	86.5
Frsky Archer Plus Sr12+ Receiver	23	Jeti Duplex REX 7 Receiver	13	Jeti Duplex REX 7 Receiver	13
80A Maxi Blade Fuse	62	80A Maxi Blade Fuse	62	80A Maxi Blade Fuse	62
Fuse holder	91	Fuse holder	91	Fuse holder	91

GAONENG 4500mAh 6S 70C Propulsion Battery	409.8	GAONENG 4500mAh 6S 70C Propulsion Battery	409.8	Spektrum 4000mAh 6s 50C Propulsion Battery	580
ATECH Infinity 550mAh 2S 85C Receiver Battery	38	ATECH Infinity 550mAh 2S 85C Receiver Battery	38	ATECH Infinity 550mAh 2S 85C Receiver Battery	38
Hobbywing UBEC 10A-Car	34	Hobbywing UBEC 10A-Car	34	Hobbywing UBEC 10A-Car	34
KM9040MD Servo(s)	100	KM9040MD Servo(s) (for control system x5 and banner deployment x2)	140	KM9040MD Servo(s) (for control system x5 and banner deployment x2)	140
Guohua A0090 Servos (banner deploy & release)	27	Dsservo Digital Servo 35kg (banner release)	60	Dsservo Digital Servo 35kg (banner release)	60
Total	1081.7	Total	1141.7	Total	1368.3

Table 28 mass of avionics components for prototype 2,3 and 4

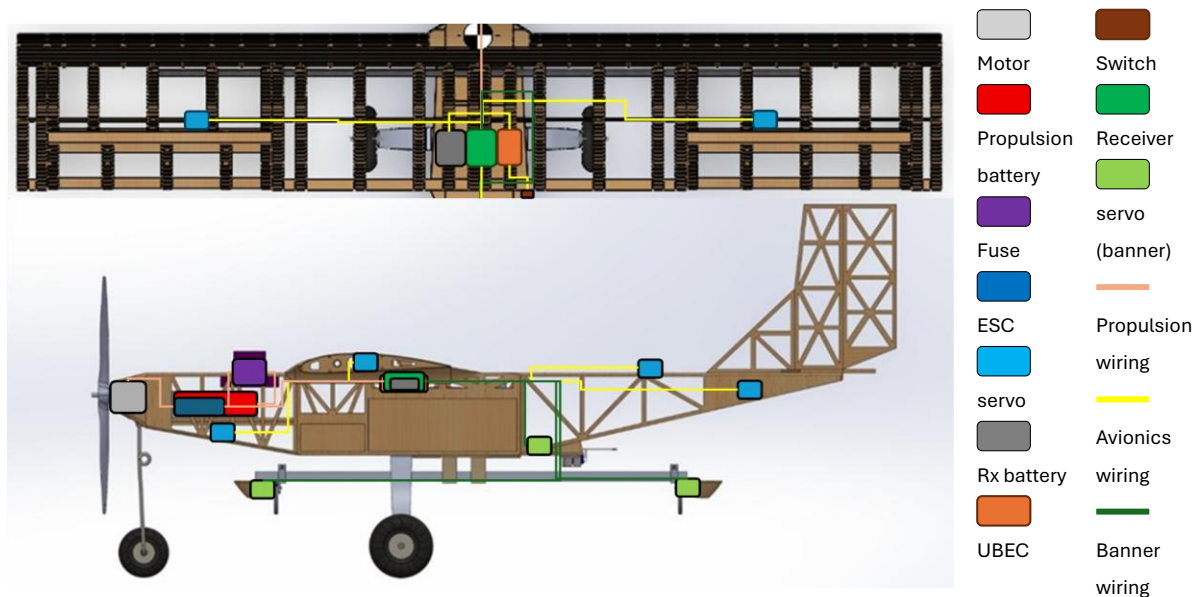


Figure 86 Prototype 2 Avionics and Wiring Diagram

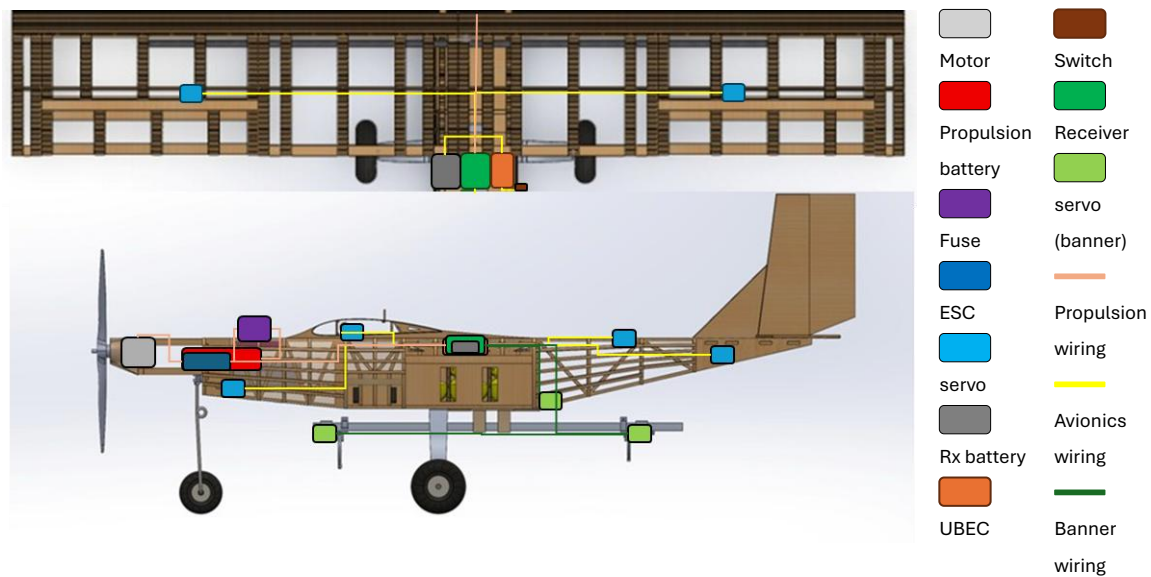


Figure 87 Prototype 3 Avionics and Wiring Diagram

5.7 Flight Performance & Mission Specific Performance

Utilising the detail design of the aircraft, the flight performance and the mission specific performance were analyzed and resulted in the table 5.6 below.

Parameters	GM	M1	M2	M3
Payload (kg)	N/A	N/A	0.23	N/A
TOW (kg)	N/A	1.61	1.77	1.81
TOFL (m)	N/A	19.8	19.8	19.8
Average Throttle (%)	N/A	100	100	100
Average Speed (m/s)	N/A	32	30.48	18
Cruise Speed (m/s)	N/A	32	32	18
Maximum Speed (m/s)	N/A	41	40.23	21
Turning Speed (m/s)	N/A	32	30.48	16.76
Battery Usage (WAh)	N/A	62	62	62
Number of Laps	N/A	3	9	5

Mission Time (sec)	25	93.75	281.25	293
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Table 29 Flight Performance Parameter Table

6 Manufacturing Plan

The team evaluated numerous manufacturing processes and materials for each major aircraft component to identify the optimal combination of weight, ease of repair, manufacturing speed, and cost. The following subsections detail the selected fabrication methods for specific subsystems.

6.1 Manufacturing Processes Investigated

6.1.1 Built-Up Balsa and Plywood

The team used a laser cutter to cut through low-density balsa wood boards according to finalized CAD models. Balsa was chosen for its high strength-to-weight ratio and easy repairability during the prototyping stages of the manufacturing process. The resulting balsa wood parts were glued together using cyanoacrylate (CA) adhesive to form the precise balsa truss structure of the fuselage with lightening holes, the wing ribs and spar webs, dihedral braces of the wing, and the star-like pattern of the empennage.

For critical structural components such as the landing gear mount plate and the fuselage formers/bulkhead, aircraft-grade plywood was used to withstand the concentrated loads, the impact/vibration, and the shear/torque generated by bolts and screws these components experience. For the firewall and the nose of the plane, a sandwich composite of balsa reinforced on either side by two pieces of plywood was considered for maximum strength to withstand propeller and landing impact forces.

6.1.2 Carbon Fiber and Stainless Steel

For load-bearing components in the model aircraft, carbon fibre and stainless-steel offer lightweight performance and localised durability. Carbon fibre was the preferred choice for primary structures such as the main landing gear strut thanks to its excellent strength-to-weight ratio, crucial damping characteristics, and ability to be sourced at specific desired dimensions, reducing the need for extensive post-processing. Its filleted leading edge also helped lower

aerodynamic drag, a feature that was difficult to find in struts made of other materials like aluminium alloys.

Stainless steel, although slightly heavy, was selectively employed for high-stress, small-volume components like axles, control linkages, and the nose landing gear strut, where its superior hardness, wear resistance, and ability to withstand concentrated point loads without deformation were critical.

6.1.3 3D Printing

Professional-grade 3D printers were mainly used for the rapid prototyping of complex shapes that would otherwise be difficult to manufacture through conventional additive manufacturing techniques, such as a smooth cowling for the propeller mount, interior compartments for the passenger and cargo, and fairings for the main landing gear wheels

6.1.4 Reinforcements

Carbon fibre strips, bonded with CA glue and epoxy, were employed to reinforce high-stress regions such as wing joints and fuselage truss intersections. Chopped carbon fibre "hair", mixed with epoxy, created lightweight reinforcements over abnormal shapes such as the nose gear mounting. Triangular wooden gussets were installed at all major structural intersections to resist racking and torsion. Fiberglass tape laminated with epoxy provided additional shear strength over seams and joints and, in some cases, was used as hinges for compartment doors. Finally, a heat-shrink film covered the entire airframe, delivering a smooth, aerodynamic surface that encased all avionics and compartments.

6.2 Manufacturing Processes Selected

The team's manufacturing plan (Figure 88) was tightly integrated with design and simulation to ensure performance and reliability. The manufacturing plan's strength lay in its logical progression and the integration of both additive and subtractive manufacturing techniques, tailored to the specific requirements of each subsystem. This linear flowchart was driven by a series of critical iterative loops essential for managing risk and elevating the team's synergy.

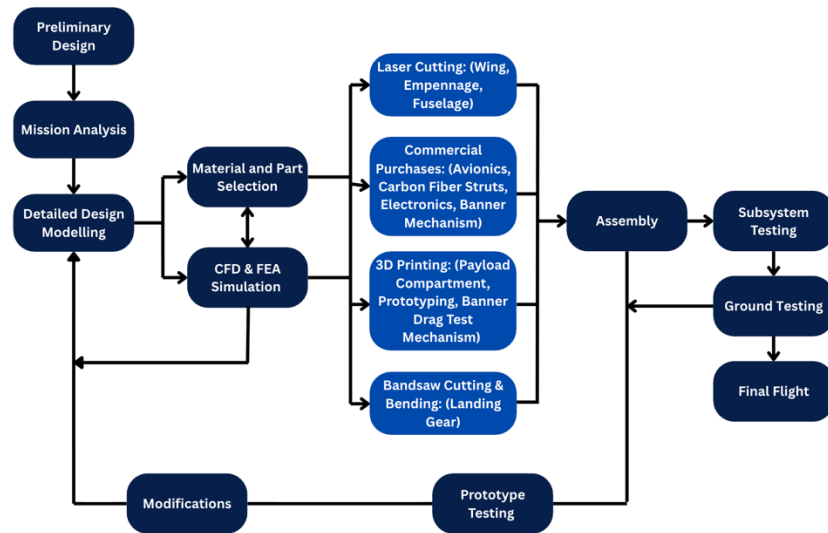


Figure 88 Closed-loop manufacturing plan showing the chosen manufacturing processes amidst the team's design strategies

6.2.1 Fuselage

Fuselage truss members were designed in AutoCAD and precision laser-cut from 3 mm balsa wood. As seen in Figure 89, each component featured integrated notches that allowed the pieces to interlock like a puzzle, ensuring accurate alignment during assembly. For the side panels, a 1:1 scale template of the fuselage profile was printed and secured to the work surface with box tape. Each balsa member was positioned and glued on this template, enabling precise, repeatable, and simple tool-free assembly. Laser-cut 5 mm balsa formers with rounded corners were glued to the side panels, integrating with the pre-assembled passenger and cargo boxes. Metal square blocks maintained precise parallel alignment during curing to ensure consistent fuselage width. Wooden stringers were then applied along the former edges to provide a mounting surface for the heat-shrink film covering.

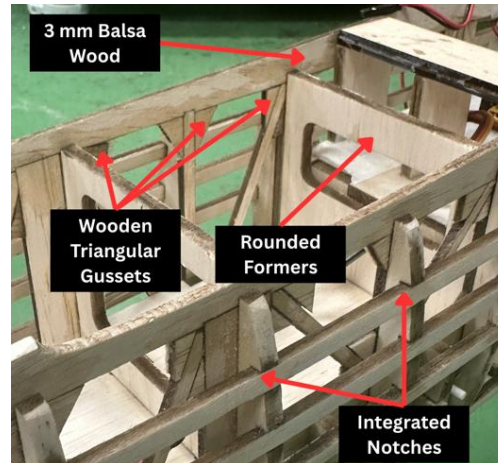


Figure 89 Structural framework of a fuselage section constructed from 3 mm balsa wood, showing triangular gussets, rounded formers, and integrated notches

6.2.2 Wing

The wing, shown in Figure 90 (a) and (b), Figure 90 was fabricated from laser-cut 3 mm balsa ribs, spars, webbings, and reinforcement rings exported from DXF files. Assembly followed a jig-based method: after the rib combs were positioned over a printed planform, the ribs were inserted in the combs, and the front upper spar was installed to establish alignment before gluing. Shear webs were glued between spars to complete the primary box structure. The leading edge was sanded to the final airfoil contour, and a D-box was formed by gluing pre-moistened 1 mm balsa sheeting over the forward section. The I-beam reinforcement was achieved by bonding 15 mm long strips of 1 mm thick balsa along the rib surfaces. The ailerons were assembled separately using balsa strips and attached via Monokote heat-shrink film, which also served as the hinge. The entire wing was covered with Monokote, the two halves of the wing were joined with structural adhesive and a carbon fibre rod, and lastly, servo cut-outs were made for the final integration with avionics.

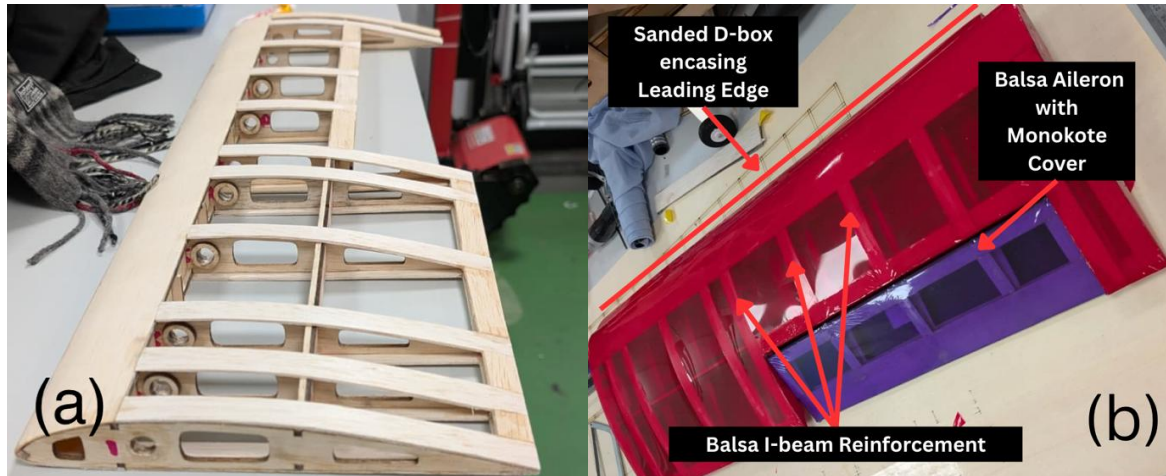


Figure 90 Construction sequence of a model aircraft wing, showing (a) the open balsa framework with ribs and spars, and (b) the finished assembly featuring a sanded D-box leading edge, an I-beam reinforcement, and a Monokote-covered aileron.

6.2.3 Empennage

The entire empennage was divided into separate parts and exported as DXF files for laser cutting from 5 mm low-density balsa sheets. These parts were assembled using an engraved plywood plate, which served as a jig or reference based on projections of both the horizontal and vertical stabilizers, and then fixed in place with CA glue. Specific high-stress areas identified in section 5.3.3.3 were reinforced with carbon fibre strips and CA glue to minimize deformation and enhance overall structural integrity as illustrated in Figure 94.

The attachment method utilizes multiple mortise-and-tenon joints in Figure 91 and Figure 92: the tenon of the vertical stabilizer passes through the mortise in the horizontal stabilizer and fuselage tail, extending through a support plate designed to reduce the degree of freedom. The horizontal stabilizer is permanently bonded to the fuselage using epoxy resin. To ensure perpendicular alignment between the vertical and horizontal stabilizers, a triangular plywood strip as shown in Figure 93 is adhered to the junction gap with epoxy, eliminating installation misalignment.

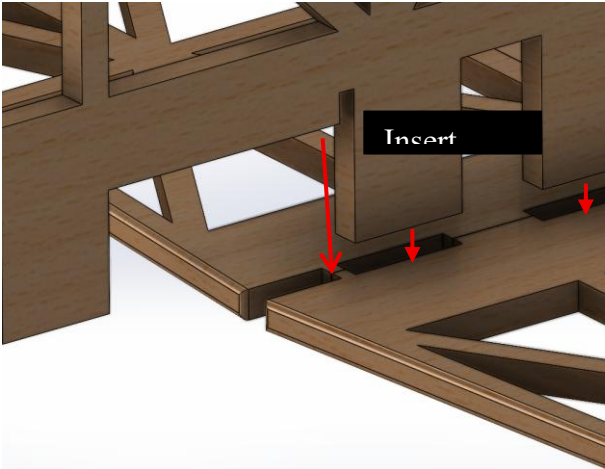


Figure 91 Empennage's Mortise-and-Tenon Joints
Installation on CAD

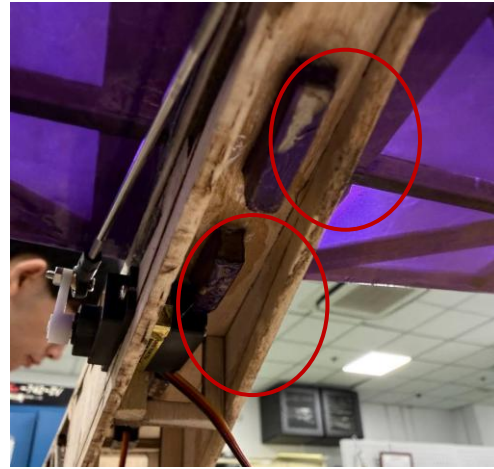


Figure 92 Empennage's Mortise-and-Tenon Joints
Installation on Prototype

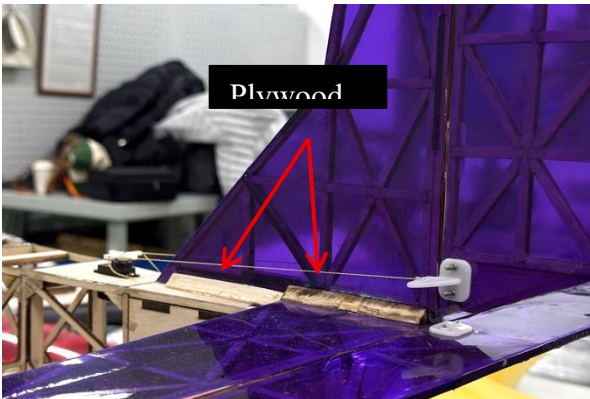


Figure 93 Empennage's Plywood Reinforcement

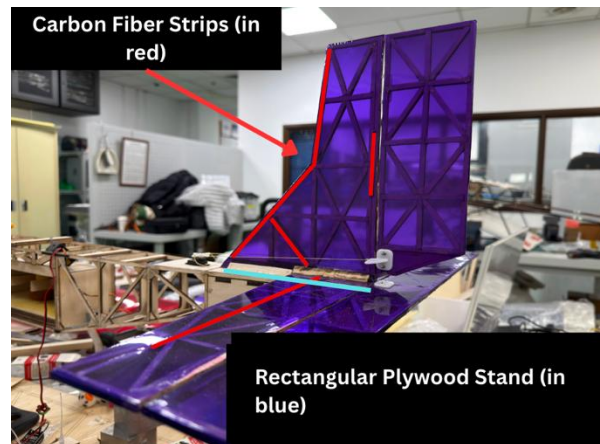


Figure 94 Empennage's CFRP Reinforcement

6.2.4 Landing Gear

In prototypes 1 and 2, using metal cutting lubricant and a vertical bandsaw machine donning a bi-metal blade, the 6061-T6 aluminium alloy strut was cut to the desired length, and the cut edges were deburred using a manual file to remove any stress-inducing sharpness. 6 mm diameter axle holes were drilled at the strut ends using HSS twist drill bits to round and widen the holes. The strut was then manually bent using a bench vise with soft jaws to prevent surface marring, with a steel pipe employed as a lever. Bending was performed incrementally, with frequent angle checks using a protractor to account for the natural springback that aluminium experiences when bent. The manufactured assembly is shown below in Figure 95.

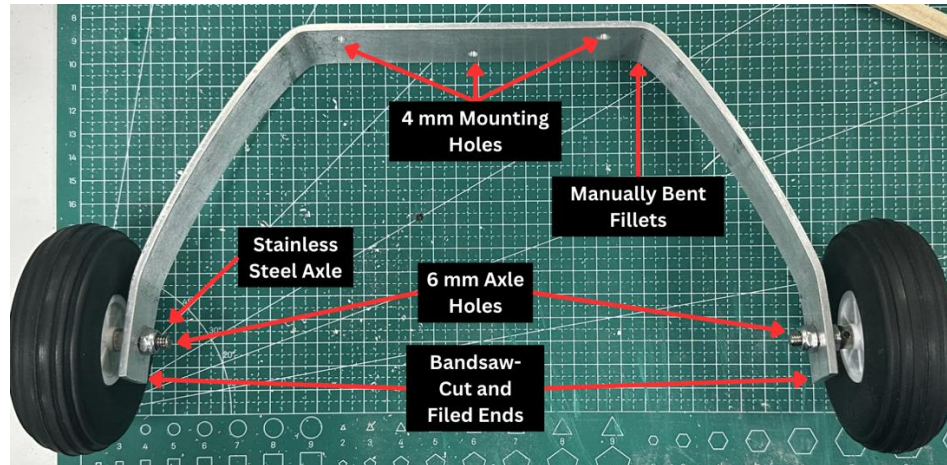


Figure 95 Frontal view of major manufactured elements within the 6160-T6 aluminium strut, including drilled mounting holes, manually bent fillets, and bandsaw-cut ends

A manufacturing interference issue arose when the attachment bolts, adhering the landing gear to the fuselage bottom, protruded into the passenger compartment. This was resolved, as shown in Figure 96, by using an additional 5 mm balsa wood layer that was laser-cut with recesses to fully embed the protruding nuts, restoring a flush interior surface without compromising structural integrity or passenger space.

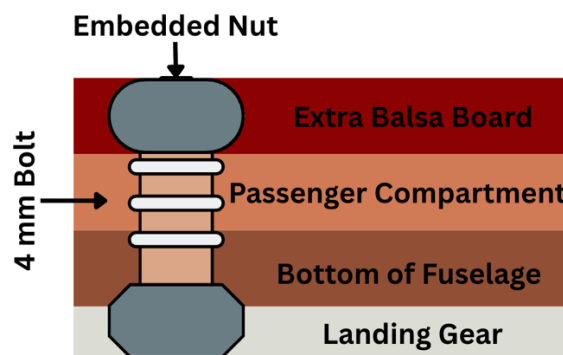


Figure 96 Cross-sectional schematic of the landing gear attachment interference solution using an extra balsa wood layer to recess protruding bolts

For prototypes 3 and 4, the main carbon fiber gear, as witnessed in Figure 73, had 4 mm mounting and axle holes drilled under running water using solid carbide twist bits. The strut was bent in a bench vise with soft jaws, using a steel pipe lever, overbent by 2° to compensate for springback. HDPE fairings were drilled and attached to the axle with self-locking nuts and epoxy.

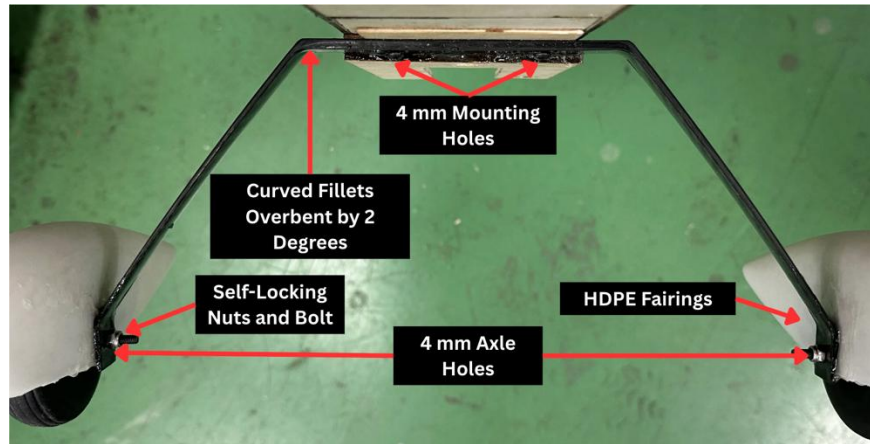


Figure 97 Frontal view of major manufactured elements within the carbon fiber strut, including overbent fillets, self-locking nuts and bolts, and HDPE fairings

The nose gear in Figure 98Figure 98 comprises an embedded coil spring fabricated from 4 mm stainless steel rods, cut to length using industrial bolt cutters and an angle grinder. A D-shaped flat was filed onto the rod to accommodate wheel collars, enabling integration with a nylon servo mount. In Figure 98In Figure 98(b), a rigid metal pushrod links the servo arm to the steering mechanism, the optimal position of which was empirically determined through iterative testing to maximize the rotational range of motion.

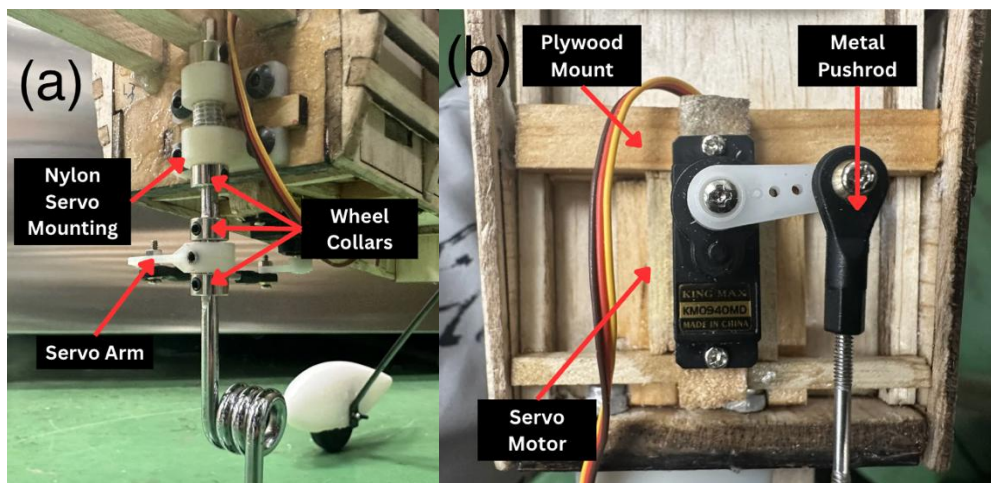


Figure 98 Nose gear manufacturing process illustrated through (a) the servo-controlled landing gear assembly with the nylon servo mount, wheel collars, and a servo arm and (b) a view of the plywood mount and servo on the underside of the fuselage

6.2.5 Banner Deployment/Detachment Mechanism

The banner mechanism, presented in Figure 99, consists of a lightweight poly pongee banner mounted on a carbon fibre rod with a 3D printed PLA payload affixed to the rod's lower end. This payload helps the banner maintain a vertical orientation during flight through a gravitational restoring moment. The entire assembly is then attached underneath the fuselage, held by two 3D-printed PLA grippers that are actuated by two high-torque servos mounted on a slender, lightweight aluminium rectangular tube. The wires from which the banner hangs are made of lightweight steel which are retracted in a commercially sourced badge reel.



Figure 99 Rearward view of the loaded banner deployment mechanism, showcasing the placement of the badge reel relative to the carbon fiber-attached pongee banner within actuated grippers

6.2.6 Passenger and Cargo Restraint System

The passenger and cargo restraint system feature holder boxes and their respective doors with pusher foams attached. The rectangular passenger and cargo holder boxes were constructed using puzzle-like panels laser cut from 5 mm balsa wood. This interlocking design allowed the boxes to hold their shape even without adhesive, facilitating alignment during assembly. However, to ensure sufficient rigidity and the secure retention of the payloads, thin CA glue was applied to all joints to permanently bond the panels together. For the doors, they were laser cut with built-in holes to allow for visual inspection of payload securement. Pusher foam blocks were attached to the doors using simple double-sided tape. The assembled doors were then installed on the fuselage using glass fibre tape serving as the hinge and a spring latch mechanism functioning as the door lock to secure the compartment during flight. The final component breakdown is shown in Figure 100.

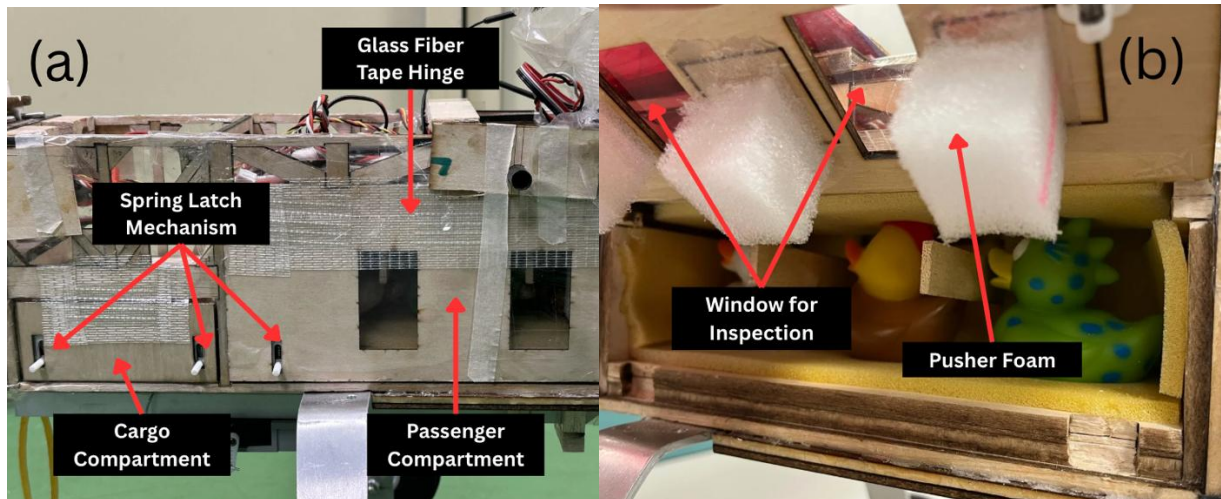


Figure 100 Fuselage payload restraint system shown from (a) the side view and (b) from inside the passenger compartment

6.2.7 Firewall

The firewall in Figure 101, was constructed following a sandwich structure, consisting of two 3mm plywood boards pressed into either side of a 3mm balsa wood core. Eight tee nuts were embedded in this composite structure to accommodate screw-mounted attachments, allowing two motors to be installed on this firewall. Ventilation holes were incorporated in the centre of the assembly, allowing for airflow to cool the battery housed within the head compartment.

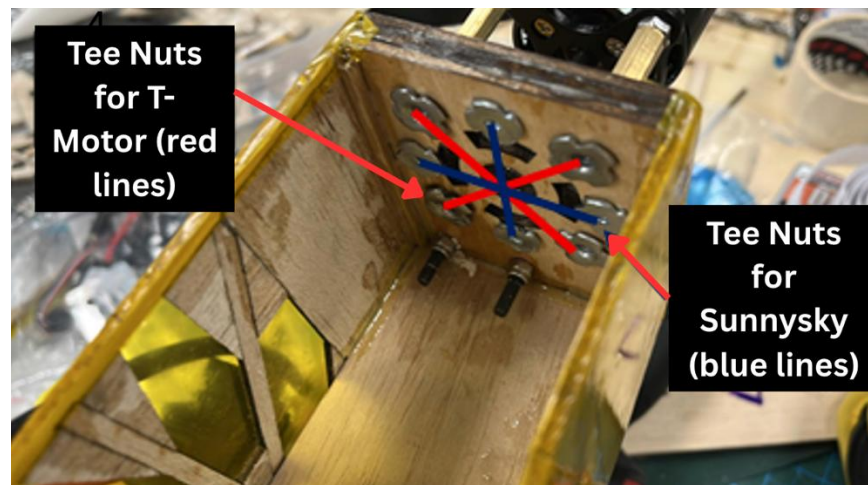


Figure 101 Firewall configuration showing four tee nuts for T-motor installation and four tee nuts for Sunnysky motor

6.2.8 Control horn/servo attachment

Three kinds of control-horn-to-servo attachment methods had been tested throughout four prototypes. Z-bend push rods were used in the ailerons of prototype 1 (Figure 102 (a)). However, this configuration necessitated a very precise bending of the push rod. Furthermore, the control horn and servo arm must be aligned within two planes which complicated their integration with the elevator due to there being limited space for attachment. Therefore, all other servo systems were replaced by adjustable ball-bearings push rods as seen in Figure 102 (b). A pull-pull wire system was selected for the rudder because of the long servo-to-horn distance (Figure 103). Kevlar wire was used in prototype 3 and 4 because of its high tensile strength.

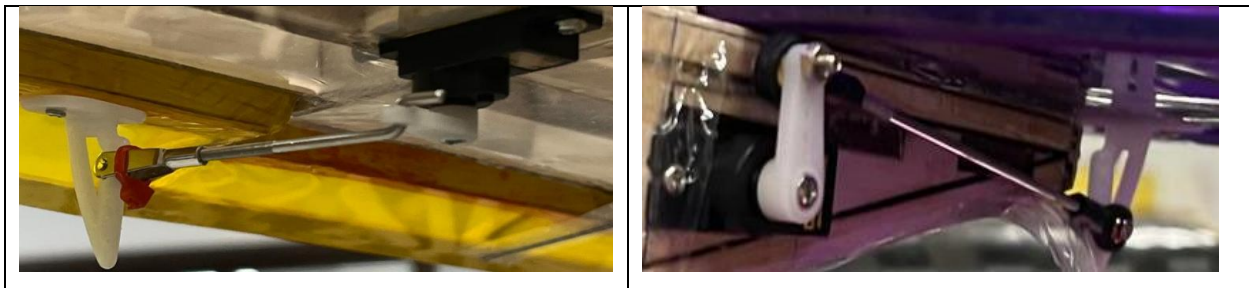


Figure 102 push rod systems: (a) z-bend pushrod (b) ball-bearings push rod

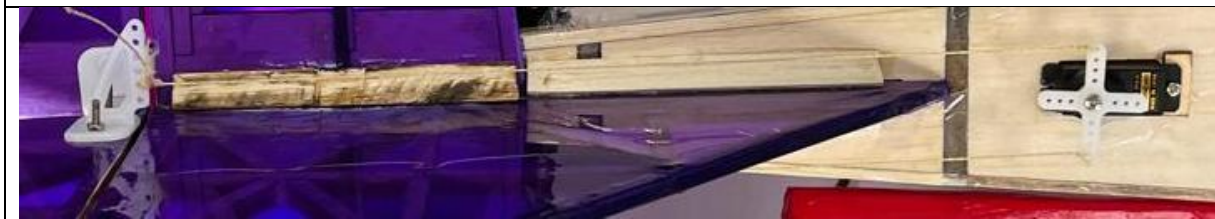


Figure 103 Pull-pull wire system

6.3 Manufacturing Milestones

A manufacturing Gantt chart was developed prior to prototype 1 to establish a clear schedule and facilitate cross-functional coordination. Progress is tracked by the manufacturing lead to ensure all milestones are met. The schedule presented in Figure 79 reflects both the proposed and actual production timeline of all four prototypes. The template was made to remain consistent across all iterations and to represent a typical 20-day manufacturing cycle for a single aircraft. However, as the fuselage design for prototype 2 became much smaller and simpler compared to prototype 1, the manufacturing process was sped up and was recalibrated to a new manufacturing timeframe of around 10 days, which the team also followed for prototypes 3 and 4. This timeline includes

allocated time for component testing and redesign, supporting the iterative manufacturing process described in Section 6.2.

Manufacture Process	Day 1-4				Day 4-8				Day 8-12				Day 12-16				Day 16-20			
Laser Cut Parts	////	////								##	##									
Fuselage Construction			////	////	////	////	////	////	////	////	////	////	////	////	////	////	////	////	////	////
Banner Mechanism Design				////	////	////	////	////	////	////	////	////	////	////	////	////	////	////	////	////
Wing Construction						////	////	////	////	////	////	////	////	////	////	////	////	////	////	////
Empennage Construction							////	////	////	////	////	////	////	////	////	////	////	////	////	////
Landing Gear Building								////	////	////	////	////	////	////	////	////	////	////	////	////
Insert Avionics			Prototype 1 Planned Timing										////	////	////	##	##			
Insert Servos and Wiring			Prototype 2, 3 & 4 Planned Timing										////	////	////	##	##			
Monokote		////	Prototype 1 Actual Timing													////	////	##	##	##
Final Assembly		##	Prototype 2, 3 & 4 Actual Timing													////	////	////	////	////

Figure 104 Manufacturing Gantt chart showing the planned and actual manufacturing timeline of prototype 1 (brown) and prototype 2, 3, and 4 (beige) across all subsystems and major manufacturing processes.

7 Testing Plan

This section explains the test objectives, test schedule, sub-system tests, flight checklist, and flight tests conducted, along with their results and discussion.

7.1 Testing Objectives and Schedule

The testing objectives and the schedule are outlined in the following Table 30 and Table 31.

Division	Method
Avionics	Assess thrust, battery current, power, and RPM across various propeller and motor configurations; obtain static thrust values.
Ground Test	
Structure, CAD	Adding payloads to the restraint systems and record the highest holdable amount.
Aerodynamic, Avionics	Perform the release mechanism of the banner using a transmitter, receiver, and servos.
Aerodynamic, Avionics	Test and conduct actual banner drag test on a prototype with flight controller.
Structure, CAD	Lift the aircraft at its MTOW vertically by the wingtips to verify that it remains level throughout the mission.
Manufacturing, structure, avionics	Simulate each ground mission and log the corresponding times required for completion.
Avionics	Connect all electronic components and validate their functionality.
Aerodynamics	XFLR5 and Ansys to assess the aircraft's stability characteristics and banner drag performance.
Manufacturing, structure	Perform a loading test and inspect the landing gear and associated connections for any damage.
Structure	FEA methods to assess individual components and whole aircraft's structural rigidity.
Flight Test	
All leads	Perform M1, M2, M3 and GM, check the installation of all parts and assembly.

Table 30 Testing Objectives

The team conducted one full-day flight test using a commercial aircraft to examine the performance of the banner mechanism for M3 before the aircraft prototype was manufactured, and two full-day tests for each prototype. All ground tests mentioned in Table 30 were conducted during each flight test.

Date	Aircraft	Objectives
9 th Jan 2026	Commercial aircraft	Banner mechanism testing
21 st Jan 2026	Prototype 1	Testing of M1 & M3 Aerodynamic performance and Trimming
4 th Feb 2026	Prototype 1	Testing of M1 & M3
11 th Feb 2026	Prototype 2	Testing of M1 & M3 Aerodynamic performance and Trimming
13 th , 27 th , Feb 2026; 6 th , 13 th , Mar 2026	Prototype 2	Banner mechanism testing
1 st Apr, 2026	Prototype 3	Testing M1, M2 and M3 Aerodynamic performance and Trimming
8 th Apr, 2026	Prototype 4	Testing M1, M2 and M3 and GM Aerodynamic performance and Trimming
14 th Apr, 2026	Prototype 3 & 4	Testing M1, M2 & M3 in US airfield and weather condition Familiarise with the flight path
15 th Apr, 2026	Prototype 4	Testing for new battery
19 th Apr, 2026	Prototype 4	Collecting data during flight

Table 31 Flight Testing Schedule

7.2 Sub-system Testing

7.2.1 Static Thrust Test

The static thrust test measured power usage, thrust, and motor temperature to guide design decisions (Figure 105). The setup followed competition rules, including motor, propeller, battery, ESC (with BEC disabled), and an 80 A fuse, simulating a five-minute mission. Safety measures included a net, belt barriers, an arming switch, and an emergency switch to prevent accidental activation and allow quick shutdown.

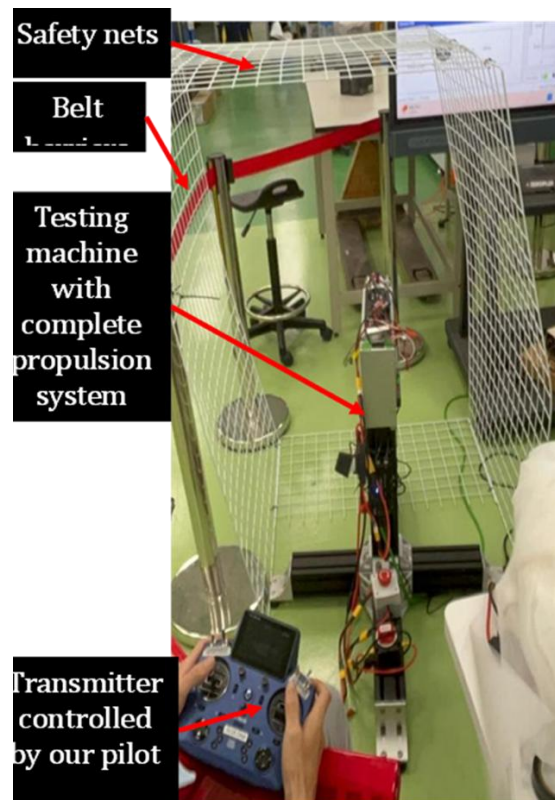


Figure 105 Static thrust test device setup

Table 32 shows the results of the propulsion test. The EOLO 13×7 propeller, being a high-efficiency propeller, generated 777 gf more thrust than the Sail 13×7 (wooden propeller). Therefore, the EOLO propeller was chosen. Moreover, when the pilot simulated the flight test and ran the propulsion system for five minutes using the EOLO 12×6.5 propeller, the actual energy consumption (61.90 Wh) was found to be close to the specification's energy consumption (63.43 Wh) at 80% of the maximum current at the battery's nominal voltage (22.2 V). Based on the above, the battery capacity selection for the EOLO 13×7 was refined to 4500 mAh based on the 95.74

Wh prediction. The linear prediction was larger than the actual energy consumption; the reason could be the difference in using different ranges of non-linear discharging curves for different battery capacities. Since the Sunnysky motor burned out during the five-minute test of the EOLO 13×7, the EOLO 12×6.5 propeller was chosen for Prototype 3 as it remained intact after the same test. For Prototype 4, the thrust generated by the T-Motor AT3520 KV585 with the EOLO 13×7 was found to be 3842 gf, which fell within the range of maximum current and thrust of the two recommended propellers (one with a larger diameter and another with a smaller pitch).

Motor	Propeller	Battery	Experimental			Theoretical		E (Wh) For 5 mins
			Maximum Thrust (T_{max}) (gf)	I_{max} (A)	E (Wh) For 5 mins	Maximum Thrust (T_{max}) (gf)	I_{max} (A)	
Sunnysky X3120 V3 KV585	Sail 13x7	ATECH Infinity 3300mAh 6s battery 55C	3594	47.56	N.A.	5040	65	81.29
	EOLO 13x7	GAONENG 4500mAh 6s battery 100C	4371	52.77	72.1	5040	65	95.7
	EOLO 12x6.5	GAONENG 3300mAh 6s battery 100C	3710	43.16	61.9	4320	50.5	63.43
T- MOTOR AT3520 Long Shaft KV550	EOLO 13x7	GAONENG 4500mAh 6s battery 70C	3842	40.43	62.4	$3575 < T_{max} < 4116$	$34.89 < I_{max} < 44.28$	N.A.

Table 32 Propulsion test results

During the propulsion test, an infrared camera was used to monitor the motor temperature to ensure that the motor did not overheat during the five-minute test period (Figure 106). The measurement results are recorded in Table 33.



Figure 106 Infrared camera for motor temperature monitoring

Motors	Sunnysky X3120 V3 KV585	Sunnysky X3120 V3 KV585	T-MOTOR AT3520 Long Shaft KV550
Propeller	12x6.5E	13x7E	13x7E
Motor temperature (°C) (After 5 mins operation)	123	210 (burnt) (Figure 107)	72

Table 33 Motor temperatures recorded for different motor-propeller combinations after five minutes of operation



Figure 107 Motor overheated during the static trust test

7.2.2 Electronics Test

Before manufacturing, a standalone control system was tested on a foam board to verify servos, UBEC, receiver, and battery responses to radio commands. (Figure 108). The ailerons were set to move oppositely with exponential response, while the rudder and front landing gear were synchronized. Radio range and fail-safe tests were also conducted. For the range test, one member walked 30 steps away to simulate 900 steps in flight, while another checked signal strength and control surface movement (Figure 109). The fail-safe test confirmed that surfaces followed competition rules when the signal was lost. All the control surfaces were checked again before flight tests were conducted at the airfield (Figure 110). These electronics tests were conducted during the first day of competition in the technical inspection.

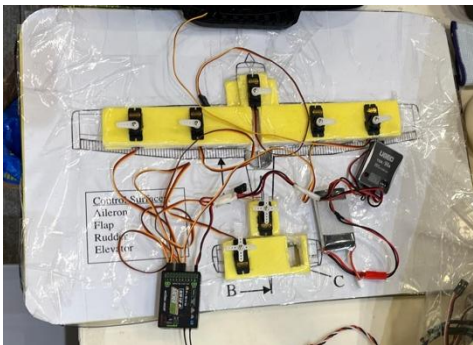


Figure 108 Electronics test (early stage)



Figure 109 Radio range test



*Figure 110
Electronics test
(airfield)*

7.2.3 Landing Gear Testing

To validate landing gear reliability, the team executed a multi-stage testing campaign progressing from basic inspections to structural and functional tests, identifying all potential failure modes before the fly-off.

First, structural integrity and alignment checks were carried out before every flight. This included visual inspection of all struts, axles, and mounts for micro-cracks or weak points. Concurrently, the team verified the security of every fastener, D-cut screw, wheel collar, and bolt to ensure no disassembly occurred. Next, as shown in Figure 111, using precision tools, the track width and wheelbase were measured to ensure symmetry, with toe and camber angles set to zero to eliminate drift or ground-looping during high-speed ground operations.



Figure 111 Example of a manual pre-flight alignment check of the wheel toe/camber angles and landing gear track width

Second, to simulate a real landing event, the team performed a controlled drop test, as shown in Figure 112(a). Assuming a stopping distance of 40 mm, the fully loaded aircraft was dropped from a height of 80 mm to simulate a 5g landing. This test was repeated multiple times to verify the gear's smooth and predictable operation in a dynamic flight environment. Lastly, a steering response test was performed on the nose gear connected to a makeshift firewall to verify that it turned freely and accurately responded to control inputs from the servo motor without hesitation. This makeshift firewall is demonstrated in Figure 112(b).

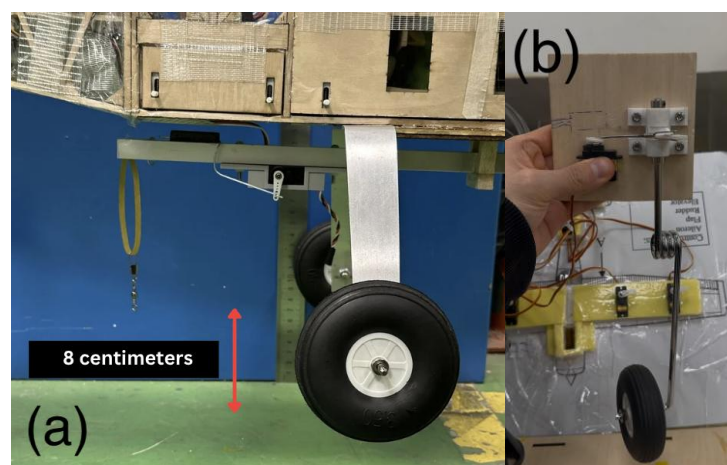


Figure 112 Prototype landing gear dynamic testing showing (a) wheel clearance measurement during controlled drop trials and (b) servo-linked nose gear on a makeshift firewall, used to verify smooth impact absorption and accurate steering response under simulated 5g landing conditions.

7.2.4 Banner Mechanism Deployment Testing

As mentioned in Section 5.3, the initially desirable deployment mechanism was the four-bar linkage bucket. As it was not cantered underneath the fuselage, it shifted the aircraft's centre of gravity toward one side, as shown in Figure 113(a). In the rubber band mechanism, the team discovered that the plastic wires were trapped by the hooks on the metal rod, which prevented the banner from deploying, as shown in Figure 113(b). In the cage design, the mechanism was not rigid enough to hold the banner, as shown in Figure 113(c). For the spool release mechanism, the gap between the pin and the wooden blocks was difficult to reduce to prevent the spool from vibrating vigorously. The metal pin was too soft to be pulled out safely from the blocks. The gripper mechanism was determined to be the final design for the banner deployment mechanism.

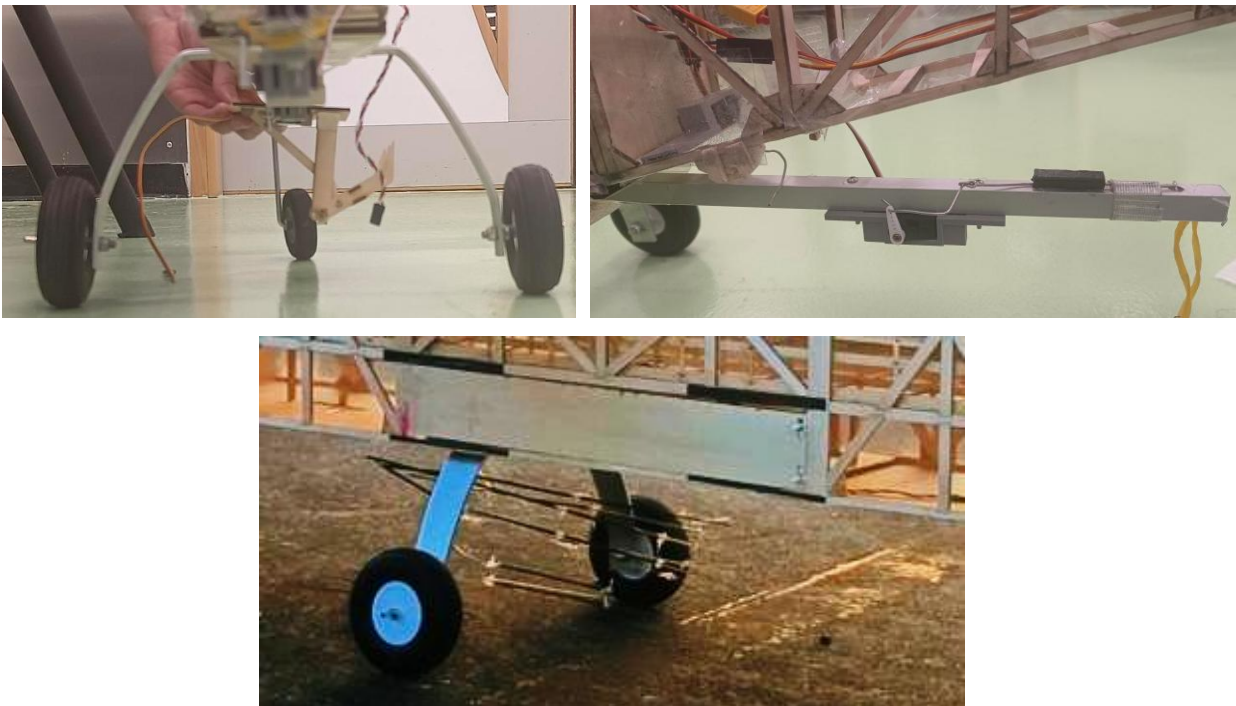


Figure 113 a) Bucket, b) Rubber-band, c) Cage

The team also focused on the impact induced by the drag of the banner during flight and compared different types of materials for the fixture rod attached to the edge of the banner. As shown in Figure 114, the banner was deployed in the air during flight and released as shown in Figure 115. Two flight tests were conducted on 14th and 15th April 2026, with over six banner mission attempts. Except for the first attempt, the banner was successfully deployed in all attempts, with a total number of laps ranging from one to three. The team aimed to fly five laps in the actual flight mission.



Figure 114 Banner fully deployed and towed in the air



Figure 115 Banner release stage. The banner was released on the last lap of flight.

7.2.5 Banner Payload Testing

Both the cylinder-shaped container and the fabric bag container were built and tested in flight, as shown in Figure 116. The cylinder took much longer to make because it required several steps: 3D printing, measuring tolerances, updating the CAD file, and printing again. This process was repeated multiple times until the tolerances were correct. In contrast, the fabric bag was simply bought from a store. As a result, the team chose the fabric bag container for the final competition prototype.



Figure 116 Payload containers: (a) Cylinder-shape container (b) fabric bag shape container

7.3 Flight Test

Several flight tests were conducted to verify the performance of the GRIFFIN. Different telemetry sensors, such as GPS, altimeter, and airspeed sensors, were used to record data during most flights for both prototypes.

Sensors	Usages
Frsky ASS70 ADV Airspeed sensor	Monitor airspeed, ground speed, wind speed and altitude
Frsky GPS ADV sensor	
Radiolink GPS TS100	Perform aerodynamic analysis
Built-in accelerometer in receivers	Obtain the g-forces act on the wing and the landing gear
Frsky FLVS ADV Voltage Sensors	Check the battery usage and warn the pilot when the battery voltage is low
Frsky FAS100 ADV Current Sensors	Obtain energy consumption and check battery capacity used

Table 34 Sensors used in flight tests and their functions

For the GPS sensors, the flight route was recorded and could be imported into Google Earth. The team measured and compared its diameter to the actual diameter (1000 ft). Using Equations (2) and (3), the team refined the prediction of the number of laps the aircraft could fly:

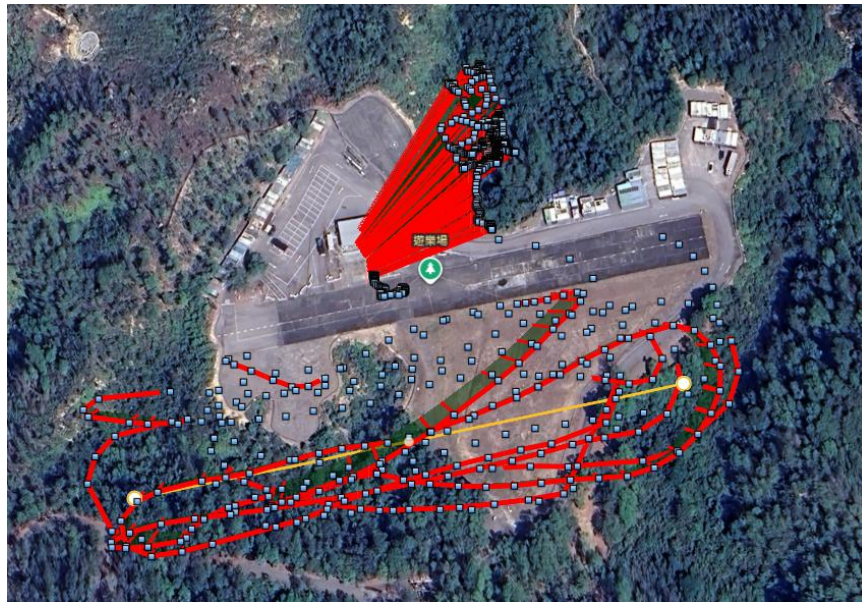


Figure 117 *Flight route recorded by GPS*

$$laps_{predicted} = laps_{observed} \times \frac{lap\ slot\ diameter(ft)}{1000ft} \times \frac{5\ mins}{t_{flight\ test}(min)} \quad (2)$$

$$laps_{predicted} = \frac{distance\ travelled(m)}{predicted\ lap\ distance} \times \frac{5\ mins}{t_{flight\ test}(min)} \quad (3)$$

As required by the AIAA 2025-2026 DBF rules, GPS was used only for data collection during flight tests for design optimization and was not used during mission attempts at the 2026 competition in the USA.

7.4 Flight Checklist

The HKUDBF team used the checklist in Table 35 to ensure that every flight test was safe and valuable. The checklist was updated after each flight test when new design requirements and challenges were discovered. The checklist included a pre-flight checklist, and a similar post-flight checklist was used after each flight test. The team checked, verified, and confirmed the accuracy and thoroughness of the data collected from each flight test. Data were stored in a shared cloud folder among the team to support study and design refinement of the aircraft.

Date:					
Power-off Preflight			Power-on Preflight		
Fuselage	Aircraft skin tear-free	Y/N	Fuse out	No fuse, Propulsion system off	Y/N
	Servos, linkages and horns secured	Y/N	Fuse in, receiver switch off	No response to command	Y/N
	Control surfaces stiffness	Y/N	Range check	Good signal quality at 30 steps away	Y/N
	Landing gears secured	Y/N	Fail-safe	Throttle closed	Y/N

Motors	Motor & firewall secured (check by wrench)	Y/N		Full up elevator	Y/N
Propeller	Prop undamaged	Y/N		Full right rudder	Y/N
	Prop direction (words to front)	Y/N		Full right aileron	Y/N
Batteries	Propulsion battery voltage		Motor	Rotating direction	Y/N
	Receiver battery charged?	Y/N	Take off		
	GPS battery charged	Y/N	Round		
	Remote controller charged	Y/N		Computer connected to GPS?	Y/N
Avionics	ESC/batteries/sensors secured	Y/N	Data Collection	All sensors' data on transmitter?	Y/N
	Servo moved correctly	Y/N		Cameraman ready	Y/N
	No loosen/uninsulated wires	Y/N		Start	
	All wires secured	Y/N		Number of Laps	
M1	No payload	Y/N		End	
	Take off weight			Duration	
M2	Passengers and cargoes placed?	Y/N	Post Flight		
M3	Parts' position	Y/N	Disarm	Transmitter throttle cut	Y/N
	Clamp locked?	Y/N		Propulsion battery disarmed	Y/N
Wing tip test	Wing strength	Y/N		Battery Voltage	
	Cg correct	Y/N	Data Collection	GPS disconnected?	Y/N
	No of weight added (+: nose; -: tail)				
Total take off weight			Visual Inspection	Damage observed	Y/N

Table 35 Flight checklist

7.5 Complete Aircraft Flight Performance

Table 36 compares the complete performance of the four prototypes of Griffin:

Prototype 1 (48 passengers +16 cargoes;4x0.8m banner)				
Properties	M1	M2	M3	GM
Payload (kg)	N/A	0.23	N/A	N/A
Banner (kg)	N/A	N/A	0.25	N/A
Take-off Weight (kg)	3.71	7.39	3.87	N/A
Cruise Speed (m/s)	25.5	N/A	18	N/A
Number of Laps	3	N/A	5	N/A
Mission Time (sec)	86.47	N/A	293	505
Prototype 2 (3 passengers + 1 cargo;4x0.8m banner)				
Properties	M1	M2	M3	GM
Payload (kg)	N/A	0.509	N/A	N/A
Banner (kg)	N/A	N/A	0.25	N/A
Take-off Weight (kg)	2.40	2.63	2.70	N/A
Cruise Speed (m/s)	32	32	18	N/A
Number of Laps	3	9	5	N/A
Mission Time (sec)	93.75	281.25	293	40
Prototype 3 (3 passengers + 1 cargo;4x0.8m banner)				
Properties	M1	M2	M3	GM
Payload (kg)	N/A	0.509	N/A	N/A
Banner (kg)	N/A	N/A	0.25	N/A
Take-off Weight (kg)	2.01	2.24	2.31	N/A
Cruise Speed (m/s)	24	31	15.6	N/A
Number of Laps	3	10	5	N/A
Mission Time (sec)	106	274	272	40
Prototype 4 (3 passengers +1 cargo; 4x0.8m banner)				
Properties	M1	M2	M3	GM
Payload (kg)	N/A	0.23	N/A	N/A

Banner (kg)	N/A	N/A	0.25	N/A
Take-off Weight (kg)	1.92	2.15	2.42	N/A
Cruise Speed (m/s)	32	32	17	N/A
Number of Laps	3	9	3	N/A
Mission Time (sec)	96	250	200	30

Table 36 Complete aircraft flight performance comparison

For Prototype 1, Mission 2's performance could not be recorded because Prototype 1 was lost due to sudden damage to the electronic equipment during a flight test examining the aircraft's performance for Mission 3. For Prototype 2, it completed the predicted number of laps from the preliminary design stage. Both prototypes were tested at an altitude of 20 m. The maximum g-force the prototype withstood during flight testing was 6g. Prototype 3 exhibited similar performance to Prototype 2. The M1 speed for Prototype 3 was lower than expected because calibration was performed during the flight.

Prototype 4 was tested once at the flight site at HKMEC (Hong Kong Model Engineering Club) and tested several more times on two days at the Wichita Radio Control Club in the United States before the competition began. The team was able to perform M1, M2, and M3 and match the expected targets. The expected net income for Mission 2, calculated using the formula from AIAA DBF 2026 rules [1], was estimated as follows:

$$Income = (3 * (6 + (2 * 9))) + (1 * (10 + (8 * 9)))$$

$$Income = 154$$

$$Cost = 9 * (10 + (3 * 0.5) + (1 * 2)) * 0.908$$

$$Cost = 110.322$$

$$Net\ income = Income - Cost$$

$$Net\ income = 43.678$$

8 Competition Results and Discussion

From 10th to 16th April 2026, the team participated in the competition at the Textron Aviation Employee's Flying Club in Wichita, Kansas, USA. The team conducted technical inspection on the first day, Mission 1 on the second day, Mission 2 on the third day, and Mission 3 on the final day. Before each mission began, the team was required to remain in an area called the staging box, where only the assembly crew member could touch the aircraft. The assembly crew member had to load the payload, place the battery and fuse into the aircraft during the five-minute window in the staging box for Mission 2, and lay the banner out flat on the ground and fold it after the time started counting for Mission 3.

The team successfully completed technical inspection on day one, which verified that the aircraft configuration and mechanisms fulfilled the competition requirements. The team completed Mission 1 on day two and flew nine laps in Mission 2 on day three, which matched the estimated number of laps introduced in Section 4. However, the aircraft began landing before it passed the finishing line; therefore, the final lap was not counted. The team received a score of 1.02 out of 2. In Mission 3 on day four, unfortunately, the banner was not stowed appropriately under the aircraft during the five-minute window in the staging box. After the aircraft took off, the banner was released unintentionally from the grippers without pilot input. The banner rod snapped in half in the air under high cruising speed, causing it to behave like a parachute. Therefore, Mission 3 was not successful.

The ground mission was a timed mission in which only the ground crew member could touch the aircraft, load and unload the payloads, fold the banner, and stow it under the aircraft. The team had two attempts on day four, and the best record was 36 seconds, which matched the estimated time for the ground mission mentioned in the MDO framework result in Section 4. As the fastest team completed the entire mission in under seven seconds, the team received a score of 0.13 out of 1, which ranked 16th out of 98 teams.

After adding all the flight mission scores and multiplying by the report score, the team received a total score of 178.77 and ranked 32nd worldwide, second in Hong Kong, and third in Asia. Table 37 compares the performance of Prototype 4 on the final flight test in Hong Kong on 8th April

2026 with the competition performance. The length of one lap was close to 1000 m, which differs from the estimated length mentioned in Section 4, as the actual flight path was much longer than the flight site at HKMEC (Hong Kong Model Engineering Club).

Prototype 4 Competition Performance (3 passengers +1 cargo; 4x0.8m banner)				
Properties	M1	M2	M3	GM
Payload (kg)	N/A	0.23	N/A	N/A
Banner (kg)	N/A	N/A	0.25	N/A
Take-off Weight (kg)	1.92	2.15	2.42	N/A
Cruise Speed (m/s)	29.6	30.6	N/A	N/A
Number of Laps	3	9	N/A	N/A
Mission Time (sec)	98	270	N/A	36

Table 37 Prototype 4 flight test performance and competition performance

Several reasons may have led to the failure of Mission 3 during the competition. First, there was a significant difference in material choices for the banner rod during testing in Hong Kong and in the USA. The team did not carefully pack the banner rod into the shipping box; it was wrapped outside the box and lost during transit between flights. The team had to find other materials at a local hobby shop in the USA to replace the lost carbon fibre tubes. Ultimately, the team used an airfoil-shaped aluminium tube as the banner stand, which may have improved aerodynamic performance and reduced drag. However, carbon fibre rods provide 80% higher tensile stiffness than aluminium tubes [22]. When the banner deployed under high cruising velocity (22 m/s), the drag force endured by the banner rod can be calculated as:

$$\text{Forward force(Thrust)} - \text{Backward force(Drag)} = \text{Force endured by the banner}$$

$$3.5 \times 10 - 6.5 = 29.5N$$

If the aluminium tube could not withstand such force, it would snap in half. However, the team tested the banner mission several times on the two days before the competition started, and the banner was successfully deployed and towed. Therefore, the structural strength of the rod should have been sufficient to withstand the pulling force from the banner. Nevertheless, it is suspected that the structural rigidity of the banner rod may have been damaged over multiple flight trials, causing it to snap in half during the competition flight, as shown in Figure 118.



Figure 118 Snapped banner after flight mission 3

Second, the towing line of the banner may have become entangled on the release mechanism. The towing line was a fishing wire rated to withstand 36.2 kg of force, as shown in Figure 119. There was no issue with the strength of the towline. However, during the five-minute aircraft preparation in the staging box, there was insufficient time for the assembly crew member to check the position of the wire, and the folded banner was not stowed appropriately. After take-off, the banner fell out of the gripper mechanism by itself before the aircraft completed the first lap, and the wire became entangled on the release mechanism, which prevented the banner from opening. Consequently, the banner rod had to withstand not only the pulling force from the aircraft but also the additional tension from the entangled towline. As a result, it snapped in half and turned into a parachute-like object, as shown in Figure 120. With precise control from the pilot, the badge reel and the towline were able to open completely, but the banner was already damaged. Therefore, Mission 3 was counted as a failure.



Figure 119 Fishing wire used for the banner towline in Mission 3, rated to withstand 36.2 kg of force



Figure 120 Parachute-like banner during **Mission 3** after the rod snapped in half

9 Conclusion

The GRIFFIN features a single-motor tractor configuration with tricycle landing gear, a high rectangular wing, and a conventional thin-plate backward-tapered tail on a truss-structured fuselage. The single motor reduces overall weight. The tricycle landing gear provides greater ground clearance for Mission 3 banner towing and improves manoeuvrability. The high wing optimizes payload and avionics compartment design. The conventional tail offers structural simplicity and excellent stability. The team targeted carrying three passengers and one cargo in Mission 2 to maximize speed, and a 4-meter long, 0.8-meter-wide banner in Mission 3 to maximize

the banner score. Construction utilized plywood, balsa, and CFRP strips and rods; the banner mechanism employed lightweight tricot fabric.

The team progressed through conceptual, preliminary, and detailed design stages. The conceptual design established the methodology, mission requirements, and configuration. The preliminary design addressed problem statements, sizing, trade studies, performance simulations, and mission score predictions for different prototypes. The detailed design specified all aircraft parts and mechanism working principles. Prototype manufacturing and testing followed to select the optimal design.

Throughout the process, failsafe analysis and uncertainties prompted iterative redesigns, simulations, and prototype modifications. Prototype 1 prioritized maximum payload carriage (maximum hockey pucks and plastic ducks) for the Mission 2 score. However, MDO results shifted Prototype 2 toward weight minimization to increase speed across all missions. Prototype 3 introduced a curved fuselage, but simulations and flight tests confirmed that Prototype 2 outperformed it in manoeuvrability, speed, and Mission 3 performance. Prototype 2's simpler manufacturing allowed completion within 1.5 weeks and final flight validation, making it the basis for the final prototype.

The team completed numerous analyses, manufacturing, and testing throughout the nine-month project and conducted a four-day competition in the USA in April 2026. The models and analysis results constructed by the team will serve as useful resources for the next AIAA DBF team in 2027. The team's work will continue to provide valuable support for future generations to iterate and develop more innovative, efficient, and high-performing aircraft.

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